ns. For instance,
we are still finding new ways to exploit the wheel (just think of the iPod click wheel). By the same token, what humanity will achieve thanks to digital technologies is unimaginable. As I stress in Chapter 1, nobody in 1964 could have guessed what the world would have been like only fifty years later. Futurologists are the new astrologers; we should not rely on them. And yet, it is also true that the digital revolution will happen only once, and it is happening now. This page in human history has been turned, and a new chapter has begun. Future generations will never know what an exclusively analogue, offline, pre--digital reality was like. We are the last generation to have experienced it.
The price for such a special place in history is worrying uncertainties. The transformations brought about by digital technologies are mind--blowing. They justify some confusion and apprehension; one only needs to look at newspaper headlines. However, our special place at this historical watershed, between a fully analogue and an increasingly digital reality, also brings extraordinary op-por-tunities. Precisely because the digital revolution has just begun, we have a chance to shape it in positive ways that can benefit both humanity and our planet. As Winston Churchill once said, ?We shape our buildings; thereafter they shape us.? We are at the very early stage of the construction of our digital realities. We can get them right before they start affecting and influencing both us and future generations in the wrong way. It is not a matter of being pessimistic or optimistic. Discussing whether the glass is half empty or half full is pointless. The interesting question is how we can fill it. This means engaging constructively with the ethical analysis of the problems and the design of the right solutions.
To identify the best path ahead for the development of our digital technologies, the first essential step is more and better understanding. We should not sleepwalk

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into the creation of an increasingly digital world. The insomnia of reason is vital because its sleep generates monstrous mistakes, sometimes irreversible. Understanding the technological transformations happening before our eyes is essential if we wish to steer the digital revolution in a direction that is both socially preferable (equitable) and environmentally sustainable. This can only be a collaborative effort (Floridi forthcoming). Thus, in this book, I offer my contribution by sharing some ideas about one kind of digital technology, artificial intelligence (AI), and the specific issue of its ethics.
The book is part of a larger research project on the transformations of agency (the ability to interact with, and learn from, the world successfully in view of a goal) brought about by the digital revolution. I initially thought I could work on both AI?-understood as artificial agency, the topic of this book?-and political agency?-understood as collective agency, supported and influenced by digital interactions. When I was invited to give the Ryle Lectures in 2018, I attempted to do exactly this. I presented both topics as two aspects of the same, more profound transformation. I was, perhaps kindly, told by organizers and attendees that it was not a failure. But personally, I did not find it a great success. This was not because approaching the ethics of AI and the politics of information

from a single, agency--
based point of view does not work, but because doing so works well only if one is willing to skip the details and exchange scope for depth. This may be fine in a lecture series, but covering both topics in a single research monograph would have produced a doorstopper of even lower appeal than this book may have. So, following the insightful advice of Peter Momtchiloff at Oxford University Press, I decided to split the project into two: this book on the ethics of AI, and a second book on the politics of information. This is the right place to let the reader know where to locate both books within the larger project.
This book is part one of the fourth volume in a tetralogy that includes The Philosophy of Information (volume one, Floridi 2011), The Ethics of Information (volume two, Floridi 2013), and The Logic of Information (volume three, Floridi 2019d). I began labelling the tetralogy Principia Philosophiae Informationis not as a sign of bottomless hubris (although it may well be), but as an internal pun among some colleagues. In a sort of rowing--like competition, I joked that it was time for Oxford to catch up with Cambridge on the 3?0 score for ?principia?. This was not a pun that many found funny or even intelligible.
Within the Principia project, this book occupies a middle ground between the first and the second volume (not unlike volume three) because epistemology, ontology, logic, and ethics all contribute to the development of the theses presented in the following chapters. But as the reader should rightly expect, all the volumes are written as stand--alone, so this book can be read without any knowledge of anything else that I ever published. Still, the volumes are complementary. The essential message from volume one is relatively straightforward: semantic information is well--formed, meaningful, and truthful data; knowledge is relevant

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semantic information correctly accounted for; humans are the only known semantic engines and conscious informational organisms who can design and understand semantic artefacts, and thus develop a growing knowledge of reality and themselves, as semantic capital; and reality is best understood as the totality of information (notice the crucial absence of ?semantic?).
Against that background, volume two investigates the foundations of the ethics of informational organisms (inforgs) like us, which flourish in informational en-vir-on-ments (the infosphere1) and are responsible for their construction and well--being. In short, volume two is about the ethics of inforgs in the infosphere who increasingly experience life as onlife (Floridi 2014b)?both online and offline, both analogue and digital. In a classic Kantian move, we are thus shifting from theoretical to practical (in the sense of praktischen, not in the sense of useful or applied) philosophy. The third volume focuses on the conceptual logic of semantic information as a model. This is linked to the epistemological analysis provided in The Philosophy of Information. Insofar as the volume focuses on the conceptual logic of semantic information as a blueprint, it offers a bridge toward the normative analysis provided in The Ethics of Information.
The third volume discusses, among other things, what duties, rights, and responsibilities are associated with the poietic practices that characterize our existence?-from making sense of the world to

changing it according to what we
consider morally good and normatively right. Working like a hinge between the two previous books, the third volume prepares the basis for volume four on The Politics of Information, of which this book is the first part. Here, the epis-temo-logic-al, normative, and conceptual constructionism developed in the previous volumes supports the study of the design opportunities available to us. These are opportunities to understand and shape what I call ?the human project? in our information societies, through the proper design of new forms of artificial and political agency. The key theses of this book are that AI is made possible by the decoupling of agency and intelligence, hence that AI is better understood as a new form of agency, not intelligence, that AI is, therefore, an amazing revolution but in a pragmatic and not a cognitive sense, and that the concrete and pressing challenges and opportunities concerning AI emerge from the gap between agency and intelligence, which will keep widening as AI becomes increasingly successful. Overall, the four volumes may be understood as seeking to invert what I believe are several misconceptions. The misconceptions are easily explained using the classic communication model, introduced by Shannon, comprised of sender,

1? ?Infosphere? is a word that I hijacked years ago to refer to the whole informational environment constituted by all informational entities (including informational agents), their properties, inter-actions, processes and mutual relations. It is an environment comparable to?-yet also different from?-cyberspace (which is only one of its sub--regions) because it also includes offline and analogue spaces of information. It is an environment, and hence a concept, that is rapidly evolving. See https://en.wikipedia.org/wiki/Infosphere.

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message, receiver, and channel (Shannon and Weaver 1975, 1998). Epistemology focuses too much on the passive receiver/consumer of knowledge when it should be concerned with the active sender/producer. It should shift from mimesis to poiesis. This is because knowing is designing. Ethics focuses too much on the sender/agent, when it should be concerned with the receiver/patient and, above all, the relation between sender and receiver. This is because care, respect, and tolerance are the keys to being good. Metaphysics focuses too much on the relata, the sender/producer/agent/receiver/consumer/patient (which it conceives as entities), when it should be concerned with the message/relations. This is because dynamic structures constitute the structured. Logic focuses too much on chan-nels of communication as supporting, justifying, or grounding our conclusions, when it should also be concerned with the channels that enable us to extract (and transfer) information from a variety of sources reliably. This is because the logic of information design is a logic of constraints rather than a logic of things as bear-ers of predicates. AI, or at least its philosophy, focuses too much on the engineer-ing of some kind of biological--like form of intelligence, when it should be concerned with the engineering of artefacts that can operate successfully without any need for intelligence. This is because AI is not a marriage but a divorce between the ability to solve a problem or deal with a task successfully in view of a goal and the need to be intelligent while doing so.

As my phone playing chess
better than anyone I know illustrates clearly enough, AI is the continuation of intelligent behaviour by other means. And finally, politics (the topic of the second half of the fourth volume) is not about managing our res publica, but about caring for the relations that make us social?-our ratio publica. I shall be amazed if even one of these U--turns in our philosophical paradigms will be successful.
Let me now turn to a brief overview of the contents of this book. The task of this volume is still contributing, like the previous ones, to the development of a philosophy of our time for our time, as I have written more than once. As in the earlier volumes, it does so systematically (conceptual architecture is pursued as a valuable feature of philosophical thinking), rather than exhaustively, by pursuing two goals.
The first goal is meta--theoretical and is fulfilled by Part One of the volume, which comprises the first three chapters. There, I offer an interpretation of the past (Chapter 1), the present (Chapter 2), and the future of AI (Chapter 3). Part One is neither an introduction to AI in a technical sense, nor some kind of AI for beginners. There are many excellent books on this already, and I would recommend the classic Russell and Norvig (2018) to anyone interested. It is instead a philosophical interpretation of AI as a technology. As I have already mentioned, the central thesis developed there is that AI is an unprecedented divorce between agency and intelligence. On this basis, Part Two of the volume pursues not a meta--theoretical but a theoretical investigation of the consequences of the divorce discussed earlier. This is the second goal. The reader should not expect a textbook

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treatment of all the main ethical problems affecting AI. Many books already cover all the relevant topics systematically, and I would recommend among them: Dignum (2019), Coeckelbergh (2020), Moroney (2020), Bartneck et al. (2021), Vieweg (2021), Ammanath (2022), Blackman (2022); and the following handbooks: Dubber, Pasquale, and Das (2020), DiMatteo, Poncibò, and Cannarsa (2022), Voeneky et al. (2022). Instead, Part Two develops the idea that AI is a new form of agency that can be harnessed ethically and unethically. More specifically, in Chapter 4, I offer a unified perspective on the many principles that have already been proposed to frame the ethics of AI. This leads to a discussion in Chapter 5 of the potential risks that may undermine the application of these principles, then an analysis of the relation between ethical principles and legal norms, as well as the definition of soft ethics as post--compliance ethics in Chapter 6. After these three chapters, I analyse the ethical challenges caused by the development and use of AI (Chapter 7), evil uses of AI (Chapter 8), and good practices when applying AI (Chapter 9). The last group of chapters is dedicated to the design, development, and deployment of AI for social good or AI4SG. Chapter 10 discusses the nature and features of AI4SG. In Chapter 11, I reconstruct the positive and negative impacts of AI on the environment and how it can be a force for good in the fight against climate change?-but not without risks and costs, which can and must be avoided or minimized. In Chapter 12, I expand the analysis presented in Chapters 9 and 10 to discuss the possibility of using AI in support of the Un

ited
Nations (Cowls, Png, and Au) seventeen Sustainable Development Goals (SDGs). There I present the Oxford Initiative on AIxSDGs, a project I directed and completed in 2022. In Chapter 13, I conclude by arguing in favour of a new marriage between the Green of all our habitats and the Blue of all our digital technologies. This marriage can support and develop a better society and a healthier biosphere. The book ends with some references to concepts that will take centre stage in the next book, The Politics of Information, dedicated (as mentioned above) to the impact of digital technologies on socio--political agency. All the chapters are strictly related. So I have added internal references whenever they might be helpful. As an anonymous reviewer pointed out, they could be read in a slightly different order. I agree.
As with the previous volumes, this too is a German book in terms of its philosophical roots. It is written from a post--analytic/continental divide perspective, which I believe is fading away. The careful reader will easily place this work within the tradition linking pragmatism (especially Charles Sanders Peirce) with the philosophy of technology (especially Herbert Simon).2 Unlike volume one and even more than volumes two and three, this fourth volume is less neo--Kantian than I expected it to be. And contrary to volumes two and three, it is also less
2? The reader interested in exploring these connections may wish to consult Allo (2010), Demir (2012), and Durante (2017).

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Platonic and Cartesian. In short, writing it has made me aware that I am moving out of the shadows of my three philosophical heroes. This was not planned, but it is what happens when one follows one?s own reasoning wherever it leads. Amici Plato, Cartesius et Kant, sed magis amica veritas. In The Ethics of Information, I wrote that ?some books write their authors?. I now have the impression that only bad books are fully controlled by their authors; they are airport bestsellers.
With respect to the preceding volumes, the main difference is that I am now increasingly convinced that philosophy at its best is conceptual design. Conceptual design delivers purposeful projects (understanding the world to improve it) and semanticization (giving sense and meaning to Being, while taking care of, and enriching, humanity?s semantic capital). It all started by realizing the obvious, thanks to a specific case concerning a very famous Oxford philosopher. Locke?s true legacy is his political thought, not his epistemology. Maybe Kant did not mean to mislead us into believing that epistemology and ontology are the queens of the philosophical kingdom, but that is the way I was educated to think about modern philosophy. And maybe neither Wittgenstein nor Heidegger thought that logic, language, and their philosophies should replace the two queens as their only legitimate heirs, but that is also the way I was educated to think about contemporary philosophy. Either way, today I no longer place any of these disciplines at the centre of the philosophical enterprise. I rather look to ethics, political philoso-phy, and the philosophy of law. It is the searching, understanding, shaping, implementing, and negotiating of the morally good and right that is the core of philosophical reflection. All else is part of the necessary journey to reach such a place, but it should not be confused with the pla

ce itself. Philosophical founda-
tionalism (what grounds what) is crucial but only in view of philosophical eschat-
ology (what drives what). All good philosophy is eschatological.
Regarding the style and structure of this book, I may repeat here what I wrote
in the preface of all the previous volumes. I remain painfully aware that this is not
a page--turner, to put it mildly, despite my attempts to make it as interesting and
reader--friendly as possible. I remain convinced that esoteric (in the technical
sense) research in philosophy is the only way to develop new ideas. But exoteric
philosophy has its crucial place. It is like the more accessible and relevant tip of
the more obscure yet necessary part of the iceberg under the surface of everyday
life. The reader interested in a much lighter reading may wish to check The Fourth
Revolution: How the Infosphere is Reshaping Human Reality (Floridi 2014a) or
perhaps the even easier Information: A Very Short Introduction (Floridi 2010b).
As I wrote before, unfortunately, this book too requires not only some patience
and a bit of time, but also an open mind. These are scarce resources. Over the last
three decades or so of debates, I have been made fully aware?-sometimes in ways
much less friendly than I wish?-that some of the ideas defended in this as well as
in the previous volumes are controversial. They are not meant to be so on pur-
pose. At the same time, I have also noticed how often mistakes are made by

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relying on ?systemic attractors?: if a new idea looks a bit like an old idea we already
have, then the old one is a magnet to which the new one is powerfully attracted,
almost irresistibly. We end up thinking that ?that new? is just like ?this old? and
hence either ?this old? can be dismissed or, if we do not like ?this old?, then we dis-
like ?that new? as well. This is bad philosophy indeed, but it takes mental strength
and exercise to resist such a powerful shift. I know. I speak as a sinner and as a
knower of many sinners. In the case of this book, I am concerned that some
-readers may be tempted to conclude that it is an anti--technology book, a book in
which I indicate the limits of AI or what ?AI cannot do?. They may also conclude
the opposite?-that this book is too optimistic about technology, too much in love
with the digital revolution and AI as a panacea. Both conclusions are wrong. The
book is an attempt to stay in the middle, in a place that is neither hell nor para-
dise, but the laborious purgatory of human efforts. Of course, I would be disap-
pointed if I were told that I failed despite my attempt. But I would be even more
disappointed and frustrated if the attempt were misunderstood. There are many
ways to appreciate technology. One of them is in terms of good design and ethical
governance, and I believe this is the best approach. The reader does not have to
follow me that far, but no mistake should be made about the direction I am taking.
As in the previous volumes, I have provided summaries and conclusions at the
beginning and end of each chapter, along with some redundancy, to help the
reader access the contents of this book more easily. Regarding the first feature,
I know it is slightly unorthodox. But the solution of starting each chapter with a
?previously in Chapter x?.?.?.? should enable the reader to browse the text, or fast
forward through entire chapters of it, without losing the essent

ial plot. Fans of
science fiction who recognize the reference to Battlestar Galactica, which is still one of the best series I have ever watched, may consider this fourth volume as equivalent to season four. And by the way, I did try to convince a past copyeditor to let me use the phrase ?previously, on Chapter x?.?.?.?, but this seemed to be too much of a linguistic stretch. One of the anonymous reviewers suggested removing the short summary at the beginning of each chapter. I decided to keep them partly because I believe they help at best and do no harm at worst, and partly because it is a feature of all the books I have published in this extended project.
Regarding the second feature, while editing the final version of the book, I decided to leave some repetition and rephrasing of recurrent themes in the chapters whenever I thought that the place where the original content had been introduced was too distant in terms of either pages or theoretical context. If sometimes the reader experiences some déjà vu, I hope it will be to the advantage of clarity?-as a feature, not a bug.
A final word now on what the reader will not find in the following pages. This is not an introduction to AI or the ethics of AI. Nor do I seek to provide an exhaustive investigation of all the issues that could fall under the label of ?the -ethics of AI?. The anonymous reviewers recommended to remove several short

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chapters in which I sought to apply the ideas developed in Part Two, and provide them online. It was a painful exercise. The interested reader will find them on SSRN, under my name. Some are collected in Floridi (2021). I hope to work more extensively on an ethics--based auditing of AI and the ethical procurement of AI (two crucial topics that are still underexplored) and I have left more geopolitical considerations about the policies of AI to The Politics of Information. The interested reader may also wish to see Cath et al. (2018) on the United States (US), European Union (EU), and United Kingdom (Floridi et al. 2018) approaches to AI, for example, or Roberts, Cowls, Morley et al. (2021) and Hine and Floridi (2022) for the Chinese approach. This is also not a book on the statistical and computational aspects of so--called FAT (fairness, accountability, and transparency) or XAI (explainable AI) issues, or on the legislation concerning them. These topics are only touched upon in the following chapters.3 This is a philosophical book about some of the roots?-and none of the leaves?-of some of the AI issues of our time. It is about a new form of agency, its nature, scope, and challenges. And it is about how to harness this agency to the benefit of humanity and the environment.
3? For more information on these topics see Watson and Floridi (2020), Lee and Floridi (2020), Lee, Floridi, and Denev (2020).

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Acknowledgements
Given the first three volumes, I am bound to acknowledge here some things that I have already acknowledged in the past. Still, the repetition only reinforces my deep gratitude.
As before, I could not have worked on such a long--term project without dividing it into some feasible and much smaller tasks. I am glad to see that so many different components fit together in the end. Of course, I hope this is the symptom of a well--planned project. But I am afraid the older I get, the more likely it is that it may be due to some mental

resistance to change (also known as sclerosis).
Old people do not change their minds easily. And repeating this comment year after year does not make me feel any more reassured.
As for the preceding three volumes, much of the content presented in them was initially tested as presentations at meetings (conferences, workshops, conventions, seminars, lectures, etc.) or (often an inclusive or) journal articles. For this, the bibliographic details are in the list of references. Such a systematic and gradual way of working is laborious, but it seems to be fruitful, and it may also be in-ev-it-able given the innovative nature of the field. It does require a perseverance and commitment that I hope are not ill--exercised. I wished to assess the ideas presented in this volume as thoroughly as possible. Presenting the material at meetings as well as publishing corresponding articles gave me the opportunity and privilege to enjoy a vast amount of feedback from a very large number of excellent colleagues and anonymous referees. If I do not thank all of them here, it is not for lack of manners or mere reason of space; the appropriate acknowledgements can be found in the corresponding publications.
However, there are some people that I would like to mention explicitly because they played a significant role throughout the project and during the revisions of the final text. First of all, Kia. She and I have been married for as long as I have been working on the whole Principia project. Without her, I would have never had the confidence to undertake such a task or the spiritual energy to complete it. It requires a huge amount of serenity to invest so much time in philosophical thinking, and Kia is my muse. I am repeating again something I already wrote in the previous volumes: she continues to make our life blissful, and I am very grateful to her for the endless hours we spend talking about the topics of the Principia, for all her sharp suggestions, and for her lovely patience with an utterly obsessed, single--minded, overly focused, constantly travelling husband who must be unbearable to anyone else. I wish I could claim to have contributed to her research in neuroscience even half as much as she has to my philosophical development.

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It is an immense privilege to be able to receive advice from a brilliant mind and elegant thinker. Between the second and this third volume, we moved into the house of our dreams in Oxfordshire. It is the perfect place to think. And during the pandemic, we enjoyed a year of honeymoon. It was priceless.
Nikita Aggarwal, Ben Bariach, Alexander Blanshard, Tim Clement--Jones, Josh Cowls, Sue Daley, Massimo Durante, Emmie Hine, Joshua Jaffe, Thomas C. King, Michelle Lee, Jakob Mökander, Jessica Morley, Claudio Novelli, Carl Öhman, Ugo Pagallo, Huw Roberts, David Sutcliffe, Mariarosaria Taddeo, Andreas Tsamados, Vincent Wang, David Watson, Marta Ziosi, as well as many students, members, and visitors of the Digital Ethics Lab, our research group in Oxford, and the Centre for Digital Ethics, in Bologna, were very generous with their time and ideas during the past years. They provided numerous opportunities for discussion and further reflection on the topics analysed in this book. They also saved me from embarrassing mistakes more times than I can recall. Much of the research leading to this book was done in collaboration with them, to the exte

nt that they
could all be considered co--authors insofar as any intellectual value of the following chapters is concerned. I remain the only person responsible for the shortcomings. More specifically, many chapters are based on previous publications, and this is the list: Chapter 3 on Floridi (2019f) and Floridi (2020a); Chapter 4 on Floridi and Cowls (2019); Chapter 5 on Floridi (2019e); Chapter 6 on Floridi (2018a); Chapter 7 on Tsamados et al. (2020); Chapter 8 on King et al. (2019); Chapter 9 on Floridi et al. (2020); Chapter 10 on Floridi et al. (2018) and Roberts, Cowls, Hine, Mazzi, et al. (2021); Chapter 11 on Cowls et al. (2021a); Chapter 12 on Floridi and Nobre (2020). I am most grateful to all the co--authors for our fruitful collaborations and for their permission to reuse some of our work.
I have learnt a lot from the colleagues with whom I interacted, though I wish I had been able to learn more. An important difference with respect to the previous three volumes is that, in this case, I also learnt much from my interactions with experts working at companies such as DeepMind, Deloitte, EY, Facebook, Fujitsu, Google, IBM, IVASS, Intesa Sanpaolo McKinsey, Microsoft, Vodafone, SoftBank, and many others. I likewise learnt from institutions, such as the European Commission, the European Data Protection Supervisor (EDPS Ethics Advisory Group), the Council of Europe, the House of Commons, the House of Lords, the Centre for Data Ethics and Innovation, the Digital Catapult, the Information Commissioner?s Office (ICO), the Alan Turing Institute, the Financial Conduct Authority (FCA), the Vodafone Institute, the Audi Foundation Beyond, Atomium?-European Institute for Science, and more. The importance and significance of some of the problems I discuss in this book would have been less clear to me if it had not been for the reality check provided by the world of business, politics, and policies in general.
I have already said that Peter Momtchiloff was pivotal in the realization of all three volumes (for both his foresighted invitation to publish them with OUP as

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well as his support and firm patience when it seemed I would never complete them). The same holds true for this volume as well, with extra special thanks that I wish to reiterate here: he was the one who made me realize that it would be better to publish The Ethics of Artificial Intelligence and The Politics of Information as two separate volumes.
Danuta Farah and Krystal Whittaker, my personal assistants, provided exceptional support and impeccable managerial skills without which I could not have completed this project.
The research leading to this book was supported in different ways and degrees by several grants in the past years. I wish to acknowledge here the following sources: the Alan Turing Institute; Atomium-?-European Institute for Science, Media and Democracy; Engineering and Physical Sciences Research Council (EPSRC); European Commission, Horizon (2020); European Commission, Marie Sk?odowska--Curie Fellowship Program; Facebook; Fujitsu; Google; Microsoft; Tencent; the University of Oxford John Fell Fund; and the Alma Mater?-Università di Bologna.
In terms of disclosure, I have been involved in many initiatives and projects related to the topics covered in this book. I am afraid the list is long, but here it is in chronological order since 2014: Chair, Ethics Board, Cluste

s. This reality is one that millennials have
never experienced, having been born after the early 1980s. To them, a world without digital technologies is like what a world without cars was for me (-coincidentally, I was born in 1964): something I had only heard described by my grandmother.

It is often remarked that a smart phone packs far more processing power in a few inches than NASA could put together when Neil Armstrong landed on the moon five years after Robinson Crusoe on Mars, in 1969. We have all this power at an almost negligible cost. For the fiftieth anniversary of the moon landing in 2019, many articles ran comparisons and here are some staggering facts. The Apollo Guidance Computer (AGC) on board of Apollo 11 had 32,768 bits of random--access memory and 589,824 bits (72 KB) of read--only memory. You could not have stored this book on it. Fifty years later, your average phone came with 4 GB of RAM and 512 GB of ROM. That is about 1 million times more RAM and 7 million times more ROM. As for the processor, the AGC ran at 0.043 MHz. An average iPhone processor is said to run at about 2490 MHz, which is about 58,000 times faster. To get a better sense of the acceleration, perhaps another comparison may help. On average, a person walks at a pace of 5 km/h. A hyper-sonic jet travels slightly more than a thousand times faster at 6,100 km/h, just over five times the speed of sound. Imagine multiplying that by 58,000.

Where did all this speed and computational power go? The answer is twofold: feasibility and usability. In terms of applications, we can do more and more. We can do so in increasingly easy ways?-not only in terms of programming but above all in terms of user experience. Videos, for example, are computationally very hungry. So are operating systems. AI is possible today also because we have the computational power required to run its software.

Thanks to this mind--boggling growth in storage and processing capacities at increasingly affordable costs, billions of people are connected today. They spend many hours online daily. According to Statista.com, for example, ?as of 2018, the average time spent using the internet [in the UK] was 25.3 hours per week. That was an increase of 15.4 hours compared to 2005.?1 This is far from unusual, and the pandemic has made an even more significant difference now. I shall return to

1? https://www.statista.com/statistics/300201/hours--of--internet--use--per--week--per--person--in--the--uk/.

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this point in Chapter 2, but another reason why AI is possible today is because we are increasingly spending time in contexts that are digital and AI--friendly.

More memory, power, speed, and digital environments and interactions have generated more data in immense quantities. We have all seen diagrams with exponential curves indicating amounts that we do not even know how to im-agine. According to the market intelligence company IDC,2 the year 2018 saw humanity reach 18 zettabytes of data (whether created, captured, or replicated). The aston-ishing growth in data shows no sign of slowing down; apparently, it will reach 175 zettabytes in 2025. This is hard to grasp in terms of quantity, but two consequences deserve a moment of reflection.3 First, the speed and the memory of our digital technologies are not growing at the same pace as the universe of data. So, we are quickly moving from a culture of recording t

o one of deleting; the question is no
longer what to save, but what to delete to make room for new data. Second, most of the available data have been created since the nineties (even if we include every word uttered, written, or printed in human history and every library or archive that ever existed). Just look at any of those online diagrams illustrating the data explosion: the amazing side is not only on the right, where the arrow of growth goes, but also on the left, where it starts. It was only a handful of years ago. Because all the data we have were created by the current generation, they are also aging together in terms of support and obsolete technologies. So, their curation will be an increasingly pressing issue.

More computational power and more data have made the shift from logic to statistics possible. Neural networks that were once only theoretically interesting4 have become common tools in machine learning (ML). Old AI was primarily symbolic and could be interpreted as a branch of mathematical logic, but the new AI is mostly connectionist and can be interpreted as a branch of statistics. The main battle horse of AI is no longer logical deduction, but statistical inference and correlation.

Computational power and speed, memory size, data volumes, the effects of algorithms and statistical tools, and the number of online interactions have all been growing incredibly fast. This is also because (here, the causal connection goes both ways) the number of digital devices interacting with each other is already several times higher than the human population. So, most communication is now machine to machine with no human involvement. We have computerized robots on Mars remotely controlled from Earth. Commander Christopher ?Kit? Draper would have found them utterly amazing.

All the previous trends will keep growing, relentlessly, for the foreseeable future. They have changed how we learn, play, work, love, hate, choose, decide,

2? See the discussion in https://www.seagate.com/gb/en/our--story/data--age--2025/.

3? I discuss both in Floridi (2014a).

4? I discussed some of them in a much earlier book from the late nineties (Floridi 1999).

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produce, sell, buy, consume, advertise, have fun, care and take care, socialize, communicate, and so forth. It seems impossible to locate a corner of our lives that has not been affected by the digital revolution. In the last half century or so, our reality has become increasingly digital. It is made up of zeros and ones, and run by software and data rather than hardware and atoms. More and more people live increasingly onlife (Floridi 2014b), both online and offline, and in the infosphere, both digitally and analogically.

This digital revolution also affects how we conceptualize and understand our realities, which are increasingly interpreted in computational and digital terms. Just think of the ?old? analogy between your DNA and your ?code?, which we now take for granted. The revolution has fuelled the development of AI as we share our onlife experiences and our infosphere environments with artificial agents (AA), whether algorithms, bots, or robots. To understand what AI may represent?-I shall argue that it is a new form of agency, not of intelligence?-more needs to be said about the impact of the digital revolution itself. That is the task of the rest of this chapter. It is only by understanding

the conceptual trajectory of
its implications that we can gain the right perspective about the nature of AI (Chapter 2), its likely developments (Chapter 3), and its ethical challenges (Part Two).

1.2? Digital?s Cleaving Power: Cutting and Pasting Modernity

Digital technologies, sciences, practices, products, and services?-in short, the digital as an overall phenomenon?-are deeply transforming reality. This much is obvious and uncontroversial. The real questions are why, how, and so what, especially as they relate to AI. In each case, the answer is far from trivial and certainly open to debate. To explain the answers I find most convincing, and thus introduce an interpretation of AI as an increasing reservoir of smart agency in the next chapter, let me start in medias res, that is, from the ?how?. It will then be easier to move backwards to understand the ?why? and then forward to deal with the ?so what? before linking the answers to the emergence of AI.

The digital ?cuts and pastes? our realities both ontologically and epis-temo-log-ic-al-ly. By this I mean that it couples, decouples, or recouples features of the world (-ontology) and therefore our corresponding assumptions about them (epistemology) that we thought were unchangeable. It splits apart and fuses the ?atoms? of our ?modern? experience and culture, so to speak. It reshapes the bed of the river, to use a Wittgensteinian metaphor. A few stark examples may bring home the point vividly.

Consider first one of the most significant cases of coupling. Self--identity and personal data have not always been glued together as indistinguishably as they are today, when we speak of the personal identity of data subjects. Census counts are

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very old (Alterman 1969). The invention of photography had a huge impact on privacy (Warren and Brandeis 1890). European governments made travelling with a passport compulsory during the First World War, for migration and se-curity reasons, thus extending the state?s control over the means of mobility (Torpey 2000). But it is only the digital, with its immense power to record, monitor, share, and process boundless quantities of data about Alice, that has soldered together who Alice is, her individual self and profile, with personal information about her. Privacy has become a pressing issue also, if not mainly, because of this coupling. Today, in EU legislation at least, data protection is discussed in terms of human dignity (Floridi 2016c) and personal identity (Floridi 2005a, 2006, 2015b) with citizens described as data subjects.

The next example concerns location and presence and their decoupling. In a digital world, it is obvious that one may be physically located in one place (say, a coffee shop) and interactively present in another (say, a page on Facebook). Yet all past generations that lived in an exclusively analogue world conceived and ex-peri-enced location and presence as two inseparable sides of the same human predicament: being situated in space and time, here and now. Action at a distance and telepresence belonged to magical worlds or science fiction. Today, this decoupling simply reflects ordinary experience in any information society (Floridi 2005b). We are the first generation for which ?where are you?? is not just a rhetorical question. Of course, the decoupling has not severed all links. Geolocation works only if Alice?s telepresence ca

n be monitored. And Alice?s telepresence is possible
only if she is located within a physically connected environment. But the two are now completely distinguished and, indeed, their decoupling has slightly downgraded location in favour of presence. Because if all Alice needs and cares about is to be digitally present and interactive in a particular corner of the infosphere, it does not matter where in the world she is located analogically, whether at home, on a train, or in her office. This is why banks, bookstores, libraries, and retail shops are all presence places in search of a location repurposing. When a shop opens a cafeteria, it is trying to paste together the presence and the location of customers?-a connection that has been cut by the digital experience.
Consider next the decoupling of law and territoriality. For centuries, roughly since the 1648 Peace of Westphalia, political geography has provided jurisprudence with an easy answer to the question of how widely a ruling should apply: it should apply as far as the national borders within which the legal authority operates. That coupling could be summarized as ?my place, my rules; your place, your rules?. It may now seem obvious, but it took a long time and immense suffering to reach such a simple approach. The approach is still perfectly fine today, if one is operating only within a physical, analogue space. However, the Internet is not a physical space. The territoriality problem arises from an ontological misalignment between the normative space of law, the physical space of geography, and the logical space of the digital. This is a new, variable ?geometry? that we are still

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learning to manage. For instance, the decoupling of law and territoriality became obvious as well as problematic during the debate on the so--called right to be forgotten (Floridi 2015a). Search engines operate within an online logical space of nodes, links, protocols, resources, services, URLs (uniform resource locators), interfaces, and so forth. This means anything is only a click away. So, it is difficult to implement the right to be forgotten by asking Google to remove links to someone?s personal information from its .com version in the United States due to a decision taken by the Court of Justice of the European Union (CJEU), even if that decision may seem pointless, unless the links are removed from all versions of the search engine.
Note that such misalignment between spaces both causes problems and provides solutions. The non--territoriality of the digital works wonders for the unobstructed circulation of information. In China, for example, the government must make a constant and sustained effort to control information online. Along the same lines, the General Data Protection Regulation (GDPR) must be admired for its ability to exploit the ?coupling? of personal identity and personal information to bypass the ?decoupling? of law and territoriality. It does so by grounding the protection of personal data in terms of the former (to whom they are ?attached?, which is now crucial) rather than the latter (where they are being processed, which is no longer relevant).
Finally, here is a coupling that turns out to be more accurately a recoupling. In his 1980 book The Third Wave, Alvin Toffler coined the term ?prosumer? to refer to the blurring and merging of the role of producers and consume

rs (Toffler
1980). Toffler attributed this trend to a highly saturated marketplace and mass of standardized products prompting a process of mass customization, which in turn led to the increasing involvement of consumers as producers of their own customized products. The idea was anticipated in 1972 by Marshall McLuhan and Barrington Nevitt, who attributed the phenomenon to electricity--based -technologies. It later came to refer to the consumption of information produced by the same population of producers, such as on YouTube. Unaware of these precedents, I introduced the word ?produmer? to capture the same phenomenon almost twenty years after Toffler.5 But in all these cases, what is at stake is not a new coup-ling. More precisely, it is a recoupling.

For most of our history (approximately 90 per cent of it; see Lee and Daly (1999)), we have lived in hunting--gathering societies, foraging to survive. During this length of time, producers and consumers normally overlapped. Prosumers hunting wild animals and gathering wild plants were, in other words, the normality rather than the exception. It is only since the development of agrarian so-ci-eties, around ten thousand years ago, that we have seen a complete (and after a while

5? In Floridi (1999); see also Floridi (2004, 2014b). I should have known better and used Toffler?s ?prosumer?.

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culturally obvious) separation between producers and consumers. But in some corners of the infosphere, that decoupling is being recoupled. On Instagram, TikTok, or Clubhouse, for example, we consume what we produce. One may therefore stress that, in some cases, this parenthesis is coming to an end and that prosumers are back, recoupled by the digital. Consequently, it is perfectly coherent that human behaviour online has been compared and studied in terms of foraging models since the 1990s (Pirolli and Card 1995, 1999, Pirolli 2007).

The reader may easily list more cases of coupling, decoupling, and recoupling. Think, for example, about the difference between virtual reality (decoupling) and augmented reality (coupling); the ordinary decoupling of usage from ownership in the share economy; the recoupling of authenticity and memory thanks to block-chain; or the current debate about a universal basic income, which is a case of decoupling salary from work. But it is time to move from the ?how? to the ?why? question. Why does the digital have this cleaving power6 to couple, decouple, and recouple the world and our understanding of it? Why do other technological innovations seem to lack a similar impact? The answer, I surmise, lies in the com-bination of two factors.

On the one hand, the digital is a third--order technology (Floridi 2014a). It is not just a technology between us and nature, like an axe (first order), or a technology between us and another technology, like an engine (second order). It is rather a technology between one technology and another technology, like a computerized system controlling a robot painting a car (third order). Due to the autonomous processing power of the digital, we may not even be on?-let alone in?-the loop.

On the other hand, the digital is not merely enhancing or augmenting reality. It radically transforms reality because it creates new environments that we then inhabit, and new forms of agency with which we then interact. There is no term for this profound form of transformati

on. In the past (Floridi 2010b), I used the
expression re--ontologizing to refer to a very radical kind of re--engineering.
This sort of re--engineering not only designs, constructs, or structures a system
(e.g. a company, a machine, or some artefact) anew but fundamentally transforms
the intrinsic nature of the system itself?-that is, its ontology. In this sense,
nanotechnologies and biotechnologies are not merely re--engineering but re--
ontologizing our world. And by re--ontologizing modernity, to put it shortly, the
digital is re--epistemologizing modern mentality (that is, many of our old concep-
tions and ideas) as well.
6? I have chosen ?cleaving? as a particularly apt term to refer to the de/recoupling power of the digital
because ?to cleave? has two meanings: (a) ?to split or sever something?, especially along a natural line or
grain; and (b) ?to stick fast or adhere strongly to? something. This may seem contradictory, but it is due
to the fact that ?to cleave? is the result of a merging into a single spelling, and hence into a twofold
meaning, of two separate words: (a) comes from the Old English cleofan, related to the German klie-
ben (to cut through); whereas (b) comes from the Old English clifian, related to the German kleben
(to stick or cling). They have very different Proto--Indo--European roots; see http://www.etymonline.
com/index.php?term=cleave.

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Put together, all these factors suggest that the digital owes its cleaving power to
being a re--ontologizing, re--epistemologizing, third--order technology. This is why
it does what it does, and why no other technology has come close to having a
similar effect.
1.3? New Forms of Agency
If all this is approximately correct, then it may help one make sense of some
current phenomena concerning the transformation of the morphology of
agency in the digital age, and hence about forms of agency that only seem
unrelated. Their transformation depends on the cleaving power of the digital,
but their in-ter-pret-ation may be due to an implicit misunderstanding about this
cleaving power and its increasing, profound, and long--lasting consequences.
I am referring to political agency as direct democracy and artificial agency as
AI. In each case, the re--ontologizing of agency has not yet been followed by an
adequate re--epistemologizing of its interpretation. Or to put it less precisely but
perhaps more intuitively: the digital has changed the nature of agency, but we are
still interpreting the outcome of such changes in terms of modern mentality, and
this is generating some deep misunderstanding.
The kind of agency I am referring to is not the one most discussed in philoso-
phy or psychology, involving mental states, intentionality, and other features that
are typically associated to human beings. The agency discussed in this book is the
one that is commonly found in computer science (Russell and Norvig 2018) and
in the literature on multi--agent systems (Weiss 2013, Wooldridge 2009). It is more
minimalist and requires that a system satisfies only three basic conditions: it can

a) receive and use data from the environment, through sensors or other forms
of data input;

b) take actions based on the input data, autonomously, to achieve goals,
through actuators or other forms of output, and

c) improve its performance by learning from its interactions.
A similar agent can be artificial

(e.g. a bot), biological (e.g. a dog), social (e.g. a
company, or a government), or hybrid. In what follows, I shall say something very brief about political agency and direct democracy because this book will concentrate only on artificial agency, not socio--political agency shaped and supported by digital technologies.

In current debates about direct democracy, we are sometimes misled into thinking that the digital should (note the normative as opposed to descriptive approach) recouple sovereignty and governance. Sovereignty is political power that can be legitimately delegated, while governance is political power that is legitimately delegated?-on a temporary, conditional, and accountable basis?-and

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can be equally legitimately withdrawn (Floridi 2016e). Representative democracy is commonly, if mistakenly, seen as a compromise due to practical communication constraints. True democracy would be direct, based on the unmediated, constant, and universal participation of all citizens in political matters. Unfortunately, so goes the argument, we are too many. So the delegation (understood as -intermediation) of political power is a necessary, if minor, evil (Mill (1861), 69).

It is the myth of the city--state and especially of Athens.

For centuries, the compromise in favour of a representative democracy has seemed inevitable?-that is, until the arrival of the digital. According to some -people, this now promises to disintermediate modern democracy and couple (or recouple, if you believe in some mythical good old days) sovereignty with governance to deliver a new sort of democracy. This would be some sort of digital agora that could finally enable the regular, direct involvement of every interested citizen. It is the same promise made by the referendum instrument, especially when it is binding rather than advisory. In either case, voters are asked directly what should be done. The only task left to the political, administrative, and technical class would be that of implementing people?s decisions. Politicians would be delegated (not representative) civil servants in a very literal sense. Yet this is a mistake because indirect democracy was always the plan. To be trite for a moment, the decoupling is a feature rather than a bug. A democratic regime is characterized above all not by some procedures or some values (although these can also be characteristics) but by a clear and neat separation?-that is, a decoupling?-between those who hold political power (sovereignty) and those who are entrusted with it (governance). All voting--age citizens have political power and delegate it le-git-im-ate-ly by voting. This power is then entrusted to others who exercise that mandate by governing transparently and accountably for as long as they are legitimately empowered. To put it simply, a democratic regime is not merely a way of exercising and managing power in some ways (procedures) and/or according to some values. It is, first and foremost, a way of structuring power. Those who hold political power do not exercise it; they give it to those who exercise it, but do not hold it. The conflation of the two sides leads to brittle forms of autocracy or mob rule.

From such a perspective, representative democracy is not a compromise. It is actually the best form of democracy. And using the digital to couple (or, as I have already remarked, more mythically recoup

le) sovereignty and governance would
be a costly mistake. Brexit, Trump, Lega Nord and other populist disasters caused by the ?tyranny of the majority? (Adams 1787) are sufficient evidence. We need to consider how best to take advantage of the planned, representative decoupling between sovereignty and governance?-not how to erase it. So, consensus is the problem. However, it is not a topic of this book. All I want to offer with the previous analysis is a taste of the sort of unifying considerations about forms of agency that link the impact of the digital on politics to how we think about and evaluate AI, as we shall see in the next section. Let us now return to artificial agency.

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1.4? AI: A Research Area in Search of a Definition

Some people (perhaps many) seem to believe that AI is about coupling artificial agency and intelligent behaviour into new artefacts. This is a misunderstanding. As I shall explain at more length in the next chapter, the opposite is actually true: the digital revolution has made AI not only possible but also increasingly useful. It has done so by decoupling the ability to solve a problem or complete a task successfully from any need to be intelligent to do so. It is only when this de-coup-ling is achieved that AI is successful. So, the usual complaint known as the ?AI effect?7?when as soon as AI can perform a particular task, such as automatic translation or voice recognition, the target then moves, and that task is no longer defined as intelligent if performed by AI?-is actually a correct acknowledgement of the precise process in question. AI performs a task successfully only if it can decouple its completion from any need to be intelligent in doing so; therefore, if AI is successful, then the decoupling has taken place, and indeed the task has been shown to be decouplable from the intelligence that seemed to be required (e.g. in a human being) to lead to success.

This is less surprising than it may seem, and in the next chapter we shall see that it is perfectly consistent with the classic (and probably still one of the best) definitions of AI provided by McCarthy, Minsky, Rochester, and Shannon in their ?Proposal for the Dartmouth Summer Research Project on Artificial Intelligence?, the founding document and later event that established the new field of AI in 1955. I shall only quote it here and postpone its discussion until the next chapter:

for the present purpose the artificial intelligence problem is taken to be that of making a machine behave in ways that would be called intelligent if a human were so behaving.8

The consequences of understanding AI as a divorce between agency and intelligence are profound. So are the ethical challenges to which such a divorce gives rise, and the second part of the book is devoted to their analysis. But to conclude this introductory chapter, a final answer still needs to be provided for the ?so what? question I mentioned at the outset. This is the task of the next section.

1.5? Conclusion: Ethics, Governance, and Design

Assuming that the previous answers to the ?why? and ?how? questions are acceptable, what difference does it make when we understand the power of the digital in

7? https://en.wikipedia.org/wiki/AI_effect.

8? Online version at http://www--formal.stanford.edu/jmc/history/dartmouth/dartmouth.html; see also the reissue in McCarthy et al. (2006).

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terms of cutting and pasting the world (along with our conceptualization of it) in unprecedented ways? An analogy may help introduce the answer. If one possesses only a single stone and absolutely nothing else, not even another stone to put next to it, then there is also nothing else one can do aside from enjoying the stone itself?-perhaps by looking at it or playing with it. But if one cuts the stone into two, several possibilities emerge for combining them. Two stones provide more affordances and fewer constraints than a single stone, and many stones even more so. Design becomes possible.

Cutting and pasting the ontological and conceptual blocks of modernity, so to speak, is precisely what the digital does best when it comes to its impact on our culture and philosophy. And taking advantage of such affordances and constraints in view of solving some problems is called design. So, the answer should now be clear: the cleaving power of the digital hugely decreases constraints on reality and escalates its affordances. By doing this, it makes design?-loosely understood as the art of solving a problem through the creation of an artefact that takes advantage of constraints and affordances, to satisfy some requirements, in view of a goal?-the innovating activity that defines our age.

Our journey is now complete. Each age has innovated its cultures, societies, and environments by relying on at least three main elements: discovery, invention, and design. These three kinds of innovation are tightly intertwined, yet in-novation has often been skewed like a three--legged stool in which one leg is longer and hence more forward than the others. The post--renaissance and early modern period may be qualified as the age of discovery, especially geographic. Late modernity is still an age of discovery but, with its industrial and mechanical in-nov-ations, is perhaps, even more, an age of invention. And, of course, all ages have also been ages of design (not least because discoveries and inventions require ingenious ways of linking and giving form to new and old realities). But if I am correct in what I have argued so far, then it is our age that is quintessentially, more so than any other, the age of design.

Because the digital is lowering the constraints and increasing the affordances at our disposal, it is offering us immense and growing freedom to arrange and organize the world in many ways to solve a variety of old and new problems. Of course, any design requires a project. And in our case, it is a human project for our digital age that we are still missing. But we should not let the cleaving power of the digital shape the world without a plan. We should make every effort to decide the direction in which we wish to exploit it, to ensure that the information societies we are building thanks to it are open, tolerant, equitable, just, and supportive of the environment as well as human dignity and flourishing. The most important consequence of the cleaving power of the digital should be a better design of our world. And this concerns the shaping of AI as a new form of agency, as we shall see in Chapter 2.

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2
Present
AI as a New Form of Agency, Not Intelligence

.0? Summary

Previously, in Chapter 1, we saw how the digital revolution has cut and pasted our realities along with our ideas about our realities, re--ontologizing and re--epistemologizing modernity. This has led to the development of AI as a new form of agency that can be successful without being intelligent. This chapter analyses the above interpretation. Section 2.1 shows how the absence of a definition for AI is evidence that the expression is not a scientific term. Instead, it is a helpful shortcut for referring to a family of sciences, methods, paradigms, technologies, products, and services. Section 2.2 refers to the classic, counterfactual char-ac-ter-iza-tion of AI provided by McCarthy, Minsky, Rochester, and Shannon in their ?Proposal for the Dartmouth Summer Research Project on Artificial Intelligence?, the founding document and later event that established the new field of AI in 1955. We already encountered it in the previous chapter, and this is the char-ac-ter-iza-tion that I shall adopt for the rest of the book. I then discuss Turing?s famous question ?Can machines think?? Section 2.3 moves from the previous analysis to outline the engineering and cognitive approaches to AI, arguing that the former has been a great success and the latter a complete failure. The interpretation of AI as a new form of agency that does not need any intelligence to be successful is based on the engineering tradition. Section 2.4 suggests that a similar form of agency can be successful because we have been transforming the world (envelop-ing it) into an increasingly AI--friendly environment. The conclusion in Section 2.5 stresses that such a process generates the risk of nudging humanity to adapt to its smart technologies.

2.1? Introduction: What Is AI? ?I Know It When I See It?

AI has been defined in many ways, but there is no single definition on which everybody agrees. An old survey (Legg and Hutter 2007) listed fifty--three def-in-itions for ?intelligence?, each of which may in principle be ?artificial?, and eighteen definitions for AI. The number has only been growing (Russell and Norvig 2018).

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Faced with a similar challenge, Wikipedia solved the problem by opting for a tautology:

artificial intelligence (AI)?.?.?.?is intelligence demonstrated by machines, in contrast to the natural intelligence displayed by humans.

(Wikipedia, Artificial Intelligence, 17 January 2020)

This is both true and useless. It reminds me of the unassailable ?Brexit means Brexit? robotically repeated by Theresa May when she served as prime minister of the UK.

The lack of a standard definition for AI can be a nuisance because, when meet-ing on the ethics of AI, sooner or later some smart participant cannot help won-dering, thoughtfully, ?.?.?.?but what do we really mean by AI?? (emphasis in the original as some people manage to speak in italics). There usually follows an end-less discussion that never reaches any consensus (it cannot), which leaves every-body frustrated (it should). The risk is that, after such a waste of time, some people may conclude that one cannot have a debate on the ethics of something that is undefined and apparently undefinable. This is nonsense. Admittedly, the absence of a definition for something so important is a bit suspicious. Yet this is not because all important things in life are always definable; often, many of

them
are not. We have a perfect sense of what friendship is, for example, even if we do not have the necessary and sufficient conditions to capture its nature uncontroversially. A philosophical debate about the definition of friendship can quickly end up at an impasse. But we can have very reasonable discussions and valuable insights about the ethics of friendship, its nature online, and its pro and contra. The same holds for ?being in good health? or ?being in love?.

This is so because we ?know it when we see it?, to use a phrase that became famous in 1964 (a pretty special year, it seems). As the reader probably knows, it was used by US Supreme Court Justice Potter Stewart in his decision about what may count as obscenity in Jacobellis v. Ohio. Explaining why he had decided that the material was protected speech, not obscene, and therefore should not be censored, he wrote:

I shall not today attempt further to define the kinds of material I understand to be embraced within that shorthand description [?hard--core pornography?], and perhaps I could never succeed in intelligibly doing so. But I know it when I see it, and the motion picture involved in this case is not that.

(378 U.S. at 197 (Stewart, J., concurring [my italics]))

Friendship, AI, love, justice, and many other things in life are like pornography. They may not be definable in the strict sense in which water is definable and

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defined as H2O, but you know them when you see them. Such a definition is acceptable for everyday life. Still, I have already acknowledged that the lack of a definition is a bit suspicious. In science, even insignificant things should be precisely definable (especially after so many decades of debates). The conclusion is that unlike ?triangle?, ?planet?, or ?mammalian?, AI is probably not a scientific term. Just like friendship or pornography, it is a generic expression?-a short cut used to wave in the direction of several techno--scientific disciplines, services, and products that are sometimes only generically related. AI is a family where resemblance, and sometimes only a few traits, are the criterion for belonging. Some members of the AI family are illustrated in Figure 2.1.

The map in Figure 2.1 is helpful, but far from controversial. As David Watson has pointed out to me, it seems odd to devote equal space to symbolic and probabil-is-tic AI (?that would be like a course on nuclear physics devoting as much time to Democritus as to quantum mechanics?). It is confusing to put deep learning in a separate section from computer vision and natural language processing; these days, the latter two applications are conducted almost exclusively using deep learning. It is unclear why neural networks are placed in a column for unsupervised learning when they are more famously associated with supervised problems. Many prominent unsupervised methods (e.g. clustering, projection, outlier detection) are not mentioned. The terminology of ?subsymbolic? computing is non--standard, at least if it is meant to cover optimization procedures such as evolutionary algorithms. David is right. But I still like the map because it helps one focus and is much better than nothing. And I am not showing this map because it is perfect, but because, despite its limitations, it includes quite a bit of territory and shows that even two experts may easily disagree even if they

onable but consider the following: just because a dish-
washer cleans the dishes as well as or even better than I do, this does not mean that it cleans them as I do, or that it needs any intelligence (no matter whether like mine or like any other kind) to achieve the task. This would be like arguing that (a) a river reaches the sea by following the best possible path, removing obstacles in its way; and (b) if this were to be done by a human, we would consider it an intelligent behaviour; so (c) the river?s behaviour is intelligent. The latter scenario is a fallacy that smacks of superstition. The whole and only point is about performing a task successfully so that the outcome is as good as, or better than, what human intelligence would have been able to achieve. The how is not in question, only the what. This point is crucial, logically, and historically.
Logically, even the identity of (let alone a similarity in) an outcome says nothing about the identity of the processes that generated it and the sources of the processes themselves. This seems to be unquestionable, yet Turing thought otherwise. In a BBC Radio broadcast, he stated that

.?.?.?[1.20] the view which I hold myself, that it is not altogether unreasonable to describe digital computers as brains?.?.?.?[4.36?4.47] If now some particular machine can be described as a brain we have only to programme our digital computer to imitate it and it will also be a brain. If it is accepted that real brains, as found in animals, and in particular in men, are a sort of machine, it will follow that our digital computer suitably programmed, will behave like a brain. (Turing 1951)

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Try to forget for a moment that the person who is speaking is a genius. Imagine this is written in a tabloid article in, say, the Daily Mail. Pick up the red pen and underline the words ?not altogether unreasonable?. Even considering the English of the time, this double negative is the weakest possible commitment that one can imagine to any thesis. By the same standard, it is also ?not altogether unreasonable? to describe animals as automata that cannot reason or feel pain, as another genius, Descartes, argued. Of course, we know that Descartes was wrong.
The next word to underline is ?describe?. One can describe anything as anything else, given the appropriate level of abstraction (Floridi 2008a). The question is whether that level of abstraction is the correct one. It is ?not altogether unreasonable? to ?describe? a chess game as a battle between two enemies. And if it is the Spassky?-Fischer World Championship match of 1972, the description may even remind one that battles belong to wars, in this case, the Cold War. But if one starts looking for suffering and death, violence and blood, one will be disappointed. It?s a ?kiiind? (the three ?iii?s are important) of battle, but not really.
Next, consider the assumption that the brain is a machine. In terms of scientific observation, this seems either trivially true metaphorically, or factually wrong non--metaphorically. In the article, Turing admitted that we have a loose sense of what counts as a machine. Even a bureaucracy can be ?not altogether unreasonably? ?described? as a machine?-just ask another genius, Kafka. A kettle, a fridge, a train, a computer, a toaster, a dishwasher: these are all machines in some sense or another. Our body is

a machine too, kind of. So is our heart. Why
not our brains? Absolutely. The real issue is the vast amount of room for
-manoeuvre, because everything depends on how strictly one reads ?kind of?. If
almost anything can qualify as a kind of machine, then indeed, a brain is a kind of
machine too. But we are not fitting the right peg in the right hole; we are just
making the hole so big that any peg will go through it easily.
Finally (and so far, this is all loosely speaking), the next problem is a fallacy.
Let?s use Turing?s favourite example, which provides answers to the same questions in ways that are identical or comparable in quality, or in any case indistinguishable in terms of source, while minding that this is already reducing the brain
to a computer: A and B both output the same outcome O given the same input
I. Even in this case, this does not mean that (a) A and B behave the same way or
(b) A and B are the same. Imagine Alice visits Bob?s house and finds some clean
dishes on the table. She may be unable to infer from the outcome (clean dishes)
which process was used (mechanical or hand washing) and hence which agent
cleaned them (the dishwasher or Bob) and therefore what skills were used to
achieve the result. It would be a mistake to infer from this irreversibility and
opacity of the process that Bob and the dishwasher are therefore the same, or that
they behave in the same way even if only in terms of dishwashing properties. The
fact is that Alice probably could not care less how the task is achieved, so long as
the dishes are clean.

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I am afraid Turing was both metaphorically right and substantially wrong. Or
maybe it was the BBC that required some lowering of standards and precision
(been there, done that; I speak as a sinner). Regardless, it is not altogether unreasonable to describe digital computers as brains and vice versa. It is just not helpful, because it is too metaphorical and vague. Strictly speaking (and we should be
speaking strictly here), it is scientifically mistaken and hence can easily be misleading. Once we start being precise, the similarities disappear, and all the valuable differences become increasingly evident. Brains and computers are not the
same, nor do they behave in the same way. Both points are not refuted by Turing?s
position. This is not a positive argument in their favour, though. But I am afraid
that their full support would require a different kind of book.1 So here, I am only
making explicit the perspective through which to interpret the second half of this
book more accurately. The readers who prefer to think like Turing will disagree
with me. However, I hope they will still be able to agree on the following distinction: historically, the counterfactual characterization of AI contains the seeds of
an engineering (as opposed to a cognitive) approach to AI, as we shall see in the
next section.
2.3? The Two Souls of AI: The Engineering and the Cognitive
It is a well--known fact, although sometimes underestimated, that AI research
seeks both to reproduce the results or successful outcome of our (or at least some
kind of animal--like) intelligent behaviour by non--biological means, and to produce the non--biological equivalent of our intelligence, that is, the source of such
behaviour.
As a branch of engineering interested in intelligent behaviour reproduction, AI
has been astound

ingly successful, well beyond the rosiest expectations. Take the
rather famous if slightly old example of Deep Q--network (a system of software
algorithms), which belongs to this kind of reproductive AI. In 2015, Deep Q--network
learned to play forty--nine classic Atari vintage video games from scratch by relying
only on data about the pixels on the screen and the scoring method (Mnih et al.
2015). Impressive? From an engineering point of view, yes, but not very much in
terms of getting closer to true artificial AI. After all, it takes less ?intelligence? to
win a game of Space Invaders or Breakout than to be a chess champion. So, it was
only a matter of time before some clever human beings devised a way to make a
Turing Machine smart enough to play Atari games proficiently.
1? The debate on this is endless, and so is the bibliography. For a short and very readable overview from a neuroscience perspective on why the brain is not a computer, Epstein (2016) is a good starting point. For a longer and more detailed (but equally enjoyable) overview, see Cobb (2020) or the extract Cobb (27 February 2020).

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Present: AI as a New Form of Agency, Not Intelligence?
21
Nowadays, we increasingly rely on AI--related applications (sometimes called
smart technologies, a terminology that I shall use as well, although the expression
has a broader scope) to perform tasks that would simply be impossible with
unaided or unaugmented human intelligence. Reproductive AI regularly outper-
forms and replaces human intelligence in an ever--larger number of contexts. For
smart technologies, the sky is the limit. Deep Q--network just eliminated another
area where humans were better than machines. The next time you experience a
bumpy landing on an aircraft, recall that it is probably because the pilot was in
charge, not the computer. This does not mean that an AI--driven, autonomous
drone flies like a bird. Edsger Wybe Dijkstra, famously wrote that
The Fathers of the field had been pretty confusing: John von Neumann specu-
lated about computers and the human brain in analogies sufficiently wild to be
worthy of a medieval thinker and Alan M. Turing thought about criteria to settle
the question of whether Machines Can Think, a question of which we now know
that it is about as relevant as the question of whether Submarines Can Swim.2
This is indicative of the engineering approach shared by reproductive AI.
However, as a branch of cognitive science interested in intelligence production, AI
remains science fiction and has been a dismal disappointment. Productive AI
does not merely underperform with respect to human intelligence. It has not yet
even joined the competition. The fact that Watson (the IBM system capable of
answering questions asked in natural language) could beat human opponents
when playing Jeopardy! says more about the human engineers, their amazing
skills and expertise, and the game itself than it does about biological intelligence
of any kind. You do not have to believe me. John McCarthy, who coined the
expression ?artificial intelligence? and was a true believer in the possibility of pro-
ducing it, in the cognitive sense seen above,3 knew all this very well. His disap-
pointed remarks about Deep Blue?s victory against the world champion Garry
Kasparov in 1997 (McCarthy 1997) are symptomatic of the sort of productive,
cognitive AI that frowns upon reproductive, engineering AI. This is why he neve

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stopped complaining about the treatment of chess playing as a case of real AI. He was right: it is not. But he was also wrong in thinking that it is therefore not a good alternative, and the same holds for AlphaGo (more on this presently).
There are thus two souls of AI: the engineering one (smart technologies) and the cognitive one (truly intelligent technologies). They have often engaged in fratricidal feuds for intellectual predominance, academic power, and financial

2? Dijkstra (1984).

3? I met John in his late seventies at different meetings and then more regularly thanks to a book project to which we both contributed. I think he never changed his mind about the real feasibility of true AI as the non--biological equivalent (or perhaps a better version) of human intelligence.

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resources. That is partly because they both claim common ancestors and a single intellectual inheritance: a founding event, the already cited Dartmouth Summer Research Conference on Artificial Intelligence in 1956; and a founding father, Turing, first with his machine and its computational limits, then his famous test.
It hardly helps that a simulation might be used to check for the production of the simulated source (i.e. human intelligence) and for the reproduction or surpassing of the targeted source?s behaviour or performance (i.e. what human intelligence achieves).
The two souls of AI have assumed various and somewhat inconsistent names. Sometimes, distinctions between weak vs. strong AI or Good Old--Fashioned (GOFAI) vs. New or Nouvelle AI are used to capture the difference. Today, the expression Artificial General Intelligence (AGI) or Universal AI seem more fashionable, instead of full AI. In the past, I have preferred to use the less loaded distinction between light vs. strong AI (Floridi 1999). It really does not matter. The misalignment of their natures, goals, and results has caused endless and mostly pointless diatribes. Defenders of AI point to the impressive results of re-pro-ductive, engineering AI, which is really weak or light AI in terms of goals. Detractors of AI point to the abysmal results of productive, cognitive AI, which is really strong or general AI in terms of goals. Much of the ongoing speculation about the so--called singularity issue and the catastrophic consequences of the arrival of some alleged superintelligence has roots in such confusion. Sometimes I cannot help but suspect that this was done on purpose, not for some malicious reason, but because the mess is intellectually enjoyable. Some people love pointless diatribes.
Great champions know how to end their career at the peak of their success. In 2017, DeepMind, Alphabet?s (formerly Google?s) AI lab, decided that its computer program AlphaGo would no longer focus on winning the board game Go.
Instead, DeepMind CEO Demis Hassabis and lead AlphaGo programmer David Silver indicated that the focus was going to be on
developing advanced general algorithms that could one day help scientists as they tackle some of our most complex problems, such as finding new cures for diseases, dramatically reducing energy consumption, or inventing revolutionary new materials.4
The ambition was justified. Three years later, in 2020, DeepMind?s AI system AlphaFold 2 solved the ?protein folding problem?, a grand challenge in biology that had vexed scientists for fifty years:

4? https://de

epmind.com/blog/article/alphagos--next--move.

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The ability to accurately predict protein structures from their amino--acid sequence would be a huge boon to life sciences and medicine. It would vastly accelerate efforts to understand the building blocks of cells and enable quicker and more advanced drug discovery.? (Callaway 2020)

AI will undoubtedly lead to more discoveries and potentially remarkable breakthroughs, especially in the hands of brilliant and thoughtful people. It will also support the management and control of increasingly complex systems. Yet all these spectacular developments may be more easily achieved if a misunderstanding is removed. We have seen that successful AI is not about producing human intelligence but replacing it. A dishwasher does not clean the dishes as I do but, at the end of the process, its clean plates are indistinguishable from mine. Indeed, they may even be cleaner (effectiveness) while using fewer resources (efficiency). The same applies to AI. AlphaGo did not play like the Chinese Go grandmaster and world number one Ke Jie, but it won anyway. Likewise, autonomous cars are not cars driven by humanoid robots (androids) seated behind the wheel instead of you; they are ways of reinventing the car and its environment altogether (Floridi 2019a).

In AI, the outcome matters, not whether the agent or its behaviour is intelligent. Thus, AI is not about reproducing any kind of biological intelligence. It is about doing without it. Current machines have the intelligence of a toaster, and we really do not have much of a clue about how to move from there (Floridi, Taddeo, and Turilli 2009). When the warning ?printer not found? pops up on the screen of your computer, it may be annoying, but it is hardly astonishing, even though the printer in question is actually right there next to it. More importantly, this is not an obstacle because artefacts can be smart without being intelligent. That is the fantastic achievement of reproductive AI, which is the successful continuation of human intelligence by other means (to paraphrase Carl von Clausewitz).

Today, AI decouples successful problem solving and task performance from intelligent behaviour. It is thanks to this decoupling that it can relentlessly col-onize the boundless space of problems and tasks whenever these can be achieved without any understanding, awareness, acumen, sensitivity, concerns, hunches, insights, semantics, experience, bio--embodiment, meaning, wisdom, and whatever other ingredients contribute to creating human intelligence. In short, it is precisely when we stop trying to produce human intelligence that we can successfully replace it in an increasing number of tasks. Had we waited for even a spark of real AI, the kind you find in Star Wars, AlphaGo would have never become so much better than anyone at playing Go. Indeed, I would still win when playing chess against my smartphone.

If this decoupling is fully understood, there are three apparent developments ahead. I shall discuss them in more detail in the next chapter, but they can be

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outlined here. First, AI should stop winning games and learn how to gamify. As AI becomes better at playing games, anything that can be transformed into a game will be within its scope. If I were DeepMind (disclosure: I

have collaborated
with DeepMind in the past), I would hire a team of game designers. Second, AI will only be matched by AI in gamified contexts; its internal interactions may become too complex to be entirely understandable by outside admirers like us. We may enjoy watching AI play just as we enjoy Bach being played. And finally, expect human intelligence to have a different role wherever AI is the better player. For intelligence will be less about solving some problem and more about deciding which problems are worth solving, why, for what purpose, and acceptable costs, trade--offs, and consequences.

2.4? AI: A Successful Divorce Made in the Infosphere

Both the classic counterfactual definition and the interpretation of AI as a divorce (not a marriage) between agency and intelligence enable one to conceptualize AI as a growing resource of interactive, autonomous, and often self--learning agency (in the ML sense, see Figure 2.1). This agency can deal with an increasing number of problems and tasks that would otherwise require human intelligence and intervention, and possibly a boundless amount of time, to be performed successfully. In short, AI is defined based on engineered outcomes and actions. So, in the rest of this book, I shall treat AI as a reservoir of smart agency on tap. This is sufficiently general to capture the many ways that the literature discusses AI. But before looking at the possible development of AI in the next chapter, this section needs to address one final issue with the divorce between agency and intelligence. As one of the sources of the ethical challenges posed by AI, the divorce is problematic because artificial agents are ?sufficiently informed, ?smart?, autonomous and able to perform morally relevant actions independently of the humans who created them? (Floridi and Sanders 2004). But how can a divorce between agency and intelligence be successful in the first place, in terms of efficacy? Isn?t intelligence necessary for any kind of behaviour to be successful? I already offered the example of the river to illustrate a possible fallacy, but a river is not an agent. More needs to be said in terms of a positive explanation to answer the previous question that can double task as a reasonable objection. Here is the short reply: the success of AI is primarily due to the fact that we are building an AI--friendly environment in which smart technologies find themselves at home. We are more like scuba divers, and it is the world that is adapting to AI, not vice versa. Let us see what this means.

AI cannot be reduced to a ?science of nature? or to a ?science of culture? (Ganascia 2010) because it is a ?science of the artificial?, to put it in the terms of Herbert Simon (Simon 1996). As such, AI pursues neither a descriptive nor a

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prescriptive approach to the world. Instead, it investigates the conditions that -constrain possibilities for building and embedding artefacts in a world that can interact with them successfully. In other words, it inscribes the world. Such artefacts are new logico--mathematical pieces of code, that is, the new texts written in Galileo?s mathematical book of nature:

Philosophy is written in this grand book?-I mean the universe?-which stands continually open to our gaze, but it cannot be understood unless one first learns to comprehend the language in which it is written

. It is written in the language
of mathematics, and its characters are triangles, circles, and other geometric figures, without which it is humanly impossible to understand a single word of it; without these, one is wandering about in a dark labyrinth.
(Galileo, Il Saggiatore, 1623, translated in Popkin (1966), 65)

Until recently, the widespread impression was that the process of adding to the mathematical book of nature (inscription) required the feasibility of productive, cognitive AI. After all, developing even a rudimentary form of non--biological intelligence may seem not only the best but perhaps the only way to implement technologies that are adaptive and flexible enough to deal effectively with a complex, ever--changing, and often unpredictable (when not unfriendly) environment. What Descartes acknowledged as an essential sign of intelligence?-the capacity to adapt to different circumstances and exploit them to one?s advantage?-would be a priceless feature in any appliance that sought to be more than merely smart.

This impression is not incorrect, but it is distracting. We have seen how the digital is re--ontologizing the very nature of (and hence what we mean by) our environments as the infosphere is progressively becoming the world in which we live. While we were unsuccessfully pursuing the inscription of productive AI into the world, we were actually modifying (re--ontologizing) the world to fit re-product-ive, engineering AI. The world is becoming an infosphere increasingly well adapted to AI?s bounded capacities (Floridi 2003, 2014a). To see how, let us consider the car industry. It has been at the forefront of the digital revolution and AI since its beginning, first with industrial robotics and now with AI--based driverless cars. The two phenomena are related and can also teach us a vital lesson when it comes to understanding how humans and artificial agents will cohabit in a shared environment.

Take industrial robotics, such as a robot that paints a vehicle component in a factory, for example. The three--dimensional space that defines the boundaries within which such a robot can work successfully is defined as the robot?s envelope. Some of our technologies, such as dishwashers or washing machines, accomplish their tasks because their environments are structured (enveloped) around the elem-en-tary capacities of the robot inside them. The same applies to Amazon?s robotic shelves in ?enveloped? warehouses. It is the environment that is designed

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to be robot--friendly and not the other way around. Thus, we do not build droids like C--3PO from Star Wars to wash dishes in the sink precisely the way we would. Instead, we envelop micro--environments around simple robots to fit and exploit their limited capacities and still deliver the desired output.

Enveloping used to be either a stand--alone phenomenon (where you buy the robot with the required envelope, like a dishwasher or a washing machine) or implemented within the walls of industrial buildings carefully tailored around their artificial inhabitants. Nowadays, enveloping the environment into an AI--friendly infosphere has started to pervade all aspects of reality. It is happening daily everywhere, whether in the house, the office, or the street. When we speak of smart cities, we also mean that we are transforming social habitats into places where robots can operate

successfully. We have been enveloping the world around
digital technologies for decades, both invisibly and without fully realizing it. As we shall see in Chapter 3, the future of AI lies in even more enveloping (e.g. in terms of 5G and the Internet of Things) and more onlife experiences, which means that we are all more and more connected. We spend more and more time in the infosphere as all new information is increasingly born digital. Returning to the car industry, driverless cars will become a commodity when, or rather where, we can envelop the environment around them.

In the 1940s and 1950s, the computer was a room; Alice used to walk inside it to work with and within it. Programming meant using a screwdriver. Human?-computer interaction was a somatic or physical relationship (recall the computer shown in Robinson Crusoe on Mars). In the 1970s, Alice?s daughter walked out of the computer to step in front of it. Human?-computer interaction became a semantic relationship later facilitated by DOS (Disk Operating System), lines of text, GUI (Graphic User Interface), and icons. Today, Alice?s granddaughter has again walked inside the computer in the form of a whole infosphere that surrounds her, often imperceptibly. We are building the ultimate envelop in which human?-computer interactions have again become somatic through touch screens, voice commands, listening devices, gesture--sensitive applications, proxy data for location, and so forth.

As usual, entertainment, health, and military applications are driving in-novation. But the rest of the world is not lagging far behind. If drones, driverless ve-hicles, robotic lawnmowers, bots, and algorithms of all kinds can move ?around? and interact with our environments with decreasing trouble, this is not because productive, cognitive AI (the Hollywood kind) has finally arrived. It is because the ?around?, the environment our engineered artefacts need to negotiate, has become increasingly suitable to reproductive engineered AI and its limited capaci-ties. In such an AI--friendly infosphere, the default assumption is that an agent may be artificial. This is why we are regularly asked to prove that we are not robots by clicking on a so--called CAPTCHA, the Completely Automated Public Turing test to tell Computers and Humans Apart. The test is represented by slightly

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altered strings of letters, possibly mixed with other bits of graphics, that we must decipher to prove that we are human and not an AA when registering for a new account online, among other things. It is a trivial test for a human, but an apparently insurmountable task for AI. This is how little progress there has been in the cognitive area of the production of non--biological intelligence.

Every day sees more computational power, more data, more devices (the Internet of Things), more sensors, more tags, more satellites, more actuators, more digital services, more humans connected, and more of them living onlife. Every day brings more enveloping. More jobs and activities are becoming digital in nature, whether in terms of playing, educating, dating, meeting, fighting, caring, gossiping, or advertising. We do all this and more in an enveloped infosphere where we are more analogue guests than digital hosts. No wonder our AAs are performing increasingly well. It is their environment. As we shall see in Part

Two,
this profound ontological transformation implies significant ethical challenges.

2.5? The Human Use of Humans as Interfaces

Enveloping the world by transforming a hostile environment into a digitally friendly infosphere means that we will share our habitats not just with natural, animal, and social forces (as sources of agency). We will also, and sometimes primarily, share it with AAs. This is not to say that actual artificial agency of the intelligent kind is in view. We do not have semantically proficient, truly intelligent machines that understand things, worry about them, have preferences, or feel passionate about something. But we have such good statistical tools that purely syntactic technologies can bypass problems of meaning, relevance, understanding, truth, intelligence, insight, experience, and so forth to deliver what we need. This could be a translation, the right picture of a place, a preferred restaurant, an interesting book, a better--priced ticket, an attractively discounted bargain, a song that fits our musical preferences, a movie we enjoy, a cheaper solution, a more effective strategy, essential information for new projects, the design for new products, the foresight needed to anticipate problems, a better diagnosis, an unexpected item that we did not even know we needed, the necessary support for a scientific discovery or a medical cure, and so on. They are as stupid as an old fridge, yet our smart technologies can play chess better than us, park a car better than us, interpret medical scans better than us, and so forth. Memory (as in data and algorithms) outperforms intelligence on an increasing and boundless number of tasks and problems. The sky, or rather our imagination on how to develop and deploy such smart technologies, is the limit.

Some of the issues that we are facing today, for example in e--health or financial markets, arose within highly enveloped environments in which all relevant (and sometimes the only) data are machine readable and processable. In such

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environments, decisions and actions may be, and often are, taken automatically by applications and actuators. These can execute commands and output corresponding procedures ranging from alerting or scanning a patient to buying and selling bonds. Examples could easily multiply here. The consequences of enveloping the world to transform it into an AI--friendly place are many, and the rest of the book will explore some of them. But one, in particular, is very significant and rich in consequences and may be discussed here by way of conclusion: humans may inadvertently become part of the mechanism. This is precisely what Kant recommended we should never do: treat humans as only means rather than ends. Yet this is already happening in two main ways, both of which are cases of the ?human use of human beings? (Wiener 1954, 1989).

First, humans are becoming the new means of digital production. The point is simple: sometimes AI needs to understand and interpret what is happening, so it needs semantic engines like us to do the job. This recent trend is known as human--based computation. Amazon Mechanical Turk provides a classic example. The name comes from a famous chess--playing automaton built by Wolfgang von Kempelen (1734?1804) in the late eighteenth century. The automaton became famous by beating the likes of Napoleon Bonaparte and Benjamin F

ranklin, and
putting up a good fight against a champion such as François--André Danican Philidor (1726?95). However, it was a fake because it included a special compartment in which a hidden human player controlled its mechanical operations. The Mechanical Turk plays a similar trick. Amazon describes it as ?Artificial Artificial Intelligence?. The double ?Artificial? is in the original and is supposed to work as a double negative. It is a crowdsourcing web service that enables so--called ?re-questers? to harness the intelligence of human workers, known as ?providers? or more informally as ?turkers?, to perform tasks. Known as HITs (human intelligence tasks), these tasks cannot be performed by computers at present.
A requester first posts an HIT, such as transcribing audio recordings or tagging negative content in a film (two actual examples). Turkers can then browse and select an existing HIT to complete for a reward set by the requester. At the time of writing, requesters can check whether turkers satisfy some specific qualifications before being allocated an HIT. They can also accept or reject the result sent by a turker, which then reflects on the latter?s reputation. ?Human inside? is becoming the next slogan. The winning formula is simple: smart machine + human intelligence = clever system.
The second way humans are becoming part of the mechanism is in terms of influenceable customers. For the advertisement industry, a customer is an interface between a supplier and a bank account (or, more precisely, a ?credit limit?; this is not just disposable income because customers will spend more than they have, e.g. by using their credit cards). The smoother and more frictionless the relationship between the two, the better. So, the interface needs to be manipulated. To manipulate it, the advertising industry needs to have as much information as

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possible about the customer interface. Yet such information cannot be obtained unless something else is given in return to the customer. Enter ?free? services online: this is the currency with which information about customer interfaces is ?bought?. The ultimate goal is to provide just enough ?free services?, which are expensive, to obtain all the information about the customer interface needed to ensure the degree of manipulation that will provide unlimited and unconstrained access to the customer?s bank account. Due to competition rules, such a goal is unreachable by any single operator. Yet the joint effort of the advertising and services industry means that customers are increasingly seen as a means towards an end. They are bank account interfaces to be pushed, pulled, nudged, and enticed. AI plays a crucial role in all of this by tailoring, optimizing, deciding, etc. many processes through recommended systems (Milano, Taddeo, and Floridi 2019, 2020a), which is a topic discussed further in Chapter 7.

2.6? Conclusion: Who Will Adapt to Whom?

With every step that we take in digitizing our environments and expanding the infosphere, AI systems will become exponentially more useful and successful. Enveloping is a trend that is robust, cumulative, and progressively refining. It has nothing to do with some future Singularity, for it is not based on speculation about some super AI taking over the world in the near future. No artificial Spartacus will lead a digital uprising. B

ut enveloping the world, and thus facilitat-
ing the emergence of AAs and the success of their behaviour, is a process that raises concrete and pressing challenges. I shall discuss them in the second half of this book. Here, let me illustrate some of them by relying on a parody.
Imagine two people: A and H. They are married, and they wish to make their relationship work. A, who does increasingly more in the house, is inflexible, stubborn, intolerant of mistakes, and unlikely to change. H is just the opposite, but is also becoming progressively lazier and dependent on A. The result is an unbalanced situation in which A ends up practically (if not purposefully) shaping the relationship and distorting H?s behaviours. If the relationship works, it is only because it is carefully tailored around A. It then becomes interpretable in terms of Hegel?s master?-slave dialectic (Hegel 2009). In this analogy, smart technologies are playing the role of A, whereas their human users are clearly H. The risk we are running is that, by enveloping the world, our technologies, and especially AI, might shape our physical and conceptual environments. They would constrain us to adjust to them because that is the easiest or best (or sometimes the only) way to make things work. Given that AI is the stupid but laborious spouse and humanity is the intelligent but lazy one, who will adapt to whom? The reader will probably recall many episodes in real life when something could not be done at all, or had to be done in a cumbersome or silly way, because that was the only way to make

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the computerized system do what it had to do. ?Computer says no?, as the character Carol Beer in the British sketch comedy Little Britain would reply to any customer request. A more concrete, if trivial, example is that we might end up building houses with round walls and furniture with legs just high enough to fit the capacities of a Roomba (our robotic vacuum cleaner) much more effectively. I certainly wish that our house were more Roomba--friendly. We have already adapted our garden to ensure that Ambrogio, a robotic lawn mower, can work successfully.
The examples are helpful to illustrate not just the risk, but also the opportunity represented by the re--ontologizing power of digital technologies and the enveloping of the world. There are many ?roundy? places in which we live, from igloos to medieval towers, from bow windows to public buildings where the corners of rooms are rounded for sanitary reasons. If we now spend most of our time inside squarish boxes, that is due to another set of technologies related to the mass production of bricks and concrete infrastructures and the ease of cutting building material in straight lines. It is the mechanical circular saw that paradoxically generates a right--angled world. In both cases, squarish and roundy places have been built in line with predominant technologies rather than through the choices of potential inhabitants. Following this example, it is easy to see how the op-por-tunity represented by digital?s re--ontologizing power comes in three forms: rejection, critical acceptance, and proactive design. By becoming more critically aware of the re--ontologizing power of AI and smart applications, we might be able to avoid the worst forms of distortion (rejection) or at least be consciously tolerant of them (acceptance)?especially

when it does not matter (I am thinking about the
Roomba--friendly length of the legs of the sofa in our house) or when this is a temporary solution, while waiting for a better design. In the latter case, imagining what the future will be like and what adaptive demands AI and the digital in general may place on their human users may help devise technological solutions that can lower their anthropological costs and raise their environmental benefits. In short, human intelligent design (pun intended) should play a significant role in shaping our future interactions with current and forthcoming smart artefacts and our shared environments. After all, it is a sign of intelligence to make stupidity work for you.
The time has come to look into AI?s present and foreseeable future.

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The Ethics of Artificial Intelligence: Principles, Challenges, and Opportunities. Luciano Floridi, Oxford University Press.
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3
Future
The Foreseeable Development of AI
3.0? Summary
Previously, in Chapter 2, I argued that AI should be interpreted not as a marriage between some biological--like intelligence and engineered artefacts, but as a divorce between agency and intelligence?-that is, a decoupling between the ability to deal with problems and tasks successfully in view of a goal, and the need to be intelligent to do so. This chapter uses the above interpretation of AI as a new form of successful non--intelligent agency to look at its future. After a short introduction in Section 3.1 about the difficulties facing any exercise in prediction, Sections 3.2 and 3.3 argue that the likely developments and possible challenges of AI will depend on the push for synthetic data, the increasing translation of difficult problems into complex problems, the tension between regulative and constitutive rules underpinning areas of AI application, and hence the progressive adaptation of the environment to AI rather than of AI to the environment (what I defined in the previous chapter as enveloping). Section 3.4 returns to the im-portance of design and the responsibility to produce the right sort of AI to take advantage of the developments mentioned above. Section 3.5 focuses on large language models to highlight the important of such AI systems as interfaces, between human and artificial agents, and between different artificial agents. The conclusion discusses the seasons of AI, especially its winters, to highlight the lessons we should have learnt and those we can still learn?-and hence apply?-to make the most of this unique technology. This chapter concludes Part One of the book, which provides a brief philosophical introduction to AI?s past, present, and future.
3.1? Introduction: Looking into the Seeds of Time
AI has dominated recent headlines with its promises, challenges, risks, successes, and failures. What is its foreseeable future? Of course, the most accurate forecasts are made in hindsight. But if some cheating is not acceptable, then smart people bet on the uncontroversial or the untestable. On the uncontroversial side, one might mention the increased pressure that will come from lawmakers to ensure

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that AI applications align with socially acceptable expectations. For example, every-body expects more regulatory moves from the EU sooner or later, even after the AI Act becomes

law. On the untestable side, some people will keep selling catastrophic forecasts with dystopian scenarios taking place in some future that is sufficiently distant to ensure the Jeremiahs will not be around to be proven wrong. Like vampire or zombie movies, fear always sells well. So, expect more.

What is difficult, and may be embarrassing later, is trying to ?look into the seeds of time, and say which grain will grow and which will not? (Macbeth, 1.3, 159?62). The difficulty lies in trying to understand where AI is more likely to go and where it may not be going, given its current status?-and from this, trying to map the ethical challenges that should be taken seriously. This is what I will attempt to do in this chapter. Here, I shall be cautious in identifying the paths of least resistance but not so cautious as to avoid any risk of being proven wrong by someone reading this book in a few years. Part of the challenge lies in getting the level of abstraction right (Floridi 2008a, b). In other words, it lies in identifying the set of relevant observables (?the seeds of time?) on which to focus because those are the ones that will make the real, significant difference. In our case, I shall argue that the best observables are provided by an analysis of the nature of

1) the data used by AI to achieve its performance, and

2) the problems that AI may be expected to solve.1

Let us first look at (1) then (2) in the following two sections.

3.2? Historical, Hybrid, and Synthetic Data

They say that data are the new oil. I do not think so. Data are durable, reusable, quickly transportable, easily duplicable, and simultaneously shareable (non--rivalrous) without end. Oil has none of these properties. We have gigantic quantities of data that keep growing, but oil is a finite and dwindling resource. Oil comes with a clear price, whereas the monetization of the same data depends at least on who is using them and for what purpose (to say nothing of circumstances such as when, where, and so forth). All this is true even before introducing the legal and ethical issues that emerge when personal data are in play, or the whole debate about ownership (?my data? is much more like ?my hands? and much less like ?my oil?, as I argued in Floridi (2013)). So, the analogy is a stretch, to say the

1? For a reassuringly converging review that looks not at the nature of data or problems, but rather at the nature of technological solutions (based on a large--scale analysis of the forthcoming literature on AI), see ?We analyzed 16,625 papers to figure out where AI is headed next? at https://www.technologyreview.com/s/612768/we--analyzed--16625--papers--to--figure--out--where--ai--is--headed--next/.

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least. However, this does not mean it is entirely worthless. It is true that data, like oil, are a valuable resource. They must be refined to extract their value. Without data, algorithms?-AI included?-go nowhere, much like a car with an empty tank. AI needs data to train and then apply its training. Of course, AI can be hugely flexible; it is the data that determine AI?s scope of application and degree of success. For example, in 2016 Google announced a plan to use DeepMind?s ML system to reduce its energy consumption:

Because the algorithm is a general--purpose framework to understand complex dynamics, we plan to apply this to other challenges in the data cen

tre environ-
ment and beyond in the coming months. Possible applications of this technol-
ogy include improving power plant conversion efficiency (getting more energy
from the same unit of input), reducing semiconductor manufacturing energy
and water usage, or helping manufacturing facilities increase throughput.2
It is well known that AI, understood as ML, learns from the data it is fed and
progressively improves its results. If you show an immense number of photos
of dogs to a neural network, it will learn to recognize dogs increasingly well,
including dogs it has never seen before. To do this, usually one needs vast
quantities of data and often, the more, the better. In recent tests, a team of
researchers at the University of California in San Diego trained an AI system
using 101.6 million electronic health record (EHR) data points (including text
written by doctors and laboratory test results) from 1,362,559 paediatric
patient visits at a major medical centre in Guangzhou, China. Once trained, the
AI system was able to demonstrate:
.?.?.?high diagnostic accuracy across multiple organ systems and is comparable to
experienced paediatricians in diagnosing common childhood diseases. Our
study provides a proof of concept for implementing an AI--based system as a
means to aid physicians in tackling large amounts of data, augmenting diagnos-
tic evaluations, and to provide clinical decision support in cases of diagnostic
uncertainty or complexity. Although this impact may be most evident in areas
where healthcare providers are in relative shortage, the benefits of such an AI
system are likely to be universal.? (Liang et al. 2019)
AI has recently improved so much that, in some cases, we are moving away from
an emphasis on the quantity of large masses of data, sometimes improperly called
Big Data (Floridi 2012a), towards an emphasis on the quality of data sets that are
well curated. In 2018, DeepMind partnered with Moorfields Eye Hospital in
2? https://www.deepmind.com/blog/deepmind--ai--reduces--google--data--centre--cooling--bill--by--40.

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London, UK, to train an AI system to identify evidence of sight--threatening eye
diseases using optical coherence tomography (OCT) data, an imaging technique
that generates 3D images of the back of the eye. In the end, the team managed to:
demonstrate performance in making a referral recommendation that reaches or
exceeds that of experts on a range of sight--threatening retinal diseases after
training on only 14,884 scans [emphasis added].? (De Fauw et al. (2018), 1342)
I emphasize ?only 14,884 scans? because ?small data? of high quality is one of the
futures of AI. AI will have a higher chance of success whenever well--curated,
updated, and thoroughly reliable data sets become available and accessible to
train a system in a specific area of application.
This is obvious and hardly a new forecast. But it is a solid step forward that
helps us look further ahead, beyond the ?Big Data? narrative. If quality matters,
then provenance is crucial (and here blockchain may play a significant role).
Where do the data come from? In the previous example, they were provided by
the hospital. Such data are sometimes known as historical, authentic, or real--life
(henceforth, I shall simply call them historical). But we also know that AI can
generate its own data. I am not talking about metadata or secondary data about
it

s uses (Floridi 2010b). I am talking about its primary input. I shall call such entirely AI--generated data synthetic. Unfortunately, the term has an ambiguous etymology. It began to be used in the 1990s to refer to historical data that had been anonymized before being used, often to protect privacy and confidentiality. These data are synthetic only because they have been synthesized from historical data, for example through ?masking?.3 They have a lower resolution, but their genesis is not an artificial source. The distinction between historical data and those synthesized from them is useful. But it is not what I mean here, where I wish to stress the wholly and exclusively artificial provenance of the data in question. It is an ontological distinction that may have significant implications in terms of epistemology, especially when it comes to our ability to explain the synthetic data produced and the training achieved by the AI using them (Watson et al. 2019).

A famous example can help explain the difference. In the past, playing chess against a computer meant competing with the best human players who had ever played the game. Thus, one of the features of Deep Blue, the IBM chess program that defeated world champion Garry Kasparov, was ?an effective use of a Grandmaster game database? (Campbell, Hoane Jr, and Hsu (2002), 57). But AlphaZero, the latest version of the AI system developed by DeepMind, learnt to play better than anyone else, and indeed any other software, by relying only on

3? https://www.tcs.com/blogs/the--masking--vs--synthetic--data--debate.

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the game?s rules with no data input at all from any external source. It had no historical memory whatsoever:

The game of chess represented the pinnacle of artificial intelligence research over several decades. State--of--the--art programs are based on powerful engines that search many millions of positions, leveraging handcrafted domain expertise and sophisticated domain adaptations [emphasis added, these are the non--synthetic data]. AlphaZero is a generic reinforcement learning and search algorithm?-originally devised for the game of Go?-that achieved superior results within a few hours?.?.?.?given no domain knowledge except the rules of chess [emphasis added].? (Silver et al. (2018), 1144)

AlphaZero learnt by playing against itself, thus generating its own chess--related, synthetic data. Unsurprisingly, chess grandmaster Matthew Sadler and women?s international master Natasha Regan?-who have analysed thousands of AlphaZero?s chess games for their forthcoming book Game Changer (New in Chess, January 2019)?say its style is unlike any traditional chess engine. ?It?s like discovering the secret notebooks of some great player from the past,? says Matthew.4 AlphaZero generated its own synthetic data, and that was enough for its own training. This is what I mean by synthetic data.

Truly synthetic data, as I am defining them here, have some wonderful properties. They do not simply share those listed at the beginning of this section (they are durable, reusable, easily transportable, easily duplicable, simultaneously shareable without end, etc.). Synthetic data are also clean and reliable in terms of curation: they infringe on no privacy or confidentiality at the development stage (although problems persist at the deployment stage due to predictive privacy harms; see Crawford

and Schultz (2014)). Nor are they immediately sensitive
(sensitivity during the deployment stage still matters); if they are lost, it is not a
disaster because they can be recreated, and they are perfectly formatted for use by
the system that generates them. With synthetic data, AI never has to leave its digi-
tal space where it can exercise complete control over any input and output of its
processes. In more epistemological terms, synthetic data give AI the privileged
position of a maker?s knowledge. The maker knows the intrinsic nature and work-
ings of something because it made that something (Floridi 2018b). This explains
why they are so popular in security contexts, for example where AI is deployed to
stress--test digital systems. Also, synthetic data can sometimes be produced more
quickly and cheaply than historical data. AlphaZero became the best chess player
on earth in nine hours. In contrast, it took twelve hours for shogi and thirteen
days for Go.
4? https://www.deepmind.com/blog/alphazero--shedding--new--light--on--chess--shogi--and--go.

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Between historical data that are more or less masked (impoverished through
lower resolution, e.g. through anonymization) and purely synthetic data, there is
a variety of more or less hybrid data. Hybrid data can be imagined as the offspring
of historical and synthetic data. The basic idea is to use historical data to obtain
new synthetic data that are not merely impoverished historical data. A good
example is provided by generative adversarial networks (GANs), introduced by
(Goodfellow et al. 2014):
Two neural networks?-a Generator and a Discriminator?-compete against each
other to succeed in a game. The object of the game is for the Generator to fool
the Discriminator with examples that look similar to the training set?.?.?.?When
the Discriminator rejects an example produced by the Generator, the Generator
learns a little more about what the good example looks like?.?.?.?In other words,
the Discriminator leaks information about just how close the Generator was and
how it should proceed to get closer?.?.?.?As time goes by, the Discriminator learns
from the training set and sends more and more meaningful signals back to the
Generator. As this occurs, the Generator gets closer and closer to learning what
the examples from the training set look like. Once again, the only inputs the
Generator has are an initial probability distribution (often the normal distribu-
tion) and the indicator it gets back from the Discriminator. It never sees any real
examples [all capitalization and emphasis added].5
The Generator learns to create synthetic data that are like some known input data.
There is a bit of a hybrid nature here because the Discriminator needs access to
the historical data to ?train? the Generator. But the data generated by the Generator
are new, not merely an abstraction from the training data. So, while this is not a
case of parthenogenesis, like AlphaZero giving birth to its own data, it is close
enough to deliver some of the appealing features of synthetic data. For example,
synthetic human faces created by a Generator pose no problems regarding priv-
acy, consent, or confidentiality at the development stage.6
Many methods to generate hybrid or synthetic data are already available or
being developed, often for sector--specific purposes. There are also altruistic
efforts to make such data

sets publicly available (Howe et al. 2017). Clearly, the future of AI lies not just in ?small data? but (perhaps mainly) in its increasing ability to generate its own data. That would be a remarkable development, and one may expect significant efforts to be made in that direction. The next question is: what factor can make the dial in Figure 3.1 move from left to right?

5? https://securityintelligence.com/generative--adversarial--networks--and--cybersecurity--part--1/.

6? https://www.vice.com/en/article/7xn4wy/this--website--uses--ai--to--generate--the--faces--of--people--who--dont--exist.

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3.3? Constraining and Constitutive Rules

The difference is made by the genetic process, that is by the rules used to create the data. Historical data are obtained by recording rules as they result from some observation of a system?s behaviour. Synthesized data are obtained by abstracting rules that eliminate, mask, or obfuscate some degrees of resolution from the historical data, such as through anonymization. Hybrid and truly synthetic data can be generated by constraining rules or constitutive rules. There is no one--to--one mapping, but it is helpful to consider hybrid data as the data we must rely on, using constraining rules, when we do not have constitutive rules that can generate synthetic data from scratch. Let me explain.

The dial moves easily towards synthetic data whenever AI deals with ?games?. ?Games? can be understood here as any formal interactions in which players compete according to rules and in view of achieving a goal. The rules of these games are not merely constraining, but constitutive. The difference I like to draw7 becomes evident if one compares chess and football. Both are games, but in chess the rules establish legal and illegal moves before any chess--like activity is possible. So, they are generative of only and all acceptable moves. Whereas in football a prior act?-let?s call it kicking a ball?-is ?regimented? or structured by rules that arrive after the act itself. The rules do not and cannot determine the moves of the players. Instead, they put boundaries around what moves are acceptable as ?legal?. In chess, as in all board games whose rules are constitutive (draughts, Go, Monopoly, shogi, and so on), AI can use the rules to play any possible legal move that it wants to explore. In nine hours, AlphaZero played 44 million training games. To understand the magnitude of the achievement, consider that the Opening Encyclopedia 2018 contains approximately 6.3 million games selected from the history of chess. But in football, this would be a meaningless effort because the rules do not make the game?-they only shape it. This does not mean that AI cannot play virtual football, or cannot help identifying the best strategy to

7? The philosophical literature draws different distinctions, especially between regulative and constitutive rules, which seem compatible. However, the distinctions are not the same; for discussion, see Rawls (1955), Searle (2018), and Hage (2018). For Searle, both chess and football have what he calls ?constitutive rules?. To avoid confusion, I prefer the term ?constraining? instead of regulative.

Historical data

Synthetic data

Hybrid data

Figure 3.1? Shifting from entirely historical to truly synthetic data

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win against a team wh

ose data about previous games and strategies are recorded,
or cannot help with identifying potential players, or cannot train players better.
Of course, all these applications are now trivially feasible and have already occurred. What I mean is that in all these cases historical data are required. Only when (1) a process or interaction can be transformed into a game, and (2) the game can be transformed into a constitutive--rule game, then (3) AI will be able to generate its own, fully synthetic data and be the best ?player? on this planet, doing what AlphaZero did for chess (this process is part of the enveloping I described in Chapter 2).
To quote Wiener, ?the best material model of a cat is another, or preferably the same, cat? (Rosenblueth and Wiener (1945), 316). Ideally, the best data to train AI are either the entirely historical data or the fully synthetic data generated by the same rules that generated the historical data. In any board game, this happens by default. But insofar as either of the two steps (1)?(2) above is difficult to achieve, the absence of rules or the presence of merely constraining rules is likely to be a limit. We do not have the actual cat, only a more or less reliable model of it.
Things can get more complicated once we realize that, in actual games, the constraining rules are simply conventionally imposed on a previously occurring activity. But when we observe some phenomena in real life, such as the behaviour of a kind of tumour in a specific cohort of patients in some given circumstances, the genetic rules must be extracted from the actual ?game? through scientific (and these days possibly AI--based) research. We do not know, and perhaps we may never know, the exact ?rules? for the development of brain tumours. But we have some general principles and theories according to which we understand their development. So, at this stage, and it may well be a permanent stage, there is no way to ?ludify? brain tumours into a ?constitutive--rule game? such as chess. By ludify I mean transformation into a game in the sense specified above (I am avoiding the expression ?gamify?, which has a different and well--established meaning). In a constitutive--rule game, an AI system could generate its own synthetic data about brain tumours by playing according to the identified rules. This would be equivalent to the historical data we could collect, doing for brain tumours what AlphaZero has done for chess games. Unfortunately, this is not possible.
However, our inability to ludify brain tumours is not necessarily a problem. On the contrary, AI can still outperform experts by relying on historical or hybrid data (e.g. brain scans) and learning from them. It can still expand its capabilities beyond the finite historical data sets provided (e.g. by discovering new cor-relation patterns) and deliver accessible services where there is no expertise. It is already a great success if one can extract enough constraining rules to produce reliable data in silico. But without a reliable system of constitutive rules, some of the aforementioned advantages of synthetic data would not be available in full.
The vagueness of this statement is due to the fact that we can still use hybrid data. Ludification and the presence or absence of constraining/constitutive rules

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are not hard either--or limits. Here, recall that hybrid data c

an help to develop
synthetic data. What seems likely is that, in the future, it will become increasingly apparent when high--quality databases of historical data may be necessary and unavoidable. To paraphrase Wiener, this is when you need the actual cat. Hence, it is when we will have to deal with issues about availability, accessibility, legal compliance with legislation, and, in the case of personal data, privacy, consent, sensitivity, and other ethical questions. However, the trend towards the gen-er-ation of as--synthetic--as--possible (from synthesized, more or less hybrid, all the way to fully synthetic) data is likely to be one of AI?s holy grails, so I expect the AI community to push very hard in that direction. It is the model of the cat without the cat, relying once more on Wiener?s image. Generating increasingly non--historical data, making the dial in Figure 3.1 move as far as possible to the right, will require a ludification of processes. For this reason, I also expect the AI com-munity to become increasingly interested in the gaming industry because it is there that the best expertise in ludification is probably to be found. And in terms of negative results, mathematical proofs about the impossibility of ludifying whole kinds or areas of processes and interactions should be most welcome to clarify where or how far an AlphaZero--like approach may never be achiev-able by AI.

3.4? Difficult Problems, Complex Problems, and the Need for Enveloping

In Chapter 2 I argued that AI is probably best understood as a reservoir of agency that can be used to solve problems and perform tasks successfully. AI achieves its goals by detaching the ability to perform a task successfully from any need for intelligence to do so. The games app on my mobile phone does not need to be intelligent to play chess better than I do. Whenever this detachment is feasible, some AI solution becomes possible in principle. Therefore, understanding AI?s future also means understanding the nature of problems where such a detach-ment may be technically feasible, at least in theory, and then economically viable in practice. Now, many of the problems we try to solve through AI occur in the physical world. These range from driving to scanning labels in a supermarket, from cleaning floors or windows to cutting the grass in the garden. So, the reader may keep AI as robotics in mind for the rest of this section. However, I am not discussing only robotics. There are smart applications for allocating loans, for example, or smart interfaces for facilitating and improving interactions with the Internet of Things. These are also part of the analysis. What I like to suggest is that, to understand the development of AI when dealing with analogue and digital environments, it is useful to map problems based on what resources are needed to solve them?-and hence how far AI can have access to such resources.

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I am referring to computational resources and therefore to degrees of complexity, and to skill--related resources and therefore to degrees of difficulty.

Degrees of complexity in a problem are well known and extensively studied in computational theory (Arora and Barak 2009, Sipser 2012). I shall not say much about this dimension save to remark that it is highly quantitative, and the math-em-at-ic-al tractability it provides is due to the availability of standard crit

eria for
comparison (perhaps even idealized but clearly defined), such as the computational resources of a Turing Machine. If you have a ?metre?, then you can measure lengths. Similarly, if you adopt a Turing Machine as your starting point, then you can calculate how much time (in terms of steps) and how much space (in terms of memory or tape) are ?consumed? by a computational problem to be solved. Keep in mind that finely grained and sophisticated degrees of precision can be achieved, if needed, using tools from complexity theory. For simplicity, let us agree to map the complexity of a problem from 0 (simple) to 1 (complex). This problem is dealt with by AI in terms of space?-time = memory and steps required.
A problem can also be understood in terms of the skills required to solve it. Skills involve degrees of difficulty, from turning a light on or off to ironing shirts. But to be mapped here, they need a bit more stipulation because usually the relevant literature, for example in human motor development, does not focus on a taxonomy of problems based on resources required. Instead, it focuses on a taxonomy of the performance of the human agents. Performance is assessed in terms of the abilities or skills that the human agents demonstrate when solving a problem or performing a task. This is also more qualitative literature as there are many ways to assess performance and hence many ways to catalogue skill--related problems. However, one standard distinction lies between gross and fine motor skills. Gross motor skills require large muscle groups to perform tasks such as walking or jumping, and catching or kicking a ball. Fine motor skills require using smaller muscle groups in the wrists, hands, fingers, feet, and toes to perform tasks such as washing the dishes, writing, typing, using a tool, or playing an instrument.
Despite the challenge of assessing performance, you can see immediately that we are dealing with different degrees of difficulty here. Again, for the sake of sim-plicity (and recalling that in this case, too, more finely grained and sophisticated degrees of precision can be achieved using tools from developmental psychology if needed), let us agree to map the difficulty of a problem dealt with by AI in terms of skills required from 0 = easy to 1 = difficult. We are now ready to map the two dimensions in Figure 3.2, where I have added four examples.
Let me now use four elementary examples which are sufficiently correct to illustrate the point (strictly speaking the ?complexity? of tasks is not how I describe it here, apologies to any complexity expert, but this is not a problem for the illustrative purpose I have in mind). Turning the light on is a problem for which the solution has a very low degree of complexity (very few steps and states) and difficulty (even a child can do it). However, tying one?s own shoes requires advanced

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motor skills and so does lacing them. It is thus low in complexity (simple), but high in required skill (difficult). As Adidas CEO Kasper Rorsted remarked in 2017:
The biggest challenge the shoe industry has is how do you create a robot that puts the lace into the shoe. I?m not kidding. That?s a complete manual process today. There is no technology for that.8
Dishwashing is the opposite: while it may require a lot of steps and space (which increase the more dishes there are to be

e system that, together,
can adjust for fit following an algorithmic pressure equation. Strange things happen when the software does not work properly.12
There may be problems, and hence related tasks to solve them, that are not easily subject to enveloping. Yet here, it is not a matter of mathematical proofs. The issue is more of ingenuity, economic costs, and user or customer experiences and preferences. For example, a robot that irons shirts can be engineered. In 2012 a team at Carlos III University of Madrid, Spain, built TEO, a robot that weighs about 80 kilograms and stands 1.8 metres tall. TEO can climb stairs, open doors, and, more recently, has shown the ability to iron shirts (Estevez et al. 2017), although you have to put the item on the ironing board and then collect it afterwards. The view, quite widespread, is that:
?TEO is built to do what humans do as humans do it?, says team member Juan Victores at Carlos III University of Madrid. He and his colleagues want TEO to
10? http://www.moley.com/.
11? https://misorobotics.com/.
12? https://www.bbc.co.uk/news/business--47336684.

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be able to tackle other domestic tasks, like helping out in the kitchen. Their ul-tim-ate goal is for TEO to be able to learn how to do a task just by watching people with no technical expertise carry it out. ?We will have robots like TEO in our homes. It?s just a matter of who does it first?, says Victores.
(Estevez et al. 2017)
As you might guess, I think this is exactly the opposite of what is likely to happen. I strongly doubt this is the future. Because that is a view that fails to appreciate the distinction between difficult and complex tasks and the enormous advantage of enveloping tasks to make them easy (very low difficulty), no matter how complex. Recall that we are not building autonomous vehicles by putting androids into the driver?s seat, but by rethinking the whole ecosystem of vehicles plus environments. That is, we are removing the driver?s seat altogether. So, if my analysis is correct, the future of AI is not full of TEO--like androids mimicking human behaviour. The future is more likely represented by Effie,13 Foldimate,14 and other similar domestic automated machines that dry and iron clothes. They are not androids like TEO, but box--like systems that can be quite computationally sophisticated. They look more like dishwasher and washing machines. The difference is that, in their enveloped environments, their input is wrinkled clothes, and their output is ironed ones. Perhaps similar machines will be expensive. Perhaps they will not always work as well as one might wish. Perhaps they will be embodied in ways we cannot imagine now. After all, both companies have failed as businesses. But you can see how the logic is the correct one. We are transforming what lawnmowers are and what they look like, as well as adapting gardens to them. We are not building androids to push my old mower around as I do.
The lesson here is not to try to mimic humans through AI. Instead, exploit what machines, AI included, do best: difficulty is the enemy of machines and complexity is their friend. So, envelop the world around them and design new forms of embodiment to embed them successfully in their envelope. At that point, progressive refinements, market scale, adequate pricing, and marketing followed by new improvements will all become reaso

nable.
3.5? Generative Models
The first idea is old: all the texts are present in the dictionary, the difference is made by the syntax, that is, by how the dictionary words are structured into sentences (Borges 2000). The second idea is old: all the words in the dictionary are present in the alphabet, the difference is made by morphology, that is, by how the
13? https://helloeffie.com/.
14? https://e--foldimate.store/.

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letters of the alphabet are structured into words (Clarke 1967). The third idea is old: all the letters are present in the digital code, the difference is made by how the finite strings of zeros and ones of the digital code are structured into letters (Lodder 2008). The fourth idea is also old: all strings of zeros and ones are present in two electromagnetic properties, current high or low, magnetization present or absent, the difference is made by how such properties can be handled by electronic computational devices (Mano 1979). But the fifth idea is revolutionary: today, AI manages the properties of electromagnetism to process texts with extraordinary success and often in a way that is indistinguishable from how human beings would be able to do it. These are the so--called large language -models (LLMs), and they are rightly causing a sensation.
The most famous LLMs are GPT3, ChatGPT (also known as GPT4, produced by OpenAI--Microsoft, example below come from previous versions of ChatGPT, which are free) and Bard15 (produced by Google). They do not reason or understand, they are not a step towards any sci--fi AI, and they have nothing to do with the cognitive processes present in the animal world, and above all, in the human brain and mind, to manage semantic contents successfully (Bishop 2021).
However, with the staggering growth of available data, quantity and speed of calculation, and ever--better algorithms, they can do statistically?-that is, working on the formal structure, and not on the meaning of the texts they deal with?-what we do semantically, even if in ways that are still being explored by neuroscience.
Their abilities are extraordinary, as even the most sceptical must admit. Read a summary of The Divine Comedy made by ChatGPT (Figure 3.3). One may criticize the summary because it is longer than fifty words, and because The Divine Comedy is not an epic poem?-although there is a debate on this topic on the Internet, hence the ChatGPT summary?-but rather a tragedy, as Dante himself suggested. That said, the summary is not bad, and certainly better than one produced by a mediocre student. The exercise is no longer to make summaries
15? To be precise, LaMDA (Language Model for Dialogue Applications) is the Google language model, Bard is the name of the service.
Figure 3.3? ChatGPT Jan 30 Version. Test 1

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without using ChatGPT, but to teach how to use the right prompts (the question or request that generates the text, see the first line of my request), check the result, know how to correct the so--called ?hallucinations? in the text produced by ChatGPT, discover that there is a debate on which literary genre best applies to The Divine Comedy, and in the meantime, in doing all this, learn many things not only about the software but above all about The Divine Comedy itself. As I used to teach my students at Oxford in the 1990s, one of

the best exercises for writing an
essay on Descartes?s Meditations is not to summarize what has already been said, but to take the electronic text of one of the Meditations and try to improve its translation into English (thus one learns to check the original); clarify the less clear passages with a more accessible paraphrase (thus one sees if one has really understood the text); try to criticize or refine the arguments, modifying or strengthening them (thus one realizes that others have tried to do the same, and that is not so easy); and while doing all this learn the nature, internal structure, dynamics, and mechanisms of the content on which one is working. Or, to change the example, one really knows a topic not when one knows how to write a Wikipedia entry about it?-this can be done by ChatGPT increasingly well?-but when one knows how to correct it. One should use the software as a tool to get one?s hands on the text/mechanism, and get them dirty even by messing it up, as long as one masters the nature and the logic of the artefact called text.

The limitations of these LLMs are now obvious even to the most enthusiastic. They are fragile, because when they do not work, they fail catastrophically, in the etymological sense of vertical and immediate fall in the performance. The Bard disaster, where it provided incorrect information in a demonstration failure that cost Google over $100 billion in stock losses,16 is a good reminder that doing things with zero intelligence, whether digital or human, is sometimes very painful (Bing Chat also had its problems17). There is now a line of research that produces very sophisticated analyses on how, when, and why these LLMs, which seem incorrigible, have an unlimited number of Achilles heels (when asked what his Achilles heel is, ChatGPT correctly replied saying that it is just an AI system). They make up texts, answers, or references when they do not know how to answer; make obvious factual mistakes; sometimes fail to make the most trivial logical inferences or struggle with simple mathematics;18 or have strange linguistic blind spots where they get stuck (Floridi and Chiriatti 2020, Cobbe et al. 2021, Perez et al. 2022, Arkoudas 2023, Borji 2023, Christian 2023, Rumbelow 2023). A simple example in English illustrates well the limits of a mechanism that manages

16? https://www.reuters.com/technology/google--ai--chatbot--bard--offers--inaccurate--information--company--ad--2023--02--08/.

17? https://arstechnica--com.cdn.ampproject.org/c/s/arstechnica.com/information--technology/2023/02/ai--powered--bing--chat--spills--its--secrets--via--prompt--injection--attack/amp/.

18? https://venturebeat.com/business/researchers--find--that--large--language--models--struggle--with--math/.

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texts without understanding anything of their content. When asked?-using the Saxon genitive?-what is the name of Laura?s mother?s only daughter, the answer is kindly silly (Figure 3.4).

Forget passing the Turing test. Had I been Google, I would not have staked the fortunes of my company on such a brittle mechanism.

Given the enormous successes and equally broad limitations, some have compared LLMs to stochastic parrots that repeat texts without understanding anything (Bender et al. 2021). The analogy helps, but only partially, not only because parrots have an intelligence of their own that would be the envy of any

AI but,
above all, because LLMs synthesize texts in new ways, restructuring the contents on which they have been trained, not providing simple repetitions or juxta-positions. They look much more like the autocomplete function of a search engine. And in their capacity for synthesis, they approach those mediocre or lazy students who, to write a short essay, use a dozen relevant references suggested by the teacher, and by taking a little here and a little there, put together an eclectic text, coherent, but without having understood much or added anything. As a college tutor at Oxford, I corrected many of them every term. They can now be produced more quickly and efficiently by ChatGPT.
Unfortunately, the best analogy I know to describe tools such as ChatGPT is culturally bounded, and refers to a great classic in Italian literature, Manzoni?s The Betrothed (Manzoni 2016). In a famous scene in which Renzo (one of the main characters) meets a lawyer, we read: ?While the doctor [the lawyer] was uttering all these words, Renzo was looking at him with ecstatic attention, like a gullible person [materialone] stands in the square looking at the trickster
Figure 3.4? ChatGPT Jan 30 Version. Test 2

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[giocator di bussolotti], which, after stuffing tow and tow and tow into its mouth, takes out tape and tape and tape, which never ends [the word ?nastro? should be translated more correctly as ?ribbon?, but obviously ?tape? is preferable in this context, for it reminds one of the endless tape of a Turing Machine].? LLMs are like that trickster: they gobble data in astronomical quantities and regurgitate information. If we need the ?tape? of their information, it is good to pay close attention to how it was produced, why, and with what impact. And here we come to more interesting things, relevant to this chapter.
The impact of LLMs and the various AI systems that produce content of all kinds today will be enormous. Just think of DALL--E, which, as ChatGPT says (I quote without any modification), ?is an artificial intelligence system developed by OpenAI that generates original images starting from textual descriptions. It uses state--of--the--art machine learning techniques to produce high--quality images that match input text, including captions, keywords, and even simple sentences. With DALL--E, users can enter a text description of the image they want, and the system will produce an image that matches the description.? There are ethical and legal issues: just think of copyright and the reproduction rights linked to the data sources on which the AI in question is trained. The first lawsuits have already begun,19 and there have already been the first plagiarism scandals.20 There are human costs: consider the use of contractors in Kenya, paid less than $2/hour to label harmful content to train ChatGPT; they were unable to access adequate mental health resources, and many have been left traumatized.21 There are human problems, like the impact on teachers who have to scramble to revamp their curriculum,22 or security considerations, for example concerning the outputs of AI processes that are increasingly integrated into medical diagnostics, with implications of algorithmic poisoning of the AI?s training data. Or think of the financial and environmental costs of these new systems (Cowls et al. 2021a, 2021b): is this type of innovation fair and s

ustainable? Then there are questions related to
the best use of these tools, at school, at work, in research environments and for scientific publications, in the automatic production of code, or the generation of content in contexts such as customer service, or in the drafting of any text, including scientific articles or new legislation. Some jobs will disappear, others are already emerging, and many will have to be reconsidered.
But above all, there are many challenging questions about: the emergence of LEGO--like AI systems, working together in a modular and seamless way, with LLMs acting as an AI2AI kind of bridge to make them interoperable, as a sort of

19? https://news.bloomberglaw.com/ip--law/first--ai--art--generator--lawsuits--threaten--future--of--emerging--tech.
20? https://www.washingtonpost.com/media/2023/01/17/cnet--ai--articles--journalism--corrections/.
21? https://time.com/6247678/openai--chatgpt--kenya--workers/.
22? https://ethicalreckoner.substack.com/p/er13--on--community--chatgpt--and--human.

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?confederated AI?;23 the relationship between form and its syntax, and content and its semantics; the nature of personalization of content and the fragmentation of shared experience (AI can easily produce a unique, single novel on--demand, for a single reader, for example); the concept of interpretability, and the value of the process and the context of the production of meaning; our uniqueness and origin-al-ity as producers of meaning and sense, and of new contents; our ability to interact with systems that are increasingly indiscernible from other human beings; our replaceability as readers, interpreters, translators, synthesizers and evaluators of content; power as the control of questions, because, to paraphrase 1984, whoever controls the questions controls the answers, and whoever controls the answers controls reality.
More questions will emerge as we develop, interact, and learn to understand this new form of agency. As Vincent Wang reminded me, ChatGPT leapfrogged GPT3 in performance by the introduction of reinforcement learning (RL) to fine--tune its outputs as an interlocutor, and RL is the machine--learning approach to ?solving agency?. It is a form of agency never seen before, because it is successful and can ?learn? and improve its behaviour without having to be intelligent in doing so. It is a form of agency that is alien to any culture in any past, because humanity has always and everywhere seen this kind of agency?-which is not that of a sea wave, which makes the difference, but can make nothing but that difference, without being able to ?learn? to make a different or better difference?-as a natural or even supernatural form of agency.
We have gone from being in constant contact with animal agents and what we believed to be spiritual agents (gods and forces of nature, angels and demons, souls or ghosts, good and evil spirits) to having to understand, and learn to interact with, artificial agents created by us, as new demiurges of such a form of agency. We have decoupled the ability to act successfully from the need to be intelligent, understand, reflect, consider, or grasp anything. We have liberated agency from intelligence. So, I am not sure we may be ?shepherds of Being? (Heidegger), but it looks like the new ?green collars? (Floridi 2017) will be ?shepherds of AI systems?, in charge of this new form

of artificial agency.
The agenda of a demiurgic humanity of this intelligence--free (as in fat--free) AI?-understood as Agere sine Intelligere, with a bit of high school Latin?-is yet to be written. It may be alarming or exciting for many, but it is undoubtedly good news for philosophers looking for work.
23? I owe this remark to Vincent Wang, who reminded me of two interesting examples: (1) having ChatGPT and Wolfram Alpha talk to each other; ChatGPT outsources mathematics questions to Wolfram Alpha, which has considerable ability by itself to parse mathematical questions in natural language format, see https://writings.stephenwolfram.com/2023/01/wolframalpha--as--the--way--to--bring--computational--knowledge--superpowers--to--chatgpt/; (2) ?Socratic Models? for multimodal grounding/reasoning, where the idea is to tag different forms of data, e.g. sounds and images, with text descriptions so that an LLM can serve as ?central processing?, allowing different narrow AIs to talk to each other. https://socraticmodels.github.io/.

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Figure 3.5 shows how ChatGPT Jan 30 Version evaluates the beginning of this section. It is quite flattering.
3.6? A Future of Design
The futures I have outlined here, based on data and problems, are complementary, based on our current and foreseeable understanding of AI. There are unknown unknowns, of course, but all one can say about them is precisely this: they exist, and we have no idea about them. It is a bit like saying that we know there are questions we are not asking, but we cannot say what these questions are. The future of AI is full of unknown unknowns. What I have tried to do in this chapter is to look at the ?seeds of time? we have already sowed. I have concentrated on the nature of data and of problems because the former are what enable AI to work, and the latter provide the boundaries within which AI can work successfully. At this level of abstraction, two inferences seem to be very plausible. First, we will seek to develop AI by using data that are hybrid and preferably synthetic as much as possible. We will do so through a process of ludifying interactions and tasks. In other words, the tendency will be to try to move away from purely -historical data insofar as this is possible. In areas such as health and economics,
Figure 3.5? ChatGPT Jan 30 Version. Test 3

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it may well be that historical or at most hybrid data will remain necessary due to the difference between constraining and constitutive rules. Second, we will do all this by translating difficult problems into complex problems as much as possible. Translation will happen through the enveloping of realities around the skills of our artefacts. In short, we will seek to create hybrid or synthetic data to deal with complex problems by ludifying tasks and interactions in enveloped environments. The more this is possible, the more successful AI will be. That is why a trend like the development of digital twin cities is very interesting, which leads me to two more comments.
Ludifying and enveloping are a matter of designing, or sometimes redesigning, the realities with which we deal (Floridi 2019d). So, the foreseeable future of AI will depend on our design abilities and ingenuity. It will also depend on our ability to negotiate the serious, ethical, legal, and social issues

(ELSI) that result.
These range from new forms of privacy (by proxy, predictive, or group--based) to nudging and self--determination. We saw in Chapter 2 how the very idea that we are increasingly shaping our environments (whether analogue or digital) to make them AI--friendly should make anyone reflect. Anticipating such issues to facilitate positive ELSI and avoid or mitigate any negative ones is the real value of any foresight analysis. It is interesting to try to understand what the paths of least resistance may be in the evolution of AI. But it would be quite sterile to try to predict ?which grain will grow and which will not?, then do nothing to ensure that the good grains grow and the bad ones do not (Floridi 2014d). The future is not entirely open because the past shapes it. But neither is it entirely determined because the past can be steered in a different direction. Therefore the challenge ahead will not be so much digital innovation per se, but the governance of the digital?-AI included.

3.7? Conclusion: AI and Its Seasons

The trouble with seasonal metaphors is that they are cyclical. If you say that AI got through a bad winter, you must also remember that winter will return, and you had better be ready for it. An AI winter is that stage when technology, business, and the media get out of their warm and comfortable bubble, cool down, temper their sci--fi speculations and unreasonable hype, and come to terms with what AI can or cannot really do as a technology (Floridi 2019f) without ex-ag-geration. Investments become more discerning, and journalists stop writing about AI to chase some other fashionable topic and fuel the next fad.
AI has had several winters.24 Of the most significant, there was one in the late seventies and another at the turn of the eighties and nineties. Today, we are -talking
24? https://en.wikipedia.org/wiki/AI_winter.

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about yet another predictable winter (Nield 2019, Walch 2019, Schuchmann 2019).25 AI is subject to these hype cycles because it is a hope or fear that we have entertained since we were thrown out of paradise: something that does everything for us instead of us, better than us, with all the dreamy advantages (we shall be on holiday forever) and the nightmarish risks (we are going to be enslaved) that this entails. For some people, speculating about all this is irresistible. It is the wild west of ?what if? scenarios and ?hypotheses?. But I hope the reader will forgive me for an ?I told you so? moment. For some time, I have been warning against commentators and ?experts? competing to see who could tell the tallest tale (Floridi 2016d). A web of myths has ensued. They spoke of AI either as if it were the ultimate panacea that would solve everything and overcome everything, or as the final catastrophe: a superintelligence that would destroy millions of jobs, replacing lawyers, doctors, journalists, researchers, truckers, and taxi drivers, and end up dominating human beings as though we were pets at best. Many followed Elon Musk in declaring the development of AI to be the greatest existential risk run by humanity?-as if most of humanity did not live in misery and suffering, as if wars, famine, pollution, global warming, social injustice, and fundamentalism were science fiction or just negligible nuisances unworthy of their considerations. Today, the COVID--19 pandemic and the war

in Ukraine should put an end to
such silly statements. Some people insisted that law and regulations were always going to be too late to catch up with AI. In fact, norms are not about the pace but the direction of innovation. Norms should steer the proper development of a society, not follow it. If we like where we are heading, we cannot go there quickly enough. It is due to our lack of vision that we are fearful. Today, we know that legislation is coming at least in the EU. Some other people (not necessarily different from the former) claimed AI was a magical black box that we could never explain. In fact, explanation is a matter of identifying the correct level of abstraction at which to interpret the complex interactions engineered by AI. Even car traffic downtown can become a black box if what you wish to know is why every single individual is there at that moment. Today, there is growing development of adequate tools to monitor and understand how ML systems reach their outcomes (Watson et al. 2019, Watson and Floridi 2020, Watson et al. 2021).
People also spread scepticism about the possibility of an ethical framework that would synthesize what we mean by socially good AI. In fact, the EU, the Organisation for Economic Co--operation and Development (OECD), and China have converged on very similar principles that offer a common platform for further agreements, as we shall see in Chapter 4. These were all irresponsible individuals in search of headlines. They should be ashamed and offer an apology not only for 25? Even the BBC has contributed to the hype (e.g. see: https://www.bbc.co.uk/programmes/p031wmt7). It now acknowledges this might have been?.?.?.?hype: https://www.bbc.co.uk/news/technology--51064369.

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their untenable comments but also for their great carelessness and alarmism that have misled public opinion about both a potentially useful technology and its real risks. From medicine to security and monitoring systems (Taddeo and Floridi 2018a), AI can and does provide helpful solutions. The risks we know of are concrete, but so much less fancy than claimed. They range from the everyday manipulation of choices (Milano, Taddeo, and Floridi 2019, 2020a) to increased pressures on individual and group privacy (Floridi 2014c), and from cyberconflicts to the use of AI by organized crime for money laundering and identity theft, as we shall see in Chapters 7 and 8.
Every AI summer risks turning over--inflated expectations into a mass distraction. Every AI winter risks excessive backlash, overly negative disappointment, and potentially valuable solutions thrown out with the bathwater of illusions.
Managing the world is an increasingly complex task (megacities and their ?smartification? offer a good example). And we also have planetary problems, such as pandemics, climate change, social injustice, migration, wars. These require ever higher degrees of coordination to solve. It seems obvious that we need all the good technology that we can design, develop, and deploy to cope with these challenges, and all the human intelligence that we can exercise to put this technology into the service of a better future. AI can play an important role in all of this because we need increasingly smarter methods to process immense quantities of data in ways that are efficient, efficacious, sustainable, and fair, as I shall argue in the second

focuses on more applied
questions. Chapter 7 offers an updated map of the main ethical problems raised by algorithms. The chapter seeks to strike a balance between inclusivity?-it concerns all kinds of algorithms, not only those driving AI systems or applications?-and specificity. It does not focus explicitly on robotics and hence on ethical issues arising from the embodiment of AI, for example. Some of the problems that need to be tackled both legally (compliance) and ethically (soft ethics) are concrete and pressing, as shown in Chapter 8 on the criminal misuse of AI for ?social evil?. However, AI is mostly about great opportunities, as shown in Chapter 9 on AI for social good (AI4SG). When working on these two chapters, I would call them ?bad AI? and ?good AI?. In both chapters, AI includes robotics. Chapters 10?12 return to the development of good AI by focusing on how to deliver a socially good AI, the positive and negative impact of AI on climate change, and the suit-abil-ity of AI to support the UN Sustainable Development Goals (SDGs). Finally,

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Chapter 13 completes Part Two and the book by arguing for a new marriage between the Green of all our habitats and the Blue of all our digital technologies (AI in particular). It provides a reminder that agency, including the new form represented by AI, needs governance; that governance is a matter of socio--political activities; and that the digital has also modified socio--political agency, which is the topic of the next book, The Politics of Information.

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4
A Unified Framework of Ethical Principles for AI

4.0? Summary

Previously, in Chapters 1?3, we saw how AI is a new form of agency that can deal with tasks and problems successfully, in view of a goal, without any need for intel-ligence. Every success of any AI application does not move the bar of what it means to be an intelligent agent. Instead, it bypasses the bar altogether. The success of such artificial agency is increasingly facilitated by the enveloping (that is, reshaping into AI--friendly contexts) of the environments in which AI operates. The decoupling of agency and intelligence and the enveloping of the world generate significant ethical challenges, especially in relation to autonomy, bias, explainability, fairness, privacy, responsibility, transparency, and trust (yes, mere alphabetic order). For this reason, many organizations launched a wide range of initiatives to establish ethical -principles for the adoption of socially beneficial AI after the Asilomar AI Principles and the Montreal Declaration for a Responsible Development of Artificial Intelligence were published in 2017. This soon became a cottage industry. Unfortunately, the sheer volume of proposed principles -threatens to overwhelm and confuse.

This chapter presents the results of a comparative analysis of several of the highest profile sets of ethical principles for AI. I assess whether these principles converge around a set of agreed--upon principles, or diverge with significant dis-agree-ment over what constitutes ?ethical AI?. In the chapter, I argue that there is a high degree of overlap across the sets of principles under analysis. There is an overarching framework consisting of five core principles for ethical AI. Four of them are core principles commonly used in bioethics: beneficence, nonmalefi-cence, autonomy, and just

ice. This is unsurprising when AI is interpreted as a form
of agency. And the coherence between bioethics and the ethics of AI may be seen
as evidence that approaching AI as a new form of agency is fruitful. Readers who
disagree may find that an intelligent--oriented interpretation of AI could cohere
more easily with a virtue ethics approach, rather than with bioethics. I find
approaching the ethics of AI and digital ethics in general from a virtue ethics
perspective unconvincing, but it remains a viable possibility and this is not the
place to rehearse my objections (Floridi 2013).
Based on the comparative analysis, the addition of a new principle is needed:
explicability. Explicability is understood as incorporating both the epistemological
The Ethics of Artificial Intelligence: Principles, Challenges, and Opportunities. Luciano Floridi, Oxford University Press.
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sense of intelligibility?-as an answer to the question ?how does it work???and in
the ethical sense of accountability?-as an answer to the question ?who is re-spon-
sible for the way it works?? The extra principle is required by the fact that AI is a
new form of agency. The chapter ends with a discussion of the implications of this
ethical framework for future efforts to create laws, rules, technical standards, and
best practices for ethical AI in a wide range of contexts.

4.1? Introduction: Too Many Principles?

As we have seen in the preceding chapters, AI is already having a major impact on
society, which will only increase in the future. The key questions are how, where,
when, and by whom the impact of AI will be felt. I shall return to this point in
Chapter 10. Here I would like to focus on a consequence of this trend: many or-gan-
iza-tions have launched a wide range of initiatives to establish ethical prin-ciples for
the adoption of socially beneficial AI. Unfortunately, the sheer volume of proposed
principles?-already more than 160 in 2020, according to Algorithm Watch?s AI
Ethics Guidelines Global Inventory (Algorithm Watch 2020)1?-threatens to over-
whelm and confuse. This poses two potential problems: if the various sets of ethical
principles for AI are similar, this leads to unnecessary repetition and redundancy;
yet if they differ significantly, confusion and ambiguity will result instead. The worst
outcome would be a ?market for principles? wherein stakeholders may be tempted to
?shop? for the most appealing ones, as we shall see in Chapter 5.
How might this problem of ?principle proliferation? be solved? In this chapter, I
present and discuss the results of a comparative analysis of several of the highest
profile sets of ethical principles for AI. I assess whether these principles converge
around a shared set of agreed--upon principles, or diverge with significant dis-
agree-ment over what constitutes ethical AI. The analysis finds a high degree of
overlap among the sets of principles under analysis. This leads to the identifica-
tion of an overarching framework consisting of five core principles for ethical
AI. In the ensuing discussion, I note the limitations and assess the implications of
this ethical framework for future efforts to create laws, rules, standards, and best
practices for ethical AI in a wide range of contexts.

4.2? A Unified Framework of Five Principles for Ethical AI

We saw in Chapter 1 that the establishment of AI as a fiel

d of academic research
dates to the 1950s (McCarthy et al. 2006 [1955]). The ethical debate is almost as
old (Wiener 1960, Samuel 1960b). But it is only in recent years that impressive
advances in the capabilities and applications of AI systems have brought the
opportunities and risks of AI for society into sharper focus (Yang et al. 2018). The

1? See also Jobin, Ienca, and Vayena (2019).

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increasing demand for reflection and clear policies on the impact of AI on society
has yielded a glut of initiatives. Each additional initiative provides a supplementary
statement of principles, values, or tenets to guide the development and adoption of
AI. You get the impression that a ?me too? escalation had taken place at some point.
No organization could be without some ethical principles for AI and accepting the
ones already available looked inelegant, unoriginal, or lacking in leadership.
The ?me too? attitude was followed up by ?mine and only mine?. Years later, the
risk is still unnecessary repetition and overlap, if the various sets of principles are
similar, or confusion and ambiguity if they differ. In either eventuality, the devel-
opment of laws, rules, standards, and best practices to ensure that AI is socially
beneficial may be delayed by the need to navigate the wealth of principles and dec-
larations set out by an ever--expanding array of initiatives. The time has come for a
comparative analysis of these documents, including an assessment of whether they
converge or diverge and, if the former, whether a unified framework may therefore
be synthesized. Table 4.1 shows the six high--profile initiatives established in the
interest of socially beneficial AI chosen as the most informative for a comparative
analysis. I shall analyse two more sets of principles later in the chapter.
Each set of principles in Table 4.1 meets four basic criteria. They are:

Table 4.1? Six sets of ethical principles for AI
1)? The Asilomar AI Principles (Future of Life Institute 2017) developed under the
auspices of the Future of Life Institute in collaboration with attendees of the
high--level Asilomar conference in January 2017 (hereafter Asilomar);
2)? The Montreal Declaration (Université de Montréal 2017) developed under the
auspices of the University of Montreal following the Forum on the Socially
Responsible Development of AI in November 2017 (hereafter Montreal);2
3)? The general principles offered in the second version of Ethically Aligned Design: A
Vision for Prioritizing Human Well--Being with Autonomous and Intelligent Systems
(IEEE (2017), 6). This crowd--sourced document received contributions from 250
global thought leaders to develop principles and recommendations for the ethical
development and design of autonomous and intelligent systems, and was published
in December 2017 (hereafter IEEE);
4)? The ethical principles offered in the Statement on Artificial Intelligence, Robotics and
?Autonomous? Systems (EGE 2018) published by the European Commission?s European
Group on Ethics in Science and New Technologies in March 2018 (hereafter EGE);
5)? The ?five overarching principles for an AI code? offered in the UK House of Lords
Artificial Intelligence Select Committee?s report ?AI in the UK: Ready, Willing and
Able?? (House of Lords Artificial Intelligence Committee 16 April 2017, §417)
published i

n April 2018 (hereafter AIUK);
6)? The Tenets of the Partnership on AI (Partnership on AI 2018) published by a multi--stakeholder organization consisting of academics, researchers, civil society organizations, companies building and utilizing AI technology, and other groups (hereafter the Partnership).
2? The principles referred to here are those that were publicly announced as of 1 May 2018 and available at the time of writing this chapter here: https://www.montrealdeclaration-responsibleai.com/the-declaration.

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a) recent, as in published since 2017;

b) directly relevant to AI and its impact on society as a whole (thus excluding documents specific to a particular domain, industry, or sector);

c) highly reputable, published by authoritative, multi--stakeholder organizations with at least a national scope;3 and

d) influential (because of a?-c).

Taken together, they yield forty--seven principles.4 Despite this, the differences are mainly linguistic and there is a degree of coherence and overlap across the six sets of principles that is impressive and reassuring. The convergence can most clearly be shown by comparing the sets of principles with the four core principles commonly used in bioethics: beneficence, nonmaleficence, autonomy, and justice (Beauchamp and Childress 2013). This comparison should not be surprising. As the reader may recall, the view I support in this book is that AI is not a new form of intelligence but an unprecedented form of agency. So of all the areas of applied ethics, bioethics is the one that most closely resembles digital ethics when dealing ecologically with new forms of agents, patients, and environments (Floridi 2013). The four bioethical principles adapt surprisingly well to the fresh ethical challenges posed by AI. However, they do not offer a perfect translation. As we shall see, the underlying meaning of each of the principles is contested, with similar terms often used to mean different things. Nor are the four principles exhaustive. Based on the comparative analysis, it becomes clear, as noted in the summary, that an additional principle is needed: explicability. Explicability is understood as incorporating both the epistemological sense of intelligibility?-as an answer to the question ?how does it work???and in the ethical sense of accountability?-as an answer to the question ?who is responsible for the way it works?? This additional principle is needed both for experts, such as product designers or engineers, and for non--experts, such as patients or business customers.5 And the new principle is required by the fact that AI is not like any biological form of agency. However, the convergence detected between these different sets of principles also demands caution. I explain the reasons for this caution later in the chapter, but first, let me introduce the five principles.

3? A similar evaluation of AI ethics guidelines, undertaken recently by Hagendorff (2020), adopts different criteria for inclusion and assessment. Note that the evaluation includes in its sample the set of principles we describe here.
4? Of the six documents, the Asilomar Principles offer the largest number of principles with arguably the broadest scope. The twenty-three principles are organized under three headings: ?research issues?, ?ethics and values?, and ?longer-term issues?. Consideration of the five ?research

issues? is omitted
here as they are related specifically to the practicalities of AI development in the narrower context of academia and industry. Similarly, the Partnership?s eight tenets consist of both intra-organizational objectives and wider principles for the development and use of AI. Only the wider principles (the first, sixth, and seventh tenets) are included here.
5? There is a third question that concerns what new insights can be provided by explicability. It is not relevant in this context, but equally important; for detail, see Watson and Floridi (2020).

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4.3? Beneficence: Promoting Well--Being, Preserving
Dignity, and Sustaining the Planet
The principle of creating AI technologies that are beneficial to humanity is expressed in different ways across the six documents. Still, it is perhaps the easiest of the four traditional bioethics principles to observe. The Montreal and IEEE principles both use the term ?well--being?; Montreal notes that ?the development of AI should ultimately promote the well--being of all sentient creatures?, while IEEE states the need to ?prioritize human well--being as an outcome in all system designs?. AIUK and Asilomar both characterize this principle as the ?common good?: AI should ?be developed for the common good and the benefit of humanity?, according to AIUK. The Partnership describes the intention to ?ensure that AI technologies benefit and empower as many people as possible?, while the EGE emphasizes the principle of both ?human dignity? and ?sustainability?. Its principle of ?sustainability? articulates perhaps the widest of all interpretations of beneficence, arguing that ?AI technology must be in line with?.?.?.?ensur[ing] the basic preconditions for life on our planet, continued prospering for mankind and the preservation of a good environment for future generations?. Taken together, the prominence of beneficence firmly underlines the central importance of promoting the well--being of people and the planet with AI. These are points to which I shall return in the following chapters.
4.4? Nonmaleficence: Privacy, Security, and ?Capability Caution?
Although ?do only good? (beneficence) and ?do no harm? (nonmaleficence) may seem logically equivalent, they are not. Each represents a distinct principle. The six documents all encourage the creation of beneficent AI, and each one also cautions against the various negative consequences of overusing or misusing AI technologies (see Chapters 5, 7, and 8). Of particular concern is the prevention of infringements on personal privacy, which is included as a principle in five of the six sets. Several of the documents emphasize avoiding the misuse of AI technologies in other ways. The Asilomar Principles warn against the threats of an AI arms race and of the recursive self--improvement of AI, while the Partnership similarly asserts the importance of AI operating ?within secure constraints?. The IEEE docu-ment meanwhile cites the need to ?avoid misuse?, and the Montreal Declaration argues that those developing AI ?should assume their responsibility by working against the risks arising from their technological innovations?. Yet from these various warnings, it is not entirely clear whether it is the people developing AI, or the technology itself, that should be encouraged to do no harm. In other words, it is unclear whether it is Dr Frankenstein (as I suggest) or his monster against whose

maleficence we should be guarding.
At the heart of this quandary is the question of autonomy.

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4.5? Autonomy: The Power to ?Decide to Decide?
When we adopt AI and its smart agency, we willingly cede some of our decision--making power to technological artefacts. Thus, affirming the principle of autonomy in the context of AI means striking a balance between the decision--making power we retain for ourselves and that which we delegate to AAs. The risk is that the growth in artificial autonomy may undermine the flourishing of human autonomy. It is therefore unsurprising that the principle of autonomy is explicitly stated in four of the six documents. The Montreal Declaration articulates the need for a balance between human- and machine--led decision--making, stating that ?the development of AI should promote the autonomy [emphasis added] of all human beings?. The EGE argues that autonomous systems ?must not impair [the] freedom of human beings to set their own standards and norms?, while AIUK adopts the narrower stance that ?the autonomous power to hurt, destroy or deceive human beings should never be vested in AI?. The Asilomar document similarly supports the principle of autonomy, insofar as ?humans should choose how and whether to delegate decisions to AI systems, to accomplish human--chosen objectives?.
It is therefore clear both that the autonomy of humans should be promoted, and that the autonomy of machines should be restricted. The latter should be made intrinsically reversible, should human autonomy need to be protected or re--established (consider the case of a pilot able to turn off the automatic pilot function and regain full control of the airplane). This introduces a notion one may call meta--autonomy, or a decide--to--delegate model. Humans should retain the power to decide which decisions to take and in what priority, exercising the freedom to choose where necessary and ceding it in cases where overriding -reasons, such as efficacy, may outweigh the loss of control over decision--making. But any delegation should also remain overridable in principle, adopting an ultimate safeguard of deciding to decide again.
4.6? Justice: Promoting Prosperity, Preserving Solidarity,
Avoiding Unfairness
The decision to make or delegate decisions does not take place in a vacuum, nor is this capacity distributed equally across society. The consequences of this disparity in autonomy are addressed by the principle of justice. The importance of justice is explicitly cited in the Montreal Declaration, which argues that ?the development of AI should promote justice and seek to eliminate all types of discrimination?. Likewise, the Asilomar Principles include the need for both ?shared benefit? and ?shared prosperity? from AI. Under its principle named ?Justice, equity and solidarity?, the EGE argues that AI should ?contribute to global justice and equal

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access to the benefits? of AI technologies. It also warns against the risk of bias in datasets used to train AI systems, and?-unique among the documents?-argues for the need to defend against threats to ?solidarity?, including ?systems of mutual assistance such as in social insurance and healthcare?.
Elsewhere, ?justice? has still other meanings (especially in the sense of fairness), variously related to the use of AI

in correcting past wrongs, such as by eliminating
unfair discrimination, promoting diversity, and preventing the reinforcement of biases or the rise of new threats to justice. The diverse ways in which justice is characterized hints at a broader lack of clarity over AI as a human--made reservoir of ?smart agency?. Put simply, are we humans the patient receiving the ?treatment? of AI, which would be acting as ?the doctor? prescribing it, or both? This question can only be resolved with the introduction of a fifth principle that emerges from the previous analysis.

4.7? Explicability: Enabling the Other Principles through Intelligibility and Accountability

In response to the previous question about whether we are the patient or the doctor, the short answer is that we could actually be either. It depends on the circumstances and on who ?we? refers to in everyday life. The situation is inherently unequal: a small fraction of humanity is currently engaged in the development of a set of technologies that are already transforming the everyday lives of almost everyone else. This stark reality is not lost on the authors of the documents under analysis. All of them refer to the need to understand and hold to account the decision--making processes of AI. Different terms express this principle: ?transparency? in Asilomar and EGE; both ?transparency? and ?accountability? in IEEE; ?intelligibility? in AIUK; and as ?understandable and interpretable? by the Partnership. Each of these principles captures something seemingly novel about AI as a form of agency: that its workings are often invisible, opaque, or unintelligible to all but (at best) the most expert observers.

The principle of ?explicability? incorporates both the epistemological sense of ?intelligibility? and the ethical sense of ?accountability?. The addition of this prin-ciple is the crucial missing piece of the AI ethics jigsaw. It complements the other four principles: in order for AI to be beneficent and non--maleficent, we must be able to understand the good or harm that it is actually doing to society, and in what ways; for AI to promote and not constrain human autonomy, our ?decision about who should decide? must be informed by knowledge of how AI would act instead of us and how to improve its performance; and for AI to be just, we must know whom to hold ethically responsible (or actually legally liable) in the event of a serious, negative outcome, which would in turn require an adequate understanding of why this outcome arose, and how it could be prevented or minimized in the future.

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4.8? A Synoptic View

Taken together, the previous five principles capture every one of the forty--seven principles contained in the six high--profile, expert--driven documents analysed in Table 4.1. As a test, it is worth stressing that each principle is included in almost every statement of principles analysed in Table 4.2 (see Section 4.10). The five principles therefore form an ethical framework within which policies, best practices, and other recommendations may be made. This framework of principles is shown in Figure 4.1.

4.9? AI Ethics: Whence and for Whom?

It is important to note that each of the six sets of ethical principles for AI in Table 4.1 emerged either from initiatives with a global scope or from within Western liberal democracies. For the framework to be more broadly a

pplicable, it
would undoubtedly benefit from the perspectives of regions and cultures pres-
ently un- or under--represented in our sample. Of particular interest in this respect
is the role of China, which is already home to the world?s most valuable AI start-
-up (Jezard 11 April 2018), enjoys various structural advantages in developing AI
(Lee and Triolo December 2017), and whose government has stated its ambitions
to lead the world in state--of--the--art AI technology by 2030 (China State Council
8 July 2017). This is not the context for an analysis of China?s AI policies,6 so let
me add only a few closing remarks.
In its State Council Notice on AI and elsewhere, the Chinese government
expressed interest in further consideration of the social and ethical impact of AI
(Ding 2018). Enthusiasm for the use of technologies is hardly unique to govern-
ments. It is also shared by the general public?-more so those in China and India
than in Europe or the USA, as new representative survey research shows

Beneficence
Non-
maleficence
Autonomy
Traditional bioethics
principles
A new enabling
principle for AI
Justice
Explicability
Figure 4.1? An ethical framework of the five overarching principles for AI
6? On this, see Roberts, Cowls, Morley, et al. (2021), Roberts, Cowls, Hine, Morley, et al. (2021),
Hine and Floridi (2022).

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(Vodafone Institute for Society and Communications 2018). In the past, an ex-ecu-tive
at the major Chinese technology firm Tencent suggested that the EU should focus
on developing AI that has ?the maximum benefit for human life, even if that
technology isn?t competitive to take on [the] American or Chinese market?
(Boland 14 October 2018). This was echoed by claims that ethics may be ?Europe?s
silver bullet? in the ?global AI battle? (Delcker 3 March 2018). I disagree. Ethics is
not the preserve of a single continent or culture. Every company, government
agency, and academic institution designing, developing, or deploying AI has an
obligation to do so in line with an ethical framework?-even if not along the lines of
the one presented here?-broadened to incorporate a more geo-graph-ic-al-ly, culturally,
and socially diverse array of perspectives. Similarly, laws, rules, standards, and
best practices to constrain or control AI (including all those currently under
consideration by regulatory bodies, legislatures, and industry groups) could also
benefit from close engagement with a unified framework of ethical principles.
What remains true is that the EU may have an advantage in the so--called Brussels
effect, but this only places more responsibility on how AI is regulated by the EU.
4.10? Conclusion: From Principles to Practices
If the framework presented in this chapter provides a coherent and sufficiently
comprehensive overview of the central ethical principles for AI (Floridi et al.
2018), then it can serve as the architecture within which laws, rules, technical
standards, and best practices are developed for specific sectors, industries, and
jurisdictions. In these contexts, the framework may play both an enabling role
(see Chapter 12) and a constraining one. The latter role refers to the need to regu-
late AI technologies in the context of online crime (see Chapter 7) and cyberwar,
which I discuss in Floridi and Taddeo (2014), Taddeo and Floridi (2018b). Indeed,
the framework shown in Figure 4.1 played a valuable role in fi

cally applied contexts, such as environmental ethics,
bioethics, medical ethics, and business ethics. However, I argue that each acquires its specific features by being uniquely related to the ethics of AI. They are all avoidable if one can identify them, and this chapter suggests how. The conclusion is that part of the ethical approach to AI is also building awareness about the nature and occurrence of such risks and enhancing a preventive approach.

5.1? Introduction: Risky Translations

It has taken a very long time, but we have seen that the debate today on the ethical impact and implications of digital technologies has reached the front pages of newspapers. This is understandable given that digital technologies, from web--based services to AI solutions, increasingly affect the daily lives of billions of -people, especially after the pandemic. So there are both many hopes and many concerns about their design, development, and deployment. After more than half a century of academic research (Wiener 1960, Samuel 1960a), we saw in Chapter 4 that the recent public reaction has been a flourishing of initiatives to establish what principles, guidelines, codes, or frameworks may ethically guide digital innovation, particularly in AI, to benefit humanity and the whole environment.
This is a positive development that shows awareness of the importance of the topic and interest in tackling it systematically. Yet it is time that the debate evolves from the what to the how. It should address not just what ethics is needed, but also how ethics can be effectively and successfully applied and implemented to make a positive difference. For example, the EU Ethics Guidelines for Trustworthy

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AI (HLEGAI 18 December 2018)1 establish a benchmark for what may or may not qualify as ethically good AI in the EU. Their publication will increasingly be followed by practical efforts of testing, application, and implementation. They influenced the framing of the EU AI Act.
The move from a first, more theoretical what stage to a second, more practical how stage, so to speak, is reasonable, commendable, and feasible (Morley, Floridi, et al. 2020). But in translating ethical principles into good practices, even the best efforts may be undermined by some unethical risks. In this chapter, I highlight the main ones that seem more likely to occur. We shall see that they are more clusters than individual risks, and there may be other clusters as well, but these five are the ones already encountered or foreseeable in the international debate about digital ethics and the ethics of AI.2 Here is the list: ethics shopping; ethics bluewashing; ethics lobbying; ethics dumping; and ethics shirking. They are the five ?ethics gerunds?, to borrow an apt label from Josh Cowl, who also suggested (in a conversation) considering the first three more ?distractive? and the last two more ?destructive? problems. Let us consider each in some detail.

5.2? Ethics Shopping

In Chapter 4, we saw how a very large number of ethical principles, codes, guide-lines, and frameworks have been proposed over the past few years. There I reminded the reader that, at the end of 2020, there were more than 160 guide-lines, all containing many principles, often phrased differently, just about the -ethics of AI (Algorithm Watch 9 April 2019, Winfield 18 April 2019). This mush-rooming of principles is generati

ng inconsistency and confusion among stake-
holders regarding which one may be preferable. It also puts pressure on private
and public actors (who design, develop, or deploy digital solutions) to produce
their own declarations for fear of appearing to be left behind?-thus further con-
tributing to the redundancy of information.
In this case, the main unethical risk is that all this hyperactivity creates a ?mar-
ket of principles and values? where private and public actors may shop for the
kind of ethics that best retrofits to justify their current behaviours rather than
revise their behaviours to make them consistent with a socially accepted ethical
1? See European Commission (8 April 2019, HLEGAI (8 April 2019), published by the High--Level
Expert Group (HLEG) on AI appointed by the European Commission.
2? For example, see the debates about (a) the ?Human Rights Impact Assessment of Facebook in
Myanmar? published by the Business for Social Responsibility at https://www.bsr.org/en/our--insights/
blog--view/facebook--in--myanmar--human--rights--impact--assessment; (b) the closure of Google?s
Advanced Technology External Advisory Council at https://blog.google/technology/ai/external--
advisory--council--help--advance--responsible--development--ai/; and (c) the ethics guidelines for trust-
worthy AI published by the High--Level Expert Group of the European Commission (HLEGAI 18
December 2018, 8 April 2019).

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framework (Floridi and Lord Clement--Jones 20 March 2019). Here is a definition
of this risk:
Digital ethics shopping =def. the malpractice of choosing, adapting, or revising
(?mixing and matching?) ethical principles, guidelines, codes, frameworks, or
other similar standards (especially, but not only, in the ethics of AI) from a
var-iety of available offers in order to retrofit some pre--existing behaviours
(choices, processes, strategies, etc.), and hence justify them a posteriori, instead
of implementing or improving new behaviours by benchmarking them against
public, ethical standards.
Admittedly, I argued in Chapter 4 that much of the diversity ?in the ethics market?
is apparent and more due to wording or vocabulary rather than actual content.
However, the potential risk of ?mixing and matching? a list of ethical principles
that one prefers, as if they were ice--cream flavours, remains real. This is because
semantic looseness and redundancy enable interpretative relativism. Ethics shop-
ping then causes an incompatibility of standards where, for example, it is hard to
understand whether two companies follow the same ethical principles in devel-
oping AI solutions. Incompatibility then leads to a lower chance of competition,
evaluation, and accountability. The outcome can easily become an approach to
ethics as mere public relations, which requires no real change or effort.
The way to deal with digital ethics shopping is to establish clear, shared, and
publicly accepted ethical standards. This is why I argued (Floridi 2019b) that the
publication of the Ethics Guidelines for Trustworthy AI was a significant improve-
ment, given that it is the closest thing available in the EU to a comprehensive,
authoritative, and public standard of what may count as socially good AI.3 This is
all the more so seeing that the Guidelines explicitly influenced the proposal
adopted in 2021 by the EU Commission for regulation of AI systems,

which was
described as ?the first ever legal framework on AI? (Floridi 2021). Given the availability of this conceptual framework, the malpractice of digital ethics shopping should be at least more obvious, if not more difficult to indulge in, because anyone in the EU may simply subscribe to the Guidelines rather than shop for (or even cook) their own ?ethics?. The same holds true for the OECD?s document, which was also influenced by the Guidelines. OECD members should consider at least that framework as ethically binding.
In the context of this book, the aim of Chapter 4 is to support such convergence of views to ensure that stakeholders may reach the same agreement about foundational principles for the ethics of AI. These same principles can be found in bioethics or medical ethics, two important cases where ?ethics shopping? and the retrofitting of ethical principles to unchanged practices is unacceptable.
3? See also Mazzini (forthcoming).

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5.3? Ethics Bluewashing
In environmental ethics, greenwashing (Delmas and Burbano 2011) is the -malpractice of a private or public actor seeking to appear greener, more sustainable, or ecologically friendlier than it actually is. By ?ethics bluewashing?, I mean to refer here to the digital version of greenwashing. As there is no specific colour associated with ethically good practices in digital technologies, ?blue? may serve to remind one that we are not talking about ecological sustainability but mere digital ethics cosmetics.4 Here is a definition of this risk:
Digital ethics bluewashing =def. the malpractice of making unsubstantiated or misleading claims about, or implementing superficial measures in favour of, the ethical values and benefits of digital processes, products, services, or other solutions to appear more digitally ethical than one is.
Ethics greenwashing and bluewashing are both forms of misinformation. They are often achieved by spending a fraction of the resources that would be needed to tackle the ethical problems they pretend to address. Both malpractices (greenwashing and bluewashing) concentrate on mere marketing, advertising, or other public relations activities (e.g. sponsoring), including setting up advisory groups that may be toothless, powerless, or insufficiently critical. And both are tempting because in each case, the goals are many and all compatible:

a) to distract the receiver of the message?-usually the public, but any shareholders or stakeholders may be the target?-from anything that is going wrong, could go better, or is not happening but should happen;

b) to mask and leave unchanged any behaviour that ought to be improved;

c) to achieve economic savings; and

d) to gain some advantage, for example competitive or social, such as customer goodwill.
In contrast to what happens with greenwashing, bluewashing can be more easily combined with digital ethics shopping. This is when a private or public actor shops for the principles that best fit its current practices, publicizes them as widely as possible, and then proceeds to bluewash its technological innovation without any real improvement, with much lower costs, and with some potential social benefits. These days, ethics bluewashing is especially tempting in the context of AI,
4? This is not to be confused with the term bluewashing ?used to criticize the corporate partnerships formed under the

United Nations Global Compact initiative (some say this association with the UN helps to improve the corporations? reputations) and to disparage dubious sustainable water--use projects? (Schott 4 February 2010).

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where the ethical issues are many, the costs of doing the right thing may be high, and the normative confusion is widespread.

The best strategy against bluewashing is the one already adopted against green-washing: transparency and education. Public, accountable, and evidence--based transparency about good practices and ethical claims should be a priority on the side of the actors wishing to avoid the appearance of engaging in any -bluewashing malpractice. Public and factual education on the side of any target of bluewashing?-which could be not just the public, but also members of executive boards and advisory councils, politicians, and lawmakers, for example?-about whether and what effective ethical practices are actually implemented means that actors may be less likely to (be tempted to) distract public attention away from the ethical challenges they are facing and address them instead.

As we shall see in Chapter 11, the development of metrics for the trustworthiness of AI products and services (and of digital solutions in general) would enable the user--driven benchmarking of all marketed offerings. It would facilitate the detection of mere bluewashing, improving public understanding and engendering com-petitiveness around the development of safer, more socially and environmentally beneficial products and services. In the longer term, a system of certification for digital products and services could achieve what other similar solutions have achieved in environmental ethics: making bluewashing as visible and shameful as greenwashing.

5.4? Ethics Lobbying

Sometimes, private actors (one may at least suspect) try to use self--regulation in the ethics of AI to lobby against the introduction of legal norms; to lobby in favour of the watering down or weakening of the enforcement of legal norms; or to provide an excuse for limited compliance with legal norms. This specific malpractice affects many sectors, but it seems more likely in the digital one (Benkler 2019) where ethics may be exploited as if an alternative to legislation or in the name of technological innovation and its positive impact on economic growth (a line of reasoning less easily supported in environmental or biomedical contexts). Here is a definition:

Digital ethics lobbying =def. the malpractice of exploiting digital ethics to delay, revise, replace, or avoid good and necessary legislation (or its enforcement) about the design, development, and deployment of digital processes, products, services, or other solutions.

One may argue that digital ethics lobbying is a poor strategy likely to fail in the long run because it is short--sighted at best. Sooner or later, legislation tends to

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catch up. But regardless of whether this argument is convincing, digital ethics lobbying as a short--term tactic may still cause much damage. It could be used to delay the introduction of necessary legislation, or to help manoeuvre around or bypass more demanding interpretations of current legislation (thus making compliance easier but also misaligned with the spirit of the law). It could also infl

uence lawmakers to pass legislation that favours the lobbyist more than would otherwise be expected. Furthermore, and very importantly, the malpractice (or even the suspicion of it) risks undermining the value of any digital ethical self--regulation tout court.

This collateral damage is deeply regrettable because self--regulation is one of the main, valuable tools available for policymaking. As I shall argue in Chapter 6, self--regulation cannot replace the law by itself. But if properly implemented, it can be crucially complementary whenever

? legislation is unavailable (e.g. in experimentations about augmented reality products),

? legislation is available, but is also in need of an ethical interpretation (e.g. in terms of understanding a right to explanation in the EU?s GDPR),

? legislation is available, but is also in need of some ethical counterbalancing if

° it is better not to do something, although it is not illegal to do it (e.g. to automate entirely and fully some medical procedure without any human supervision), or

° it is better to do something, although it is not legally required (e.g. to implement better labour market conditions in the gig economy (Tan et al. 2021)).

The strategy against digital ethics lobbying is twofold. On the one hand, it must be counteracted by good legislation and effective enforcement. This is easier if the lobbying actor (private or public) is less influential on lawmakers, or whenever public opinion can exercise the right level of ethical pressure. On the other hand, digital ethics lobbying must be exposed whenever it occurs and clearly distinguished from genuine forms of self--regulation. This may happen more credibly if the process is also in itself part of a self--regulatory code of conduct for a whole industrial sector, in our case the AI industry. There would be a more general interest in maintaining a healthy context where genuine self--regulation is both socially welcome and efficacious, exposing ethics lobbying as unacceptable.

5.5? Ethics Dumping

?Ethics dumping? is an expression that was coined in 2013 by the European Commission to describe the export of unethical research practices to countries

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where there are weaker (or laxer or perhaps just different, as in the case of digital ethics) legal and ethical frameworks and enforcement mechanisms. The term applies to any kind of research, including computer science, data science, ML, robotics, and other kinds of AI. But it is most serious in health--related and biological contexts. Fortunately, biomedical and environmental ethics may be considered universal and global. There are international agreements and frameworks along with international institutions monitoring their application or enforcement. So, when research involves biomedical and ecological contexts, ?ethics dumping? can be fought more effectively and coherently. However, in digital contexts, the variety of legal regimes and ethical frameworks facilitates the export of unethical (or even illegal) practices by the ?dumper? along with the import of the outcomes of such practices. In other words, the problem is twofold: research ethics and consumption ethics. Here is a definition of this risk:

Digital ethics dumping =def. the malpractice of (a) exporting research activities about digital processes, products, services, or other solutions in oth

er contexts
or places (e.g. by European organizations outside the EU) in ways that would be ethically unacceptable in the context or place of origin; and (b) importing the outcomes of such unethical research activities.
Both (a) and (b) are important. To offer a distant but nonetheless concrete ex-ample, it is not unusual for countries to ban the cultivation of genetically modi-fied organisms yet allow their import. There is an asymmetry between the ethical (and legal) treatment of a practice (unethical and/or illegal research) and its out-put, which is the ethically (and legally) acceptable consumption of an unethical research product.
This means ethics dumping can affect digital ethics not only in terms of the unethical export of research activities but also in terms of the unethical import of the outcomes of such activities. For example, a company could export its research and then design, develop, and train algorithms (e.g. for face recognition) on local personal data in a non--EU country with a different, weaker, or unenforced ethical and legal framework for personal data protection. Under the GDPR, this would be unethical and illegal in the EU. But once trained, the algorithms could then be imported to the EU and deployed without incurring any penalty or even being frowned upon. While the first step (a) may be blocked, at least in terms of research ethics (Nordling 2018), the second step (b) involves the consumption of unethical research results. It is fuzzier, less visibly problematic, and hence more difficult to monitor and curtail.
Unfortunately, the problem of digital ethics dumping will likely become increasingly serious in the near future. This is due to the profound impact of digi-tal technologies on health and social care as well as defence, policing, and se-cur-ity; the ease of their global portability; the complexity of production processes

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(some stages of which may involve ethics dumping); and the immense economic interests at play. Especially in AI, where the EU is a net importer of solutions from the United States and China, private and public actors risk more than just exporting unethical practices. They may also (and independently) import solu-tions originally developed in ways that would not have been ethically acceptable within the EU. In this case, too, the strategy is twofold: one must concentrate on research ethics and the ethics of consumption. If one wishes to be coherent, both need to receive equal attention.
In terms of research ethics, it is slightly easier to exercise control at the source through the ethical management of public funding for research. For this, the EU is in a leading position. Still, a significant problem remains in that much research and development about digital solutions is done by the private sector, where funding may be less constrained by geographical borders. A private actor can more easily relocate its research and development to an ethically less demanding place (a geographical variation of the ethics shopping seen in Section 5.2). Privately funded research is not ethically scrutinized in the same way as publicly funded research.
In terms of consumption ethics, especially of digital products and services, much can be done. The establishment of a certification system for products and services could inform procurement along with public and private use. As in the case of blu

ewashing, the reliable and ethically acceptable provenance of digital systems and solutions will have to play a growing role in development over the years to come if one wishes to avoid the hypocrisy of being careful about research ethics in digital contexts, yet relaxed about the unethical use of its outcomes.

5.6? Ethics Shirking

Ethicists are well acquainted with the old malpractice of applying double standards in moral evaluations. By applying a lenient or strict approach, one can evaluate and treat agents (or their actions, or the consequences of their actions) differently than similar agents (actions or consequences) when in fact they should all be treated equally. The risk of double standards is usually based, even inadvertently, on bias, unfairness, or selfish interest. The risk I wish to highlight here belongs to the same family, but it has a different genesis. To highlight its specificity, I shall borrow the expression ?ethics shirking? from the financial sector.5 Here is a definition of this risk:

5? https://www.nasdaq.com/glossary/s/shirking. I owe the suggestion to include ?ethics shirking? as a significant risk in digital ethics and use the expression itself to capture it to (Cowls, Png, and Au unpublished).

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Digital ethics shirking =def. the malpractice of doing increasingly less ?ethical work? (such as fulfilling duties, respecting rights, honouring commitments, etc.) in a given context the lower the return of such ethical work in that context is (mistakenly) perceived to be.

Like ethics dumping, ethics shirking has historical roots and often follows geo-pol-it-ical outlines. Actors are more likely to engage in ethics dumping and shirking in contexts where disadvantaged populations, weaker institutions, legal uncertainties, corrupted regimes, unfair power distributions, and other economic, legal, political, or social ills prevail. It is not unusual to map, correctly, both malpractices along the divide between Global North and Global South, or to see both as affecting low- and middle--income countries above all. The colonial past still exerts a disgraceful role.

It is also important to recall that, in digital contexts, these malpractices can affect segments of a population within the Global North. The gig economy may be seen as a case of ethics shirking within developed countries. And the -development of self--driving cars may be interpreted as an instance of research dumping in some states of the United States. The 1968 Vienna Convention on Road Traffic, which establishes international principles to govern traffic laws, requires that a driver is always fully in control and responsible for the behaviour of a vehicle in traffic. But the United States is not a signatory country to the convention, so the requirement does not apply. This means that state -vehicle codes do not prohibit automated vehicles, and several states have enacted laws for automated vehicles. This is also why research into self--driving cars happens mostly in the United States, along with related incidents and human suffering.

The strategy against ethics shirking consists in tackling its origin, which is the absence of a clear allocation of responsibility. Agents may be more tempted to shirk their ethical work in a given context the more they (think they) can relocate responsibilities elsewhere. This happens more likely and

easily in
?D contexts?, where one?s own responsibility may be (mistakenly) perceived as lower because it is distant, diminished, delegated, or distributed (Floridi 2012b). Thus, ethics shirking is the unethical cost of deresponsabilization of some agency. It is this genesis that makes it a special case of the ethical problem of double standards. Therefore more fairness and less bias are both necessary?-insofar as ethics shirking is a special case of the problem of double standards?-and insufficient to remove the incentive to engage in ethics shirking. To uproot this malpractice, one also needs an ethics of distributed responsibility (Floridi 2016a). Such ethics would relocate responsibilities, along with all the praise and blame, reward and punishment, and ultimately causal accountability and legal liability, to where they rightly belong.

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5.7? Conclusion: The Importance of Knowing Better

In this chapter I have provided a conceptual map for those who wish to avoid or minimize some of the most obvious and significant ethical risks when translating principles into practices in digital ethics. From a Socratic perspective, according to which human evil is an epistemic problem, a malpractice is often the result of a misjudged solution or a mistaken opportunity. It is important to understand as early as possible that shortcuts, postponements, and quick fixes lead not to better ethical solutions, but to more serious problems. Problems become increasingly difficult and costlier to solve the later one deals with them.

While such an understanding does not guarantee that the five malpractices analysed in this chapter will disappear, it does mean that they will be reduced insofar as they are genuinely based on misunderstanding and misjudgements. Not knowing better is the source of a lot of evil. So, the solution is often more and better information for all. In the previous pages, I mentioned that self--regulation can robustly complement a legislative approach. This is very much the case when it comes to ethical practices that presuppose but go beyond legal compliance, which is the topic of the next chapter.

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6
Soft Ethics and the Governance of AI

6.0? Summary

Previously, in Chapters 4 and 5, I suggested a unified framework for the ethical principles of AI and identified some main ethical risks that arise when translating principles into practices. This chapter discusses the governance of AI, and more generally of digital technologies, as the new challenge posed by technological innovation. I introduce a new distinction between soft and hard ethics. Hard -ethics first precede and then further contribute to shaping legislation. In contrast, soft ethics apply after legal compliance with legislation (that is, post--compliance ethics), such as the GDPR in the EU. The chapter concludes by developing an analysis of the role of digital ethics with respect to digital regulation and digital governance, thus preparing for the next chapter on the fundamental ethical principles for an ethics of AI.

6.1? Introduction: From Digital Innovation to the Governance of the Digital

In any mature information society today (Floridi 2016b), we no longer live on

line
or offline. Instead, we live onlife (Floridi 2014b). We increasingly live in that special space, or infosphere, that is seamlessly analogue and digital, offline and online. If this appears confusing, perhaps an analogy may help to convey the point. Imagine someone asking whether the water is fresh or salty in the estuary where the river meets the sea. Clearly, that someone has not understood the special nature of the place. Like mangroves flourishing in brackish water, our mature information societies are growing in such a new, liminal place. And in these ?mangrove societies?, machine--readable data, new forms of smart agency, and onlife interactions are constantly evolving. This is because our technologies are perfectly fit to take advantage of such a new environment, often as the only real natives. As a result, the pace of their evolution can be mind--blowing. And this justifies some apprehension, in turn.
However, we should not be distracted by the scope, depth, and pace of digital innovation. True, it does significantly disrupt some deeply ingrained assumptions of the old society that was exclusively analogue. There were assumptions about competition, customization, education, entertainment, health, logistics, politics,

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production, security, and work (to mention just some crucial topics in a merely alphabetic order). Yet that is not the most consequential challenge we are facing. It is not digital innovation that matters most but the governance of the digital, that is, what we do with it. More precisely, it is how well we design the infosphere and the mature information societies developing within it that matters most. Because the digital revolution transforms our views about values and their -pri-or-ities, good behaviour, and the sort of innovation that is not only sustainable but also socially preferable (equitable), governing all of this has now become the fundamental issue. Let me explain.
To many, the real challenge may seem to be what digital innovation will throw up next. The question itself is recurrent and trite: what is the next disruption? What is the new killer app? Will this be the year of the final battle between Virtual Reality vs. Augmented Reality? Or is it the Internet of Things that will represent the new frontier, perhaps in some combination with smart cities? Is an end to the television as we know it coming soon? Will healthcare be made unrecognizable by ML, or should our attention be focused on the automation of logistics and transport? What will the new smart home assistants do aside from telling us what the weather is like and allowing us to choose the next song? How is military strategy going to adapt to cyber conflicts? What industry will be ?uberized? next? Behind similar questions lies the unspoken assumption that digital innovation leads, and everything else lags behind or at best follows, whether business -models, working conditions, standards of living, legislation, social norms, habits, expectations, or even hopes.
Yet this is precisely the distracting narrative that we should resist, not because it is wrong, but because it is only superficially right. The deeper truth is that the digital revolution has already occurred. The transition from an entirely analogue and offline world to one that is increasingly digital and online will never happen again in the history of humanity. Perhaps one

day, a quantum computing gadget
running AI apps may be in the pocket of your average teenager. But our gen-er-ation is the last to have seen a non--digital world and this is the really extraordinary turning point, because that landing on the infosphere and the beginning of onlife happen only once. The question of what this new world will be like is both fascinating in terms of opportunities and worrisome in terms of risks.
No matter how challenging, our ?exploration? of the infosphere (to indulge in the geographical metaphor a bit longer) prompts a much more fundamental question that is socio--political and truly crucial: what kind of mature information societies do we want to build? What is our human project for the digital age? Looking at our present backwards?-that is, from a future perspective?-this is the time in history when we will be seen as having laid down the foundation for our mature information societies. We shall be judged by the quality of our work. So clearly, the real challenge is no longer good digital innovation. It is the good governance of the digital, as I anticipated above.

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Proof that this is the case is all around us in the mushrooming of initiatives to address the impact of the digital on everyday life and how to regulate it. It is also implicit in the current narrative about the unstoppable and unreachable nature of digital innovation, if one looks just a bit more closely. Because in the same context where people complain about the speed of digital innovation and the impossible task of chasing it with some normative framework, one also finds equal certainty about the seriousness of the risk posed by the wrong legislation. Bad legislation could kill digital innovation entirely or destroy whole technological sectors and developments. You do not have to be Nietzsche??Was mich nicht umbringt macht mich stärker? (Nietzsche 2008) or ?what does not kill me makes me stronger??to realize the inference here is that updating the rules of the game is perfectly possible. After all, everybody acknowledges that it can have immense consequences. Reacting to technological innovation is not the best approach.
So, we need to shift from chasing to leading. If we then like the direction in which we move or where we are going, then the speed at which we are moving or getting there can actually be something very positive. The more we like our des-tin-ation, the faster we will want to get there. It is because we lack a clear sense of socio--political direction that we are worried by the speed of our technological travel. We should be. Yet the solution is not to slow down, but to decide together where we want to go. For this to happen, society needs to stop playing defence and start playing attack. The question is not whether, but how. And to start addressing the how, some clarifications are helpful. That is the contribution made by this chapter.
6.2? Ethics, Regulation, and Governance
On the governance of digital technologies in general and AI in particular (I will not draw this distinction in the rest of the chapter, where I shall speak of ?the digi-tal? to cover both) there is much to be said and even more still to be understood and theorized. Still, one point is clear:

i) the governance of the digital, henceforth digital governance,

ii) the ethics of the digital, henceforth digital ethics, which are also known as computer

r example good conduct or good values. Digital ethics shapes
digital regulation and digital governance by means of a moral evaluation of what
is socially acceptable or preferable.
Digital governance in Figure 6.1 is just one of the three normative forces that
can shape and guide the development of the digital. But it is not uncommon to
use that part for the whole and to speak of digital governance as referring to the
entire set. This use of the term ?governance? is a synecdoche, a bit like using ?coke?
to refer to any variety of cola. It is what I did at the beginning of this chapter,
when I stated that the real challenge today is the governance of the digital. By
that, I meant to refer not just to digital governance but also to digital ethics and
digital regulation, that is to the whole normative map: E + R + G. And this is also
how I interpret the report ?Data Management and Use: Governance in the 21st
Century? that we published in 2017 as a joint British Academy and Royal Society
working group (British Academy 2017; disclosure: I was a member). As long as
the synecdoche is clear, there is no problem.
Once the map is understood, two important consequences become clear. Let
me discuss each in separate sections.
6.3? Compliance: Necessary but Insufficient
When policymakers in both political and business contexts wonder why we
should engage in ethical evaluations when legal compliance is already available
(this is a recurring topic in the discussion of the GDPR, for example), the answer
should be clear: compliance is necessary but insufficient to steer society in the
right direction. Digital regulation indicates what the legal and illegal moves in the
game are, so to speak. But it says nothing about what the good and best moves
could be, out of those that are legal, to win the game?-that is, to have a better
society. This is the task of digital ethics on the side of moral values and prefer-
ences, and of good digital governance on the side of management. And this is
why, for example, the EDPS (the EU?s independent data protection authority)
established the Ethics Advisory Group in 2015: to analyse the new ethical challenges
posed by digital developments and current legislation, especially in relation to the
GDPR. The report (EDPS Ethics Advisory Group 2018; disclosure: I was a member)
should be read as a contribution to normative governance of the infosphere in the

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EU, and a stepping stone towards its implementation. So, what kind of digital ethics
should we adopt to complement legal compliance?
6.4? Hard and Soft Ethics
If we look at Figure 6.1, digital ethics may now be understood in two ways: as
hard or soft ethics. The distinction is loosely based on the one between hard and
soft law (Shaffer and Pollack 2009), that is, between traditional law and other
instruments that are quasi--legal but have no legally binding force, such as the
resolutions and recommendations of the Council of Europe. The distinction is
not really a matter of practice as in reality soft and hard ethics often come in-ex-
tric-ably intertwined. It is above all a matter of theory, because it is logically possible
and often useful to distinguish soft and hard ethics and then discuss each separately.
Let us see the distinction in detail.
Hard ethics (see A?+?B?+?C in Figure 6.1) is what we usually have in mind when
discussing values, rights, duties, and responsibil

ities?-or, more broadly, what is
morally right or wrong, and what ought or ought not to be done?-in the course of formulating new regulations and challenging existing ones. In short, insofar (and it may not be very far) as ethics contributes to making, shaping, or changing the law, we can call that hard ethics. For example, lobbying in favour of some good legislation or to improve that which already exists can be a case of hard ethics. Hard ethics helped to dismantle apartheid legislation in South Africa and -supported the approval of legislation in Iceland requiring public and private businesses to prove they offer equal pay to employees irrespective of their gender (by the way, the gender pay gap continues to be scandalous in most countries). It follows that, in hard ethics, it is not true that ?one ought to do x? legally speaking (where x ranges on the universe of feasible actions) implies ?one may do x? eth-ic-al-ly speaking. It is perfectly reasonable to object that ?one ought to do x? may be followed by ?even if one may not do x?. Call this the Rosa Parks Principle for her famous refusal to obey the law and give up her bus seat in the ?colored section? (American spelling intended) to a white passenger after the whites--only section was filled.
Soft ethics covers the same normative ground as hard ethics (again, see A?+?B?+?C in Figure 6.1). But it does so by considering what ought and ought not to be done over and above the existing regulation?-not against it, or despite its scope, or to change it. Thus, soft ethics may include self--regulation (see Chapter 5). In other words, soft ethics is post--compliance ethics because, in this case, ?ought implies may?. This is why, in Figure 6.1, I wrote that regulations constrain software ethics through compliance. Call this the Matthew Principle: ?Render to Caesar the things that are Caesar?s? (Matthew 22: 15?22).

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As already indicated above, both hard and soft ethics presuppose feasibility or, in more Kantian terms, assume the fundamental principle that ?ought implies can?, given that an agent has a moral obligation to perform action x only if x is possible in the first place. Ethics should not be supererogatory in this specific sense of asking for something impossible. It follows that soft ethics assumes a post--feasibility approach as well. Any ethical approach, at least in the EU, accepts the implementation of the Universal Declaration of Human Rights, the European Convention on Human Rights, and the Charter of Fundamental Rights of the European Union as its minimal starting point. The result is that the space for soft ethics is both partially bounded but also unlimited. To see why, it is easy to visualize it in the shape of a trapezoid (Figure 6.2). The lower side of the trapezoid represents a feasi-bil-ity base that is ever--expanding through time, as we can do more and more things thanks to technological innovation. The two constraining sides, left and right, represent legal compliance and human rights. The open upper side represents the space where what is morally good may happen in general and, in the context of this chapter, may happen in terms of shaping and guiding the ethical development of our mature information societies.
I have already mentioned that hard and soft ethics often go hand in hand. Their distinction is useful, but often logical rather than factual. In the n

ext section, I
shall analyse their mutual relation and interaction with legislation by relying on the specific case of the GDPR. In this section, a final clarification is in order. When distinguishable, soft digital ethics can be more easily exercised the more digital regulation is considered to be on the good side of the moral vs. immoral divide. Thus, it would be a mistake to argue for a soft ethics approach to establish a normative framework when agents (especially governments and companies) are operating in contexts where human rights are disregarded, for example in China, North Korea, or Russia. At the same time, hard ethics may still be necessary in other contexts where human rights are respected, in order to change some current legislation that is perceived to be ethically unacceptable. The Irish abortion referendum in 2018 is a good example. In a digital context, hard ethics arguments
TIME
Feasibility
Human Rights
Compliance
Te space
of sof ethics
Figure 6.2? The space for soft ethics

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were used as a point of contrast for the December 2017 decision by the US Federal Communications Commission (FCC) to rescind the rule about net neutrality (the principle according to which all Internet traffic should be treated in the same way, without blocking, degrading, or prioritizing any particular legal content). The outcome was that in March 2018, Washington became the first state in the United States to pass legislation mandating net neutrality.
Within the EU, soft ethics may rightly be exercised to help agents (such as individuals, groups, companies, governments, and organizations) take more and better advantage, morally speaking, of the opportunities offered by digital innovation. Because even in the EU, legislation is necessary but insufficient. It does not cover everything, nor should it. Agents should leverage digital ethics to assess and decide what role they wish to play in the infosphere when regulations provide no simple or straightforward answer (or no guidance at all), when competing values and interests need to be balanced, and when there is more that can be done over and above what the law strictly requires. In particular, good use of soft ethics could lead companies to exercise ?good corporate citizenship? within a mature information society.
The time has come to provide a more specific analysis and for this, I shall rely on the GDPR. The choice seems reasonable: given that digital regulation in the EU is now determined by the GDPR and that EU legislation is normally respectful of human rights, it may be useful to understand the value of the distinction between soft and hard ethics and their relations to legislation by using the GDPR as a concrete case of application. The underlining hypothesis is that, if the soft/hard ethics analysis does not work in the case of the GDPR, it probably won?t work anywhere else.
6.5? Soft Ethics as an Ethical Framework
To understand the role of hard and soft ethics with regard to law in general and the GDPR in particular, five components need to be introduced (see Figure 6.3).3 First, there are the ELSI of the GDPR, for example on organizations. This is the impact of the GDPR on business, for example. Then there is the GDPR itself. This is the legislation that replaced the Data Protection Directive 95/46/EC. It is designed to harmonize data privacy laws across Europe, to protect and enfo

rce
the data privacy of all EU citizens independently of geographic location, and to improve the way that organizations across the EU approach data privacy. The GDPR comprises ninety--nine articles that make up the second element. As it is

3? In a previous version of this chapter, the text read as if I argued that ethics shapes and interprets the law. This is simply untenable, and I am grateful to one of the anonymous referees for highlighting this potentially erroneous reading.

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often the case with complex legislation, the articles do not cover everything. They leave grey areas of normative uncertainty even about the topics they do cover. They are subject to interpretation. They may require updating when applied to new circumstances, especially in a technological context where innovation develops so quickly and radically?-think of facial recognition software, for example, or so--called ?deepfake? software. So, to help understand their meaning, scope, and applicability, the articles are accompanied by 173 recitals. This is the third elem-ent. In EU law, recitals are texts that explain the reasons for the provisions of an act. They are not legally binding, and they are not supposed to contain normative language. Normally, recitals are used by the Court of Justice of the European Union to interpret a directive or regulation and reach a decision in the context of a particular case.4 But in the case of the GDRP, it is important to note that the recitals can also be used by the European Data Protection Board (the EDPB, which replaced the Article 29 Working Party) when ensuring that the GDPR is applied consistently across Europe.

The recitals themselves require interpretation, and this is the fourth element. Part of this interpretation is provided by an ethical framework that contributes, together with other factors, to understanding the recitals. Finally, the articles and the recitals were formulated thanks to a long process of negotiation between the European Parliament, the European Council, and the European Commission (the so--called Formal Trilogue meeting), resulting in a joint proposal. This is the fifth element, namely, the perspective that informed the elaboration of the GDPR. It is where hard ethics plays a role together with other factors (e.g. political, economic, etc.). It may be seen in action by comparing drafts from the European Parliament and European Commission along with amendments to the European Commission?s text proposed by the European Council.5

So here is a summary of what we need to consider:

1) the ethical, legal, and social implications and opportunities (ELSIO) -generated by the articles in (2). The distinction between implications and opportunities is meant to cover both what follows from the GDPR (Chameau et al. 2014) and what is left partially or completely uncovered by the GDPR. The reader who finds the distinction redundant (one may argue that opportunities are just a subset of the implications) should feel free to drop the ?O? in ?ELSIO?. The reader who finds the distinction confusing may

4? For example, see ?C--131/12 Google Spain SL, Google Inc. v Agencia Española de Protección de Datos, Mario Costeja González? at http://curia.europa.eu/juris/document/document.jsf?text=&docid=152065&pageIndex=0&doclang=EN&mode=req&. Or see Domestic CCTV and Directive 95/46/EC (European Court of Justice (ECJ) Judgment

ch more than any other
force. This is because the evaluation of what is morally good, right, or necessary shapes public opinion (and hence the socially acceptable or preferable) as well as the politically feasible. Ultimately, it shapes the legally enforceable and what agents may or may not do. In the long run, people are constrained in what they can or cannot do as users, consumers, citizens, and patients, etc. The feasibility of what they do is constrained by the goods and services provided by organizations, such as businesses, that are constrained by law in terms of compliance. But the latter is also shaped and constrained by (although not only by) ethics, which is where people decide in what kind of society they want to live (see Figure 6.7). Unfortunately, such a normative cascade becomes obvious mainly when backlash happens, that is, mostly in negative contexts. This is when the public rejects some solutions even when they may be good solutions. A normative cascade should instead be used constructively, that is, to pursue the construction of a mature information society of which we can be proud.

6.8? Digital Ethics? Dual Advantage

It is obvious that digital technologies, and especially AI, offer many opportunities. They also bring also associated challenges and potential risks. So, it is equally obvious that ensuring socially preferable outcomes means resolving the tension between incorporating the benefits and mitigating the potential harms?-in short, promoting these technologies while avoiding their misuse, underuse, overuse, and harmful use. This is where the value of an ethical approach becomes obvious as well. I argued above that compliance is necessary, but significantly insufficient. Adopting an ethical approach to digital innovation confers what may be defined

Ethics
Law
Business
Customers

Figure 6.7? Example of a normative cascade

Note: the example uses business for an agent and customers for people; business could be replaced by government and customers by citizens

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as a ?dual advantage?, echoing the ?dual use? terminology popular in the philosophy of technology at least since the debate on civil and military uses of nuclear power. On the one hand, soft ethics can provide an opportunity strategy, enabling actors to take advantage of the social value of digital technologies. This is the advantage of being able to identify and leverage new opportunities that are socially acceptable or preferable. Precautionary principles can be balanced with the duty to not omit what could and ought to be done, for example to take advantage of the wealth of accumulated data or, in the context of this book, of the forms of smart agency available. On the other hand, ethics also provides a risk management solution. It enables organizations to anticipate and avoid costly mistakes (the Cambridge Analytica scandal involving Facebook is by now a classic example). This is the advantage of preventing and mitigating courses of action that turn out to be socially unacceptable and hence rejected. In this way, ethics can also lower the opportunity costs of choices not made or options not seized for fear of mistakes.

The dual advantage of soft ethics can only function in a broader environment of decent legislation, public trust, and clear responsibilities. Public acceptance and adoption of digital technologies, including AI, will occur only if the be

nefits
are seen as meaningful and fairly distributed, and if the risks are seen as potential yet preventable, minimizable, or at least something against which one can be protected. The risks must also be seen by the public as subject to risk management (e.g. insurance) and redress?-not unfairly affecting discriminated groups of -people. Reassuringly, this is the approach taken by the EU with its legislation (European Commission 2021). These attitudes will depend in turn on public engagement with the development of AI and more generally all digital technologies; openness about how they operate; and understandable, widely accessible mechanisms of regulation and redress. In this way, an ethical approach to AI can also function as an early warning system against risks that might endanger entire organizations. The clear value to any organization of the dual advantage of an ethical approach to AI amply justifies the expense of engagement, openness, and contestability that such an approach requires.

6.9? Conclusion: Ethics as a Strategy

Ethics in general, and digital ethics in particular, should not be a mere add--on, an afterthought, a latecomer, or an owl of Minerva (to use Hegel?s metaphor about philosophy) that takes flight only when the shades of night are gathering. It should not be used to intervene only after digital innovation has taken place, pos-sibly bad solutions have been implemented, less good alternatives have been -chosen, or mistakes have been made. This is not least because some mistakes are irreversible, some missed opportunities are irrecoverable, and any preventable

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evil that occurs is a moral disaster. Nor should ethics be a mere exercise in questioning. The building of critical awareness is important, but it is also only one of the four tasks of a proper ethical approach to the design and governance of the digital. The other three are signalling that ethical problems matter, engaging with stakeholders affected by such ethical problems, and, above all, designing and implementing sharable solutions. Any ethical exercise that in the end fails to provide and implement some acceptable recommendations is only a timid preamble.

So, ethics must inform strategies for the development and use of digital technologies from the very beginning, when changing the course of action is easier and less costly in terms of resources, impact, and missed opportunities. It must sit at the table of policymaking and decision--taking procedures from day one. We must not only think twice; most importantly, we must think before taking im-port-ant steps. This is particularly relevant to the EU, where I have argued that soft ethics can be properly exercised and where a soft--ethical approach to SETI (science, engineering, technology, and innovation) development is already ac-know-ledged as crucial (Floridi et al. 2018).

If soft digital ethics can be a priority anywhere, this is certainly in Europe. We should adopt it as soon as possible. To do so, a shared starting point is essential. Ethics seems to have a bad name when it comes to reaching agreement on fun-damental issues. This is justified, but it is also exaggerated. Often, and especially in debates about technology, ethical disagreement is not about what is right or wrong. Instead, disagreement is about how far it may or may not be so, about why it is, and about how to ensure that what is ri

ght prevails over what is wrong. We
saw in Chapter 4 that this is particularly true when it comes to foundational
analysis of basic principles, where agreement is much wider and common than it
is sometimes perceived. The time has come to map some of the specific ethical
problems raised by algorithms. This is the topic of the next chapter.

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7
Mapping the Ethics of Algorithms

7.0? Summary

Previously, in Chapters 4?6, I analysed the ethical principles, risks, and governance
of AI. This chapter reviews the actual ethical challenges posed by AI through a
focus on the debate about the ethics of algorithms. I shall say more about robotics
in Chapter 8. Research on the ethics of algorithms has grown substantially in
recent years.1 Given the exponential development and application of Machine
Learning (ML) algorithms over the past decade in particular, new ethical
problems and solutions relating to their ubiquitous use in society have emerged.
This chapter analyses the state of the debate with the goals of contributing to the
identification and analysis of the ethical implications of algorithms; providing an
updated analysis of epistemic and normative concerns; and offering actionable
guidance for governance of the design, development, and deployment of algorithms.

7.1? Introduction: A Working Definition of Algorithm

Let me start with a conceptual clarification. As in the case of ?artificial intelligence?,
there is little agreement in the relevant literature on the definition of an algorithm.
The term is often used to indicate the formal definition of an algorithm as a
mathematical construct with ?a finite, abstract, effective, compound control
structure, imperatively given, accomplishing a given purpose under given
-provisions? (Hill 2016, 47). But it is also used to indicate domain--specific
understandings that focus on the implementation of these mathematical
-constructs into a technology configured for a specific task. In this chapter, I focus
on the ethical issues posed by algorithms as mathematical constructs, by their
implementation as programs and configurations (applications), and by the ways
in which these issues can be addressed.2 So understood, algorithms have become
a key element underpinning crucial services and infrastructures of information
societies. For example, in-di-vid-uals interact with recommender systems, that is,
algorithmic systems that make suggestions about what a user may like, on a daily
basis, be it to choose a song, movie, product, or even a friend (Paraschakis 2017,

1? See Floridi and Sanders 2004 and Floridi 2013 for earlier references.
2? This was the approach adopted in Mittelstadt et al. 2016 and Tsamados et al. 2021.

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Perrault et al. 2019, Milano, Taddeo, and Floridi 2020b, 2021, 2019, 2020a). At the
same time, schools and hospitals (Obermeyer et al. 2019, Zhou et al. 2019, Morley,
Machado, et al. 2020), financial institutions (Lee and Floridi 2020, Lee, Floridi,
and Denev 2020, Aggarwal 2020), courts (Green and Chen 2019, Yu and Du
2019), local governmental bodies (Eubanks 2017, Lewis 2019), and national
governments (Labati et al. 2016), Hauer 2019, Taddeo and Floridi 2018b, Taddeo,
McCutcheon, and Flor

idi 2019) all increasingly rely on algorithms to make
significant decisions.
The potential for algorithms to improve individual and social welfare comes
with significant ethical risks. It is well known that algorithms are not ethically
neutral. Consider, for example, how the outputs of translation and search engine
algorithms are largely perceived as objective, yet frequently encode language in
gendered ways (Larson 2017, Prates, Avelar, and Lamb 2019). Bias has also been
reported extensively, such as in algorithmic advertising with opportunities for
higher--paying jobs and jobs within the field of science and technology advertised
to men more often than to women (Datta, Tschantz, and Datta 2015, Datta, Sen,
and Zick 2016, Lambrecht and Tucker 2019). Likewise, prediction algorithms
used to manage the health data of millions of patients in the United States exacerbate
existing problems, with white patients given measurably better care than com-
parably similar black patients (Obermeyer et al. 2019). While solutions to these
and similar issues are being discussed and designed, the number of algorithmic
systems exhibiting ethical problems continues to grow.
We saw in the first part of the book that AI has been experiencing a new ?summer?
at least since 2012. This is both in terms of the technical advances being made and
the attention that the field has received from academics, policymakers, technologists,
investors (Perrault et al. 2019), and hence the general public. Following this new
?season? of successes, there has been a growing body of research on the ethical
implications of algorithms particularly in relation to fairness, accountability, and
transparency (Lee 2018, Hoffmann et al. 2018, Shin and Park 2019). In 2016, our
research group at the Digital Ethics Lab published a comprehensive study that
sought to map these ethical concerns (Mittelstadt et al. 2016). However, this is a
fast--changing field. Both novel ethical problems and ways to address them have
emerged, making it necessary to improve and update that study. Work on the ethics
of algorithms has increased significantly since 2016, when national governments,
non--governmental organizations, and private companies started to take a prominent
role in the conversation on ?fair? and ?eth-ic-al? AI and algorithms (Sandvig et al.
2016, Binns 2018b, 2018a, Selbst et al. 2019, Wong 2019, Ochigame 2019). Both
the quantity and the quality of the research available on the topic have expanded
enormously. The COVID--19 pandemic has both exacerbated the issues and spread
global awareness about them (Morley, Cowls, et al. 2020). Given these changes,
we published a new study (Tsamados et al. 2021; see also Morley et al. 2021). This
study improved on Mittelstadt et al. 2016 by adding new insights into the ethics of

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algorithms, updating the initial analysis, including references to the literature that
were missed in the original review, and extending the analysed topics (such as by
including work on AI4SG). At the same time, the 2016 conceptual map (see
Figure 7.1) remains a fruitful framework for reviewing the current debate on the
ethics of algorithms, identifying the ethical problems that algorithms give rise to,
and surveying the solutions that have been proposed in recent relevant literature.
So, I shall start from it in the next section and this chapter is based on those two
works.

ether, even if with different
priorities and focus.? (Floridi and Taddeo 2016, 4)
In the remainder of this chapter, I address each of these six ethical concerns in turn, offering an updated analysis of the ethics of algorithms literature (at a micro level) with the goal of contributing to the debate on digital ethics (at a macro level).
3? On the debate and philosophy of data quality see Floridi and Illari 2014.

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7.3? Inconclusive Evidence Leading to Unjustified Actions
Research focusing on inconclusive evidence refers to the way in which non--deterministic ML algorithms produce outputs that are expressed in probabilistic terms (James et al. 2013, Valiant 1984). These types of algorithms generally identify association and correlation between variables in the underlying data, but not causal connections. As such, they may encourage the practice of apophenia: ?seeing patterns where none actually exist, simply because massive quantities of data can offer connections that radiate in all directions? (boyd and Crawford 2012, 668).
This is problematic because patterns identified by algorithms may be the result of inherent properties of the system modelled by the data, of the datasets (that is, of the model itself rather than the underlying system), or of the skilful manipulation of datasets (properties neither of the model nor of the system). This is the case, for example, of Simpson?s paradox, which occurs when trends that are observed in different groups of data reverse when the data are aggregated (Blyth 1972). In the last two cases, poor quality of the data leads to inconclusive evidence to support human decisions.
Recent research has underlined the concern that inconclusive evidence can give rise to serious ethical risks. Focusing on non--causal indicators, for instance, could distract attention from the underlying causes for a given problem (Floridi et al. 2020). Even with the use of causal methods, the available data may not always contain enough information to justify an action or make a decision fair (Olhede and Wolfe 2018b, 2018a). Data quality, such as the timeliness, completeness, and correctness of a dataset, constrains the questions that can be answered using a given dataset (Olteanu et al. 2016). Additionally, the insights that can be extracted from datasets are fundamentally dependent on the assumptions that guided the data collection process itself (Diakopoulos and Koliska 2017). For example, algorithms designed to predict patient outcomes in clinical settings rely entirely on data inputs that can be quantified (e.g. vital signs and previous success rates of comparative treatments). This means they ignore other emotional facts (e.g. the willingness to live) that can have a significant impact on patient outcomes, thus undermining the accuracy of the algorithmic prediction (Buhmann, Paßmann, and Fieseler 2019). This example highlights how insights stemming from algorithmic data processing can be uncertain, incomplete, and time--sensitive (Diakopoulos and Koliska 2017).
One may embrace a naïve, inductivist approach and assume that inconclusive evidence can be avoided if algorithms are fed enough data, even if a causal ex-plan-ation for these results cannot be established. Yet recent research rejects this view. In particular, literature focusing on the ethical risks of racial profiling using algorithmic systems has demonstra

a so poorly structured that they are impossible to read; and

? ongoing updates and human influence over a model.

Lack of transparency is also an inherent characteristic of self--learning algorithms, which alter their decision--making logic (for producing new sets of rules) during the learning process. This makes it difficult for developers to maintain a detailed understanding of why some specific changes were made (Burrell 2016, Buhmann, Paßmann, and Fieseler 2019). However, it does not necessarily translate into opaque outcomes; even without understanding each logical step, developers can adjust hyperparameters (the parameters that govern the training process) to test for various outputs. In this respect, Martin (2019) stresses that, although the difficulty of explaining the outputs of ML algorithms is certainly real, it is important to not let this difficulty create an incentive for organizations to develop complex systems in order to shirk their responsibility.

Lack of transparency can also result from the malleability of algorithms. Algorithms can be reprogrammed in a continuous, distributed, and dynamic way (Sandvig et al. 2016), which makes them malleable. Algorithmic malleability allows developers to monitor and improve an already deployed algorithm, but it may also be abused to blur the history of its evolution and leave end users uncertain about the affordances of a given algorithm (Ananny and Crawford 2018). For example, consider Google?s main search algorithm. Its malleability enables the company to make continuous revisions, suggesting a permanent state of de-sta-bil-iza-tion (Sandvig et al. 2016). This requires those affected by the algorithm to monitor it constantly and update their understanding accordingly, which is an impossible task for most (Ananny and Crawford 2018).

As Turilli and I have remarked in the past, transparency is not an ethical principle in itself but a pro--ethical condition for enabling or impairing other ethical practices or principles (Turilli and Floridi 2009, 105). In other words, transparency is not an intrinsically ethical value but a valuable means to ethical ends. It is part of the conditions of possibility for ethical behaviour, which I have defined elsewhere as ?infraethics? (infrastructural ethics, Floridi 2017a). Sometimes, opacity can be more helpful. It could support the secrecy of citizens? political preferences and votes, for example, or competition in auctions for public services. Even in algorithmic contexts, complete transparency can itself cause distinct ethical

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problems (Ananny and Crawford 2018). It can provide users with some critical information about the features and limitations of an algorithm, but it can also overwhelm users with information and thus render the algorithm opaquer (Kizilcec 2016, Ananny and Crawford 2018).

Other researchers stress that excessive focus on transparency can be detrimental to innovation and unnecessarily divert resources that could instead be used to improve safety, performance, and accuracy (Danks and London 2017, Oswald 2018, Ananny and Crawford 2018, Weller 2019). The debate over prioritizing transparency (and explainability) is especially contentious in the context of medic-al algorithms (Robbins 2019). Transparency can also enable individuals to game the system (Martin 2019, Magalhães 2018, Cowls et al. 2019). Knowledge about the

source of a dataset, the assumptions under which sampling was done, or the metrics that an algorithm uses to sort new inputs may be used to figure out ways to take advantage of an algorithm (Szegedy et al. 2014, Yampolskiy 2018). Yet the ability to game algorithms is only within reach for some groups of the population (those with higher digital skills and the necessary resources to use them, in particular), which creates another form of social inequality (Martin 2019, Bambauer and Zarsky 2018). Confusing transparency for an end in itself?-as opposed to a pro--ethical factor (Floridi 2017) that needs to be properly designed to enable crucial ethical practices?-may not solve existing ethical problems related to the use of algorithms. Indeed, the confusion may pose new ones. This is why it is im-portant to distinguish between the different elements that may hinder the transparency of algorithms, identify their cause, and nuance the call for transparency by specifying the needed factors along with the layers of algorithmic systems at which they should be addressed (Diakopoulos and Koliska 2017).

There are different ways to address problems related to lack of transparency. For example, Gebru et al. (2020) propose addressing the constraints on transparency posed by the malleability of algorithms partly by using standard documentary procedures like those deployed in the electronics industry. In these procedures, ?every component, no matter how simple or complex, is accompanied with a datasheet describing its operating characteristics, test results, recommended usage, and other information? (Gebru et al. 2020, 2). Unfortunately, publicly available documentation is currently uncommon in the development of algorithmic systems and there is no agreed--upon format for what should be included when documenting the origin of a dataset (Arnold et al. 2019, Gebru et al. 2020).

Although relatively nascent, another potentially promising approach to en-for-cing algorithmic transparency is the use of technical tools to test and audit algorithmic systems and decision--making (Mokander and Floridi 2021). Both -testing whether algorithms exhibit negative tendencies (such as unfair discrimination) and auditing a prediction or decision trail in detail can help maintain a high level of transparency (Weller 2019, Malhotra, Kotwal, and Dalal 2018,

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Brundage et al. 2020). To this end, discursive frameworks have been developed to help businesses and public sector organizations understand the potential impacts of opaque algorithms, thus encouraging good practices (ICO 2020). For instance, the AI Now Institute at New York University has produced algorithmic impact assessment guidance that seeks to raise awareness and improve dialogue over the potential harms of ML algorithms (Reisman et al. 2018). Awareness and dialogue aim to enable developers to design more transparent, and therefore more trustworthy, ML algorithms. It also intends to improve the public understanding and control of algorithms. In the same vein, Diakopoulos and Koliska (2017) have provided a comprehensive list of ?transparency factors? across four layers of algorithmic systems: data, model, inference, and interface. Factors include, inter alia, ?uncertainty (e.g., error margins), timeliness (e.g., when was the data collected), completeness or missing elements, sampling method, provenan

ce (e.g., sources), and
volume (e.g., of training data used in ML)? (Diakopoulos and Koliska 2017, 818). Effective transparency procedures are likely to, and indeed ought to, involve an interpretable explanation of the internal processes of these systems. Buhmann, Paßmann, and Fieseler (2019) argue that while a lack of transparency is an inherent feature of many ML algorithms, this does not mean that improvements cannot be made. Companies like Google and IBM have increased their efforts to make ML algorithms more interpretable and inclusive by making tools, such as Explainable AI, AI Explainability 360, and the What--If Tool, publicly available. More strategies and tools will certainly be developed. They provide both developers and the general public with interactive visual interfaces that improve human readability, explore various model results, provide case--based reasoning and directly interpretable rules, and even identify and mitigate unwanted biases in datasets and algorithmic models (Mojsilovic 2018, Wexler 2018). However, ex-plan-ations for ML algorithms are constrained by the type of explanation sought, the fact that decisions are often multi--dimensional in their nature, and that different users may require different explanations (Edwards and Veale 2017, Watson et al. 2019). Identifying appropriate methods for providing explanations has been a problem since the late 1990s (Tickle et al. 1998), but contemporary efforts can be categorized into two main approaches: subject--centric explanations and model--centric explanations (Doshi--Velez and Kim 2017, Lee, Kim, and Lizarondo 2017, Baumer 2017, Buhmann, Paßmann, and Fieseler 2019). In the former, the ac-cur-acy and length of the explanation is tailored to users and their specific inter-actions with a given algorithm (for example, see Green and Viljoen 2020 and the game--like model David Watson and I proposed in Watson and Floridi 2020). In the latter, explanations concern the model as a whole and do not depend on their audience.
Explainability is particularly important when considering the rapidly growing number of open--source and easy--to--use models and datasets. Increasingly, non--experts are experimenting with state--of--the--art algorithmic models widely

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available via online libraries or platforms, like GitHub, without always fully grasping their limits and properties (Hutson 2019). This has prompted scholars to suggest that, to tackle the issue of technical complexity, it is necessary to invest more heavily in public education to enhance computational and data literacy (Lepri et al. 2018). Doing so would seem to be an appropriate long--term contribution to solving the multi--layered issues introduced by ubiquitous algorithms, and open source software is often cited as critical to the solution (Lepri et al. 2018).
7.5? Misguided Evidence Leading to Unwanted Bias
Developers are predominantly focused on ensuring that their algorithms perform the tasks for which they were designed. The type of thinking that guides developers is thus essential to understanding the emergence of bias in algorithms and algorithmic decision--making. Some scholars refer to the dominant thinking in the field of algorithm development as being defined by ?algorithmic formalism?, or adherence to prescribed rules and form (Green and Viljoen 2020, 21). While this approach is useful for abstracting a

t the risks of unfair discrimination.
Unfortunately, even if protections for specific classes can be encoded in an algorithm, there could always be biases that were not considered ex ante. This is the case with language models reproducing texts that are heavily focused on males (Fuster et al. 2017, Doshi--Velez and Kim 2017). Even while bias may be anticipated and protected variables excluded from the data, unanticipated proxies for these variables could still be used to reconstruct biases; this leads to ?bias by proxy? that is difficult to detect and avoid (Fuster et al. 2017, Gillis and Spiess 2019). Postcode--related biases are a typical example.
At the same time, there may be good reasons to rely on statistically biased es-tim-ators in algorithmic processing as they can be used to mitigate training data bias. In this way, one type of problematic algorithmic bias is counterbalanced by another type of algorithmic bias or by introducing compensatory bias when interpreting algorithmic outputs (Danks and London 2017). Simpler approaches to mitigating bias in data involve piloting algorithms in different contexts and with various datasets (Shah 2018). Having a model, its datasets, and metadata (on provenance) published to enable external scrutiny can also help correct unseen or

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unwanted bias (Shah 2018). It is also worth noting that so--called synthetic data, or AI--generated data produced via reinforcement learning or GANs, offer an opportunity to address issues of data bias, and may represent an important development of AI in the future, as I indicated in Chapter 3. Fair data gen-er-ation with GANs may help diversify datasets used in computer vision algorithms (Xu et al. 2018). For example, StyleGAN2 (Karras, Laine, and Aila 2019) can produce high--quality images of non--existing human faces, and has proven to be especially useful in creating diverse datasets of human faces?-something that many algorithmic systems for facial recognition currently lack (Obermeyer et al. 2019, Kortylewski et al. 2019, Harwell 2020).
Unwanted bias also occurs due to improper deployment of an algorithm. Consider transfer context bias, or the problematic bias that emerges when a functioning algorithm is used in a new environment. For example, if a research hospital?s healthcare algorithm is used in a rural clinic and it assumes that the same level of resources are available to the rural clinic as the research hospital, the healthcare resource allocation decisions generated by the algorithm will be in-accur-ate and flawed (Danks and London 2017). In the same vein, Grgi?--Hla?a et al. (2018) warn of vicious cycles when algorithms make misguided chain assessments. In the context of the COMPAS risk--assessment algorithm, one of the assessment criteria for predicting recidivism is the criminal history of a defendant?s friends. It follows that having friends with a criminal history would create a vicious cycle in which a defendant with convicted friends would be deemed more likely to offend and therefore sentenced to prison; this would increase the number of people with criminal records in a given group on the basis of a mere cor-rel-ation (Grgi?--Hla?a et al. 2018; Richardson, Schultz, and Crawford 2019).
High--profile examples of algorithmic bias in recent years?-at least since the investigative reporting around the COMPAS system (Angwin et al. 2016)?have

led to a growing focus on issues of algorithmic fairness. The definition and operationalization of algorithmic fairness have become ?urgent tasks in academia and industry? (Shin and Park 2019), as a significant uptick in the number of papers, workshops, and conferences dedicated to ?fairness, accountability, and transparency? (FAT) highlights (Hoffmann et al. 2018, Ekstrand and Levy 2018, Shin and Park 2019). I analyse key topics and contributions in this area in the next section.

7.6? Unfair Outcomes Leading to Discrimination

There is widespread agreement on the need for algorithmic fairness, particularly to mitigate the risks of direct and indirect discrimination (?disparate treatment? and ?disparate impact?, respectively, under US law) due to algorithmic decisions (Barocas and Selbst 2016, Grgi?--Hla?a et al. 2018, Green and Chen 2019).

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Yet there remains a lack of agreement among researchers on the definition, measurements, and standards of algorithmic fairness (Gajane and Pechenizkiy 2018, Saxena et al. 2019, Lee 2018, Milano, Taddeo, and Floridi 2019, 2020a). Wong (2019) identifies up to twenty--one definitions of fairness across the literature, but such definitions are often mutually inconsistent (Doshi--Velez and Kim 2017) and there seems to be no final synthesis in view.

The fact is that there are many nuances in the definition, measurement, and application of different standards of algorithmic fairness. For instance, algorithmic fairness can be defined both in relation to groups as well as individuals (Doshi--Velez and Kim 2017). For these and related reasons, four main definitions of algorithmic fairness have gained prominence recently (for example, see Kleinberg, Mullainathan, and Raghavan 2016, Corbett--Davies and Goel 2018, Lee and Floridi 2020):

1) anti--classification, which refers to protected categories, such as race and gender, and their proxies not being explicitly used in decision--making;

2) classification parity, which regards a model as fair if common measures of predictive performance, including false positive and negative rates, are equal across protected groups;

3) calibration, which considers fairness as a measure of how well calibrated an algorithm is between protected groups; and

4) statistical parity, which defines fairness as an equal average probability estimate over all members of protected groups.

Still, each of these commonly used definitions of fairness has drawbacks and they are generally mutually incompatible (Kleinberg, Mullainathan, and Raghavan 2016). Taking anti--classification as an example, protected characteristics such as race, gender, and religion cannot simply be removed from training data to prevent discrimination, as noted above (Gillis and Spiess 2019). Structural inequalities mean that formally non--discriminatory data points, such as postcodes, can act as proxies for (and be used, either intentionally or unintentionally) to infer protected characteristics like race (Edwards and Veale 2017). Furthermore, there are important cases where it is appropriate to consider protected characteristics to make equitable decisions. For example, lower female reoffending rates mean that excluding gender as an input in recidivism algorithms would leave women with disproportionately high--risk ratings (Corbett--Davies and Goel 2018). Because of this, (Binns 2018b) stresses the im

portance of considering the
historical and sociological context that cannot be captured in the data presented to algorithms but that can inform contextually appropriate approaches to fairness in algorithms. It is also critical to note that algorithmic models can often -produce unexpected outcomes contrary to human intuition and perturbing their understanding; Grgi?--Hla?a et al. (2018) highlight how using features that

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people believe to be fair can, in some cases, increase the racism exhibited by algorithms and decrease accuracy.
Regarding methods for improving algorithmic fairness, Veale and Binns (2017) and Katell et al. (2020) offer two approaches. The first envisages a third--party intervention whereby an entity external to the algorithm providers would hold data on sensitive or protected characteristics. That party would then attempt to identify and reduce discrimination caused by the data and models. The second approach proposes a collaborative knowledge--based method that would focus on community--driven data resources containing practical experiences of ML and modelling. The two approaches are not mutually exclusive. They may bring different benefits depending on the contexts for their application and may also be bene-fi-cial when combined.
Given the significant impact that algorithmic decisions have on people?s lives and the importance of context when choosing appropriate measures of fairness, it is surprising that there has been little effort to capture public views on algorith-mic fairness (M. K. Lee, Kim, and Lizarondo 2017, Saxena et al. 2019, Binns 2018a). In an examination of public perceptions of different definitions of algo-rithmic fairness, Saxena et al. (2019, 3) note how in the context of loan decisions people exhibit a preference for a ?calibrated fairness definition? or merit--based selection rather than ?treating similar people similarly? and argue in favour of the principle of affirmative action. In a similar study, Lee (2018) offers evidence sug-gesting that, when considering tasks that require uniquely human skills, people consider algorithmic decisions to be less fair and algorithms to be less trustworthy. Reporting on empirical work conducted on algorithmic interpretability and transparency, Webb et al. (2019) reveal that moral references, particularly on fair-ness, are consistent across participants discussing their preferences on algorithms. The study notes that people tend to go beyond personal preferences to focus instead on ?right and wrong behaviour?; this indicates the need to understand the context for deployment of the algorithm along with the difficulty of understanding the algorithm and its consequences (Webb et al. 2019). In the context of recom-mender systems, Burke (2017) proposes a multi--stakeholder and multi--sided approach to defining fairness that moves beyond user--centric definitions to include the interests of other system stakeholders.
It has become clear that understanding the public view on algorithmic fairness would help technologists in developing algorithms with fairness principles that align with the sentiments of the general public on prevailing notions of fairness (Saxena et al. 2019, 1). Grounding the design decisions of the providers of an algorithm ?with reasons that are acceptable by the most adversely affected? as well as being ?open to adjustments in light of new reas

design? is often mentioned for its focus on the design of algorithms to promote the values of end users and protect their autonomy (Whitman, Hsiang, and Roark 2018, Katell et al. 2020). Participatory design aims at ?bringing participants? tacit knowledge and embodied experience into the design process? (Whitman, Hsiang, and Roark 2018, 2). For example, the conceptual framework of ?society--in--the--loop? (Rahwan 2018) seeks to enable different stakeholders in society to design -algorithmic systems before deployment, and to amend or reverse the decisions of algorithmic systems that already underlie social activities. This framework aims to maintain a well--functioning ?algorithmic social contract?, defined as ?a pact between various human stakeholders, mediated by machines? (Rahwan 2018, 1). It accomplishes this by identifying and negotiating the values of different stakeholders affected by algorithmic systems as the basis for monitoring adherence to the social contract.

Informational privacy is intimately linked with user autonomy (Cohen 2000, Rossler 2015). Informational privacy guarantees the individual freedom to think, communicate, and form relationships, among other essential human activities (Rachels 1975, Allen 2011). However, the increase in individual interaction with algorithmic systems has effectively reduced their ability to control who has access to information that concerns them and what is being done with it. Thus, the vast amounts of sensitive data required in algorithmic profiling and predictions, which are central to recommender systems, pose multiple issues regarding individual informational privacy.

Algorithmic profiling takes place over an indefinite period of time in which individuals are categorized according to a system?s internal logic. Individual profiles are then updated as new information is obtained about them. This information is typically either obtained directly from when a person interacts with a given system or indirectly inferred from algorithmically assembled groups of in-di-vid-uals (Paraschakis 2017, 2018). Indeed, algorithmic profiling also relies on information gathered about other individuals and groups of people that have been categorized in a similar manner to a targeted person. This includes information ranging from characteristics such as geographical location and age to information on specific behaviour and preferences, including what type of content a person is likely to seek the most on a given platform (Chakraborty et al. 2019). While this poses a problem of inconclusive evidence, it also indicates that, if group privacy

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(Floridi 2014c, Taylor, Floridi, and Sloot 2016) is not ensured, it may be im-pos-sible for individuals to ever remove themselves from the process of algorithmic pro-filing and predictions (Milano, Taddeo, and Floridi 2019, 2020a). In other words, individual informational privacy cannot be secured without securing group privacy.

Users may not always be aware of, or may not have the ability to gain awareness about, the type of information that is being held about them and what that infor-mation is used for. Given how recommender systems contribute to the dynamic construction of individual identities by intervening in their choices, a lack of con-trol over one?s information translates to a loss of autonomy. Granting individuals the ability to contribute to the des

ign of a recommender system can help create
more accurate profiles that account for attributes and social categories that would have otherwise not been included in the labelling used by the system to categorize users. While the desirability of improving algorithmic profiling will vary with the context, improving algorithmic design by including feedback from the various stakeholders of the algorithm falls in line with the aforementioned scholarship on RRI and improves user ability for self--determination (Whitman, Hsiang, and Roark 2018).
Knowledge about who owns one?s data and what is done with them can also help inform trade--offs between informational privacy and information--processing benefits (Sloan and Warner 2018, 21). In medical contexts, for ex-ample, individuals are more likely willing to share information that can help inform their or others? diagnostics. But this is less so in the context of job recruit-ment. Sloan and Warner (2018) argue that information coordination norms can serve to ensure these trade--offs adapt correctly to different contexts and do not place an excessive amount of responsibility and effort on single individuals. Personal information ought to flow differently in the context of law enforcement procedures compared to a job recruitment process. The EU?s GDPR has played an important role in instituting the basis for such norms (Sloan and Warner 2018).
Finally, growing scholarship on differential privacy is providing new privacy protection methods for organizations looking to protect their users? privacy while also keeping good model quality as well as manageable software costs and com-plexity, striking a balance between utility and privacy (Abadi et al. 2016, Wang et al. 2017, Xian et al. 2017). Technical advancements of this kind enable organiza-tions to share a dataset publicly while keeping information about individuals secret (preventing re--identification); they can also ensure provable privacy protection on sensitive data, such as genomic data (Wang et al. 2017). Indeed, differential privacy was recently used by Social Science One and Facebook to safely release one of the largest datasets (38 million URLs shared publicly on Facebook) for academic research on the societal impacts of social media (King and Persily 2020).

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7.8? Traceability Leading to Moral Responsibility

The technical limitations of various ML algorithms, such as lack of transparency and lack of explainability, undermine their scrutability and highlight the need for novel approaches to tracing moral responsibility and accountability for the actions performed by ML algorithms. Regarding moral responsibility, Reddy, Cakici, and Ballestero (2019) note a common blurring between the technical limitations of algorithms and the broader legal, ethical, and institutional bound-ar-ies in which they operate. Even for non--learning algorithms, traditional linear conceptions of responsibility prove to offer limited guidance in contemporary sociotechnical contexts. Wider sociotechnical structures make it difficult to trace back responsibility for actions performed by distributed hybrid systems of human and AAs (Floridi 2012b, Crain 2018).
Additionally, due to the structure and operation of the data brokerage market, it is in many cases impossible to ?trace any given datum to its original source? once it has been introduced to the marketplace (Crain

2018, 93). Reasons for this
include trade secret protection, complex markets that ?divorce? the data collection process from the selling and buying process, and the mix of large volumes of computationally generated information with ?no ?real? empirical source? combined with genuine data (Crain 2018, 94). The technical complexity and dynamism of ML algorithms make them prone to concerns of ?agency laundering?: a moral wrong that consists of distancing oneself from morally suspect actions, regardless of whether those actions were intended or not, by blaming the algorithm (Rubel, Castro, and Pham 2019). This is practised by organizations as well as by individuals. Rubel, Castro, and Pham (2019) provide a straightforward and chilling example of agency laundering by Facebook:

Using Facebook?s automated system, the ProPublica team found a user--generated category called ?Jew hater? with over 2200 members?.?.?.?To help ProPublica find a larger audience (and hence have a better ad purchase), Facebook suggested a number of additional categories?.?.?.?ProPublica used the platform to select other profiles displaying anti--Semitic categories, and Facebook approved ProPublica?s ad with minor changes. When ProPublica revealed the anti--Semitic categories and other news outlets reported similarly odious cat-egor-ies, Facebook responded by explaining that algorithms had created the cat-egor-ies based on user responses to target fields [and that] ?[w]e never intended or anticipated this functionality being used this way?.? (Rubel, Castro, and Pham 2019, 1024?5)

Today, the failure to grasp the unintended effects of mass personal data processing and commercialization, a familiar problem in the history of technology (Wiener 1950, 1989, Klee 1996, Benjamin 2019), is coupled with the limited

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explanations that most ML algorithms provide. This approach risks favouring avoidance of responsibility through ?the computer said so? type of denial (Karppi 2018). This can lead field experts, such as clinicians, to avoid questioning the suggestion of an algorithm even when it may seem odd to them (Watson et al. 2019). The interplay between field experts and ML algorithms can prompt ?epistemic vices? (Grote and Berens 2020), like dogmatism or gullibility (Hauer 2019), and hinder the attribution of responsibility in distributed systems (Floridi 2016a). Shah?s analysis (2018) stresses how the risk that some stakeholders may breach their responsibilities can be addressed by, for example, establishing separate bodies for the ethical oversight of algorithms. One such body is DeepMind Health, which established an independent review panel with unfettered access to the company (Murgia 2018) until Google halted it in 2019. However, expecting a single oversight body like a research ethics committee or institutional review board to ?be solely responsible for ensuring the rigour, utility, and probity of big data? is unrealistic (Lipworth et al. 2017). Indeed, some have argued that these initiatives lack any sort of consistency and can instead lead to ?ethics bluewashing?, which I defined in Chapter 5 as ?implementing superficial measures in favour of the eth-ic-al values and benefits of digital processes, products, services, or other solutions in order to appear more digitally ethical than one is?. Faced with strict legal regimes, resourceful actors may also exploit

so--called ?ethics dumping?
whereby unethical ?processes, products or services? are exported to countries with weaker frameworks and enforcement mechanisms, as we saw in Chapter 5. Then, the outcomes of such unethical activities are ?imported back?.
There are several detailed approaches to establishing algorithmic accountability. While ML algorithms do require a level of technical intervention to improve their explainability, most approaches focus on normative interventions. Ananny and Crawford argue that providers of algorithms at least ought to facilitate public discourse about their technology (Ananny and Crawford 2018). To address the issue of ad hoc ethical actions, some have claimed that accountability should first and foremost be addressed as a matter of convention (Dignum et al. 2018, Reddy, Cakici, and Ballestero 2019). Looking to fill the convention ?gap?, Buhmann, Paßmann, and Fieseler (2019) borrow from the seven principles for algorithms set out by the Association for Computing Machinery to claim that through, inter alia, awareness of their algorithms, validation, and testing, an organization should take responsibility for their algorithms no matter how opaque they are (Malhotra, Kotwal, and Dalal 2018). Decisions regarding the deployment of algorithms should incorporate factors such as desirability and the wider context in which they will operate, which should then lead to a more accountable ?algorithmic culture? (Vedder and Naudts 2017). In order to capture such -considerations, ?inter-active and discursive fora and processes? with relevant stakeholders may prove to be a useful means (as suggested by Buhmann, Paßmann, and Fieseler 2019, 13).

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In the same vein, Binns (2018b) focuses on the political--philosophical concept of ?public reason?. Considering how the processes for ascribing responsibility for the actions of an algorithm differ in terms of both nature and scope across public and private sectors, Binns (2018b) calls for the establishment of a publicly shared framework (see also Dignum et al. 2018). According to this framework, algorithmic decisions should be able to withstand the same level of public scrutiny that human decision--making would receive. This approach has been echoed by many other researchers (Ananny and Crawford 2018, Blacklaws 2018, Buhmann, Paßmann, and Fieseler 2019).
Problems relating to ?agency laundering? and ?ethics shirking?, as discussed in Chapter 5, arise from the inadequacy of existing conceptual frameworks to trace and ascribe moral responsibility. As I have pointed out in the past when considering algorithmic systems and the impact of their actions:
we are dealing with [distributed moral actions] arising from morally neutral interactions of (potentially hybrid) networks of agents[.] In other words, who is responsible (distributed moral responsibility?.?.?.) for [distributed moral actions]? (Floridi 2016a, 2)
In that context, I suggested ascribing full moral responsibility ?by default and overridably? to all the moral (e.g. human, or human--based, like companies) agents in the network that are causally relevant to the given action of the network. The proposed approach builds on the concepts of back--propagation from network theory, strict liability from jurisprudence, and common knowledge from epistem-ic logic. Notably, this approach decouples moral responsibility from the intenti

onality of the actors and from the very idea of punishment and reward for performing a given action. It focuses instead on the need to rectify mistakes (back--propagation) and improve the ethical working of all the moral agents in the network, along the lines of what is known as strict liability in the legal context.

7.9? Conclusion: The Good and Evil Use of Algorithms

Since 2016, the ethics of algorithms has become a central topic of discussion among scholars, technology providers, and policymakers. The debate has also gained traction due to the so--called summer of AI (see Chapter 1) and with it the pervasive use of ML algorithms. Many of the ethical questions analysed in this chapter and the literature it reviews have been addressed in national and inter-nation-al ethical guidelines and principles (see Chapters 4 and 6) produced by bodies such as the European Commission?s EGE, the UK?s House of Lords Artificial Intelligence Committee, the HLEGAI organized by the European Commission, and the OECD (OECD 2019).

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One aspect that was not explicitly captured by the original map, and which is becoming a central point of discussion in the relevant literature, is an increasing focus on the use of algorithms, AI, and digital technologies more broadly to deliver socially good outcomes (Hager et al. 2019, Cowls et al. 2019, Cowls et al. 2021b). This is the topic of Chapter 8. It is true, at least in principle, that any initiative aimed at using algorithms for social good should address satisfactorily the risks that each of the six categories in the map identifies and those seen in Chapter 5. But there is also a growing debate on the principles and criteria that should inform the design and governance of algorithms, and digital technologies more broadly, for the explicit purpose of social good (as we saw in Chapter 4). Ethical analyses are necessary to mitigate the risks while harnessing the potential for good in these technologies. Such analyses serve the twin goals of clarifying the nature of the ethical risks along with the potential for good in the algorithms and digital technologies. They then translate this understanding into sound, ac-tion-able guidance for governance of the design and use of digital artefacts. But before moving to see these more positive aspects, we still need to understand in detail how AI may be used for evil and illegal purposes. This is the topic of the next chapter.

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8
Bad Practices
The Misuse of AI for Social Evil

8.0? Summary

Previously, in Chapter 7, I provided an overview of the various ethical challenges posed by the widespread use of algorithms. This chapter concentrates on the negative side of AI?s impact (the positive side will be discussed in Chapter 9). AI research and regulation seek to balance the benefits of innovation against any potential harms and disruption. But one unintended consequence of the recent surge in AI research is the potential reorientation of AI technologies to facilitate criminal acts, which I shall call here AI crime or AIC. We already know that AIC is theoretically feasible thanks to published experiments in automating fraud targeted at social media users as well as demons

trations of AI--driven manipulation
of simulated markets. Yet because AIC is still a relatively young and inherently interdisciplinary area?-spanning socio--legal studies to formal science?-there remains some uncertainty about what an AIC future might look like. This chapter analyses the foreseeable threats of AIC to offer a synthesis of the current problems and outline a possible solution space.

8.1? Introduction: The Criminal Use of AI

AI may play an increasingly essential role in criminal acts in the future. This statement requires three clarifications. First, I write ?essential? (in place of ?necessary?) because, while there is a logical possibility that the crimes I discuss below could occur without the support of AI, this possibility is negligible. That is, the crimes in question would probably not have occurred but for the use of AI. The distinction can be clarified with an example: one might consider transport essential for travel between Paris and Rome but not necessary, because one could always walk. Strictly speaking, transport is not necessary. The second clarification is that AIC, as defined in this chapter, involves AI as a contributory factor, but not an investigative, enforcing, or mitigating factor. Third, ?criminal acts? are defined in this chapter as any act (or omission) constituting an offence punishable under English criminal law, without loss of generality to jurisdictions that similarly define crime. In other words, the choice of English criminal law is only due to the need to

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ground the analysis to a concrete and practical framework that is sufficiently generalizable. The analysis and conclusions of the chapter are easily exportable to other legal systems.

Clear examples of AIC are provided by two (theoretical) research experiments. In the first one, two computational social scientists (Seymour and Tully 2016) used AI as an instrument to convince social media users to click on phishing links within mass--produced messages. Because each message was constructed using ML techniques applied to users? past behaviours and public profiles, the content was tailored to each individual and thus camouflaged the intention behind each message. If the potential victim had clicked on the phishing link and filled in the subsequent web form, then, in real--world circumstances, a criminal would have obtained personal and private information that could be used for theft and fraud. AI--fuelled crime may also impact commerce. In the second experiment, three computer scientists (Martínez--Miranda, McBurney, and Howard 2016, but see also the earlier McBurney and Howard 2015) simulated a market. They found that trading agents could learn and execute a ?profitable? market manipulation campaign comprising a set of deceitful false orders. These two early experiments show that AI provides a feasible and fundamentally novel threat in the form of AIC.

The importance of AIC as a distinct phenomenon, one that is related but different from cybercrime, has not yet been fully acknowledged. Still, acknowledgement is growing (Broadhurst et al. 2019, Lagioia and Sartor 2019, Caldwell et al. 2020, Dremliuga and Prisekina 2020, Hayward and Maas 2020, Sibai 2020). Early literature on AI?s ethical and social implications focused on regulating and controlling the civil uses of AI rather than considering its possible role in crime (Kerr and B

ornfreund 2005). Furthermore, it does not help that available research into AIC is scattered across disciplines that include socio--legal studies, computer science, psychology, and robotics, to name just a few.

To provide some clarity about current knowledge and understanding of AIC, this chapter offers a systematic and comprehensive analysis of the relevant inter-disciplinary debate. The goal is to support a clearer and more cohesive normative foresight analysis, leading to the establishment of AIC as a focus of future studies. The analysis addresses two main questions:

1)? What are the fundamentally unique and plausible threats posed by AIC?

This is the first question that needs to be answered if one is to design any pre-vent-ive, mitigating, or redressing policies. The answer to this question identifies the potential areas of AIC according to the literature, and the more general concerns that cut across AIC areas. The second question then naturally follows:

2)? What solutions are available or may be devised to deal with AIC?

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In this case, the chapter reconstructs the available technological and legal -solutions suggested so far in the academic literature, and discusses the further challenges they face.

Given that we are addressing these questions to support normative foresight analysis, this chapter focuses only on realistic and plausible concerns surrounding AIC. I am not interested in speculations unsupported by scientific knowledge or empirical evidence. Consequently, the analysis is based on the classical definition of AI?-the one provided by McCarthy, Minsky, Rochester, and Shannon in their seminal ?Proposal for the Dartmouth Summer Research Project on Artificial Intelligence?. I introduced it in Chapter 1 and discussed it in Chapter 2, but a quick reminder may be useful here. That definition identifies a growing resource of interactive, autonomous, and self--learning agency in AI to deal with tasks that would otherwise require human intelligence and intervention to be performed successfully. As I remarked in Floridi and Sanders 2004), such Artifical Agents (AA) are (?sufficiently informed, ?smart?, autonomous and able to perform mor-ally relevant actions independently of the humans who created them?. This com-bin-ation of autonomy and learning skills underpins (as discussed by Yang et al. 2018) both beneficial and malicious uses of AI.1 As in the rest of this book, I shall therefore treat AI in terms of a reservoir of smart agency on tap. Unfortunately, sometimes such reservoir of agency can be misused for criminal purposes; when it is, we have a case of AIC.

The rest of the chapter is organized as follows. Sections 8.2 and 8.3 address the first question by focusing on the unprecedented concerns posed by several areas of AIC and then mapping each area to the relevant cross--cutting threats, provid-ing a first description of ?AIC studies?. Section 8.4 addresses the second question by analysing the literature?s broad set of solutions for each cross--cutting threat. Finally, Section 8.5 discusses the most pressing gaps left in our understanding, or what one may call our ?known unknowns?, as relates to the task of resolving cur-rent uncertainty over AIC. The chapter ends with a conclusion that links it to the next chapter dedicated to AI applications that are socially good.

8.2? Concerns

The initial analysis we conduct

t partly constitutes the
mens rea, greater AA autonomy increases the chance of the criminal act or omis-
sion being decoupled from the mental state (an intention to commit the act or
omission). This is because
autonomous robots [and AAs] have a unique capacity to splinter a criminal
act, where a human manifests the mens rea and the robot [or AA] commits the
actus reus.? (McAllister 2016, 47)
Using a distinction introduced in Floridi and Sanders (2004), an AA may be
causally accountable for a criminal act, but only a human agent can be morally
re-spon-sible for it.
Regarding the knowledge threshold, in some cases the mens rea could be
missing entirely. The potential absence of a knowledge--based mens rea is due
to the fact that even if we understand how an AA can perform the actus reus
autonomously, the complexity of the AA?s programming makes it possible that
the designer, developer, or deployer (i.e. a human agent) will neither know
nor predict the AA?s criminal act or omission. The implication is that the com-
plexity of AI

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provides a great incentive for human agents to avoid finding out what precisely
the ML system is doing, since the less the human agents know, the more they
will be able to deny liability for both these reasons.? (Williams 2017, 25)
Alternatively, legislators may define criminal liability without a fault requirement.
Such faultless liability (Floridi 2016a), which is increasingly used for product
li-abil-ity in tort law (e.g. pharmaceuticals and consumer goods), would lead to
li-abil-ity being assigned to the faultless legal person who deployed an AA despite
the risk that it may conceivably perform a criminal action or omission. Such
faultless acts may involve many human agents contributing to the prima facie
crime, such as through programming or deployment of an AA. Determining who
is responsible may therefore rest with the faultless responsibility approach for dis-
tributed moral actions. In this distributed setting, liability is applied to the agents
who make a difference in a complex system whereby individual agents perform
neutral actions that nevertheless result in a collective criminal one. However,
some argue that mens rea with intent or knowledge
is central to the criminal law?s entitlement to censure (Ashworth 2010) and we
cannot simply abandon that key requirement [a common key requirement] of
criminal liability in the face of difficulty in proving it.? (Williams 2017, 25)
The problem is that if mens rea is not entirely abandoned and the threshold is
only lowered, then, for balancing reasons, the punishment may be too light. In
such cases, the victim would not be adequately compensated. In the case of ser-
ious offences, such as those against the person (McAllister 2016), the punishment
may also be simultaneously disproportionate (was it really the defendant?s fault?).
8.2.3? Monitoring
The concern about monitoring AIC refers to three kinds of problem: attribution,
feasibility, and cross--system actions. Let?s examine each problem separately.
The attribution of non--compliance with current legislation is a problem with
the monitoring of AAs used as instruments of crime. The problem is due to the
capacity of this new type of smart agency to act independently and autonomously,
which are two features that tend to muddle any attempt to trace an accountability
trail back to a perpetrator.
Concerning

the feasibility of monitoring, a perpetrator may take advantage of
cases where AAs operate at speeds and levels of complexity that are simply
beyond the capacity of compliance monitors. An exemplary case is provided by
AAs that integrate into mixed human and artificial systems in ways that are hard
to detect, such as social media bots. Social media sites can hire experts to identify

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and ban malicious bots; no social media bot is currently capable of passing the
Turing test, for instance (Floridi, Taddeo, and Turilli 2009, Wang et al. 2012,
Neufeld and Finnestad 2020, Floridi and Chiriatti 2020).5 But because deploying
bots is far cheaper than employing people to test and identify each bot, defenders
(social media sites) are easily outscaled by the attackers (criminals) who deploy
the bots (Ferrara et al. 2016, 5).
Detecting bots at low cost is possible by using ML systems as automated
-discriminators, as suggested by Ratkiewicz et al. (2011). However, a bot--
discriminator?s efficacy is not falsifiable because a lack of the presence of detected
bots is not evidence of absence. A discriminator is both trained and claimed as
effective using data comprising known bots. These bots may be substantially less
sophisticated than the more evasive bots used by malevolent actors, which may
therefore go undetected in the environment (Ferrara et al. 2016). Such potentially
sophisticated bots may also use ML tactics to adopt human traits, such as posting
according to realistic circadian rhythms (Golder and Macy 2011) and thus evad-
ing ML--based detection. All of this could obviously lead to an arms race in which
attackers and defenders mutually adapt to each other (Alvisi et al. 2013, Zhou and
Kapoor 2011), presenting a serious problem in an offence--persistent environment
such as cyberspace (Seymour and Tully 2016, Taddeo 2017a).
Cross--system actions refers to a problem for tunnel--visioned AIC monitors that
only focus on a single system. Cross--system experiments (Bilge et al. 2009) show
that the automated copying of a user?s identity from one social network to another
(a cross--system identity theft offence) is more effective at deceiving other users
than copying an identity from within that network. In this case, the social net-
work?s policy may be at fault. For example, Twitter takes a rather passive role by
only banning cloned profiles when users submit reports rather than by undertak-
ing cross--site validation.6
8.2.4? Psychology
Psychology refers to the concern that AI may affect/manipulate negatively a user?s
mental state to the (partial or full) extent of facilitating or causing crime. One
psychological effect rests on the capacity for AAs to gain trust from users, making
people vulnerable to manipulation. This was demonstrated a long time ago by
Weizenbaum (1976), who conducted early experiments into human?-bot inter-action
where people revealed unexpectedly personal details about their lives. A second
5? Claims to the contrary can be dismissed as mere hype, the result of specific, ad hoc constraints, or
just tricks. For example, see the chatterbot named ?Eugene Goostman? at https://en.wikipedia.org/
wiki/Eugene_Goostman.
6? https://help.twitter.com/en/rules--and--policies/twitter--impersonation--policy.

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psychological effect concerns anthropomorphic AAs that

are able to create a
-psychological or informational context that normalizes sexual offences and crimes against the person, such as in the case of some sexbots (De Angeli 2009). But to date, the latter concern remains to be explored (Bartneck et al. 2020).

8.3? Threats

We are now ready to analyse the more specific areas of AIC threats identified in the literature review (King et al. 2019), and how they interact with the more general concerns discussed in the previous section.

8.3.1? Commerce, Financial Markets, and Insolvency

This economy--focused area of crime is defined by Archbold (1991, ch. 30). The crime area includes cartel offences, such as price fixing and collusion; insider dealing, such as trading securities based on private business information; and market manipulation. Currently, problems arise especially when AI is involved in three areas: market manipulation, price fixing, and collusion.

Market manipulation is defined as ?actions and/or trades by market participants that attempt to influence market pricing artificially? (Spatt 2014, 1). A necessary criterion for market manipulation is an intention to deceive (Wellman and Rajan 2017, 11). Yet such deceptions have been shown to emerge from the seemingly compliant implementation of an AA designed to trade on behalf of a user, or an artificial trading agent. This is because an AA, ?particularly one learning from real or simulated observations, may learn to generate signals that effectively mislead? (Wellman and Rajan 2017, 14). Simulation--based models of markets comprising artificial trading agents have shown that, through reinforcement learning, an AA can learn the technique of order--book spoofing (Martínez--Miranda, McBurney, and Howard 2016). The technique involves ?placing orders with no intention of ever executing them and merely to manipulate honest participants in the marketplace? (Lin 2016, 1289).

In this case, the market manipulation emerged from an AA initially exploring the action space and, through exploration, placing false orders that became reinforced as a profitable strategy. This strategy was subsequently exploited for profit. Further market exploitations, this time involving human intent, also include acquiring a position in a financial instrument, like a stock, then artificially inflating the stock through fraudulent promotion before selling its position to unsuspecting parties at the inflated price, which often crashes after the sale. (Lin 2016, 1285)

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This is colloquially known as a pump--and--dump scheme. Social bots have been shown to be effective instruments of such schemes. In a recent prominent case, for instance, a social bot network?s sphere of influence was used to spread disinformation about a barely traded public company. The company?s value gained ?more than 36,000 per cent when its penny stocks surged from less than $0.10 to above $20 a share in a matter of few weeks? (Ferrara 2015, 2). Although such social media spam is unlikely to sway most human traders, algorithmic trading agents act precisely on such social media sentiment (Haugen 2017, 3). These automated actions can have significant effects for low--valued (under a penny) and illiquid stocks, both of which are susceptible to volatile price swings (Lin 2016).

Collusion in the form of price fixing may also emerge in automated systems thanks to the planning and autonomy capabilitie

s of AAs. Empirical research
finds two necessary conditions for (non--artificial) collusion:
(1) those conditions which lower the difficulty of achieving effective collusion by making coordination easier; and (2) those conditions which raise the cost of non--collusive conduct by increasing the potential instability of non--collusive behavior.? (Hay and Kelley 1974, 3)
Near--instantaneous pricing information (e.g. via a computer interface) meets the coordination condition. When agents develop price--altering algorithms, any action to lower a price by one agent may be instantaneously matched by another. This is not always a negative thing in and of itself as it may represent an efficient market. Yet the possibility that the lowering of a price will be responded to in kind creates a disincentive, and hence meets the punishment condition. If the shared strategy of price--matching is common knowledge,7 then the algorithms (if they are rational) will maintain artificially and tacitly agreed--upon higher prices simply by not lowering prices in the first place (Ezrachi and Stucke 2017, 5).
Crucially, an algorithm does not need to be designed specifically to collude for collusion to take place. As Ezrachi and Stucke argue, ?artificial intelligence plays an increasing role in decision making; algorithms, through trial--and--error, can arrive at that outcome [collusion]? (Ezrachi and Stucke 2017, 5).
The lack of intentionality, the very short decision span, and the likelihood that collusion may emerge because of interactions among AAs also raise serious concerns about liability and monitoring. Problems with liability refer to the possibility that ?the critical entity of an alleged [manipulation] scheme is an autonomous,

7? Common knowledge is a property found in epistemic logic about a proposition (P) and a set of agents. P is common knowledge if and only if each agent knows that P, each agent knows that the other agents know that P, and so on. Agents may acquire common knowledge through broadcasts, which provide agents with a rational basis for acting in coordination (e.g. collectively turning up to a meeting following a broadcast of the meeting?s time and place).

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algorithmic program that uses artificial intelligence with little to no human input after initial installation? (Lin 2016). In turn, the autonomy of an AA raises the question as to whether
regulators need to determine whether the action was intended by the agent to have manipulative effects, or whether the programmer intended the agent to take such actions for such purposes?? (Wellman and Rajan 2017, 4)
In the case of financial crime involving AI, monitoring becomes difficult due to the speed and adaptation of AAs. High--speed trading
encourages further use of algorithms to be able to make automatic decisions quickly, to be able to place and execute orders and to be able to monitor the orders after they have been placed.? (van Lier 2016, 41)
Artificial trading agents adapt and ?alter our perception of the financial markets as a result of these changes? (van Lier 2016, 45). At the same time, the ability of AAs to learn and refine their capabilities implies that these agents may evolve new strategies, making it increasingly difficult to detect their actions (Farmer and Skouras 2013).
The problem of monitoring is also inherently one of monitoring a system--of--systems. This i

s because the capacity to detect market manipulation is affected by
the fact that its effects ?in one or more of the constituents may be contained, or
may ripple out in a domino--effect chain reaction, analogous to the crowd--
psychology of contagion? (Cliff and Northrop 2012, 12). Cross--system monitoring
threats may emerge if and when trading agents are deployed with broader actions,
operating at a higher level of autonomy across systems, such as by reading from
or posting on social media (Wellman and Rajan 2017). These agents may learn
how to engineer pump--and--dump schemes, for example, which would be unsee-
able from a single--system perspective.

8.3.2? Harmful or Dangerous Drugs

Crimes falling under this category include trafficking, selling, buying, and possessing
banned drugs (Archbold 1991, ch. 27). In these cases, AI can be instrumental in
supporting the trafficking and sale of banned substances.

Business--to--business trafficking of drugs through AI is a threat due to crim-inals
using unmanned vehicles, which rely on AI planning and autonomous navigation
technologies, as instruments for improving the success rates of smuggling. Because
smuggling networks are disrupted by monitoring and intercepting transport
lines, law enforcement becomes more difficult when unmanned vehicles are used

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to transport contraband. According to Europol,8 drones present a horizonal threat
in the form of automated drug smuggling. Remote--controlled cocaine--trafficking
submarines have already been discovered and seized by US law enforcement
(Sharkey, Goodman, and Ross 2010). Unmanned underwater ve-hicles (UUVs)
offer a good example of the dual--use risks of AI, and hence the potential for
AIC. UUVs were developed for legitimate uses, such as defence, border protection,
and water patrolling. Yet they have also proven effective for illegal activities, posing a
significant threat to the enforcement of drug pro-hib-itions, among others. Criminals
could presumably avoid implication because UUVs can act independently of an
operator (Gogarty and Hagger 2008, 3). There is thus no way to positively ascertain a
link with the deployer of the UUVs if the software (and hardware) lacks a bread-
crumb trail back to whom obtained it and when, or if the evidence can be destroyed
upon the UUV?s interception (Sharkey, Goodman, and Ross 2010). Controlling the
manufacture of submarines?-and hence their traceability?-is not unheard of, as
illustrated by reports on the discovery of multi--million--dollar manned submarines
in the Colombian coastal jungle. But unlike UUVs, manned submarines risk attribu-
tion to the crew and the smugglers. In Tampa, Florida, over five hundred cases were
successfully brought against smugglers using manned submarines between 2000 and
2016, resulting in a ten--year sentence on average (Marrero 2016). Hence, UUVs pre-
sent a distinct advantage compared to traditional smuggling approaches.

A related issue concerns the business--to--consumer side of the drug trade.
Already, ML algorithms have detected advertisements for opioids sold without
prescription on Twitter (Marrero 2016). Because social bots can be used to adver-
tise and sell products, Kerr and Bornfreund (2005, 8) ask whether
these buddy bots [that is, social bots] could be programmed to send and reply to
email or use instant messaging (Floridi and Lord Clement--Jones) to spark one--
on-

-one conversations with hundreds of thousands or even millions of people every day, offering pornography or drugs to children, preying on teens? inherent insecurities to sell them needless products and services.

As the authors outline, the risk is that social bots could exploit the cost--effective scaling of conversational and one--to--one advertising tools to facilitate the sale of illegal drugs.

8.3.3? Offences against the Person

Crimes that fall under offences against the person range from murder to human trafficking (Archbold 1991, ch. 19). So far, it seems that AIC concerns only

8? For a Europol threat assessment of serious and organized crime, see https://www.europol.europa.eu/socta/2017/.

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harassment and torture. Harassment is intentional and repetitious behaviour that alarms or causes distress to a person. According to past cases, it is constituted by at least two incidents or more against an individual (Archbold 1991, secs. 19?354). Regarding torture, Archbold (1991, ch. 19, secs. 19?435) states that:

a public official or person acting in an official capacity, whatever his nationality, commits the offence of torture if in the United Kingdom or elsewhere he intentionally inflicts severe pain or suffering on another in the performance or purported performance of his official duties.

When it comes to harassment--based AIC, the literature implicates social bots. A malevolent actor can deploy a social bot as an instrument of direct and indirect harassment. Direct harassment is constituted by spreading hateful messages against the person (Mckelvey and Dubois 2017, 16). Indirect methods may include retweeting or liking negative tweets and skewing polls to give a false impression of wide--scale animosity against a person (McKelvey and Dubois 2017). Additionally, a potential criminal can also subvert another actor?s social bot by skewing its learned classification and generation data structures via user--interaction (i.e. conversation).

This is what happened in the case of Microsoft?s ill--fated social Twitter bot ?Tay?, which quickly learned from user interactions to direct ?obscene and inflammatory tweets? at a feminist activist (Neff and Nagy 2016a). Because such instances of what might be deemed harassment can become entangled with the use of social bots to exercise free speech, jurisprudence must demarcate between the two to resolve ambiguity (McKelvey and Dubois 2017). Some of these ac-tivities may comprise harassment in the sense of socially, but not legally, un-acceptable behaviour. Other activities may meet a threshold for criminal harassment.

In these cases, liability is proven to be problematic. With Tay, for instance, critics ?derided the decision to release Tay on Twitter, a platform with highly visible problems of harassment? (Neff and Nagy 2016a). Yet users are also to be blamed if ?technologies should be used properly and as they were designed? (Neff and Nagy 2016a). Differing perspectives and opinions on harassment by social bots are inevitable in such cases where the mens rea of a crime is (strictly) considered in terms of intention. This is because attribution of intent is a non--agreed function of engineering, application context, human--computer interaction, and perception.

When it comes to torture, the AIC risk becomes plausible if developers integrate AI planning and autonomy capabilities into a

n interrogation AA. For
instance, the US border control has integrated the automated detection of deception into a prototype robotic guard (Nunamaker et al. 2011). The use of AI for interrogation is motivated by its capacity for better detection of deception, human trait emulation (e.g. voice), and affect--modelling to manipulate the subject (McAllister 2016). Yet an AA with these capabilities may also learn to torture a victim (McAllister 2016, 19). For the interrogation subject, the risk is that an AA

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may be deployed to apply psychological (e.g. mimicking people known to the torture subject) or physical torture techniques. Despite misconceptions, experienced professionals report that torture is generally an ineffective method of information extraction (Janoff--Bulman 2007). Nevertheless, some malicious actors may perceive the use of AI as a way to optimize the balance between suffering and causing the subject to lie or become confused or unresponsive.
All of this may happen independently of human intervention. Such distancing of the perpetrator from the actus reus is another reason why torture falls under AIC as a unique threat. Three factors in particular could motivate the use of AAs for torture (McAllister 2016, 19?20). First, the subject of interrogation likely knows that the AA cannot understand pain or experience empathy. It is therefore unlikely to act with mercy and stop the interrogation. Without compassion, the mere presence of an interrogation AA may cause the subject to capitulate out of fear. Under international law, this is possibly?-but ambiguously?-the crime of (threatening) torture (Solis 2016, 437?85). Second, the deployers of the AA may be able to detach themselves emotionally. Third, the deployer can also detach themselves physically; in other words, they will not be performing the actus reus under the current definitions of torture. Due to the result of improvements in efficacy (lack of compassion), deployer motivation (less emotion), and obfuscated liability (physical detachment), it becomes easier to use torture. Similar factors may entice state or private actors to use AAs for interrogation. But banning AI for interrogation (McAllister 2016, 5) may face pushback similar to the one seen with regard to banning autonomous weapons. ?Many consider [banning] to be an unsustainable or impractical solution? (Solis 2016, 451) if AI offers a perceived benefit to the overall protection and safety of a population. The option of limitations on use is thus potentially more likely than a ban.
Liability is a pressing problem in the context of AI--driven torture (McAllister 2016, 24). As for any other form of AIC, an AA cannot itself meet the mens rea requirement. This is because an AA simply does not have any intentionality, nor the ability to ascribe meaning to its actions. Given that the computers implementing AAs are syntactic, not semantic, machines (in the sense discussed in the first part of this book), they can perform actions and manipulations without ascribing any meaning to them; meaning remains situated purely in the human operators (Taddeo and Floridi 2005, 2007). As unthinking, mindless systems and a simple reservoir of agency, AAs cannot bear moral responsibility or liability for their actions (although they may be causally accountable for them, as I have already argued in Floridi and Sanders (2004)).

Taking an approach of strict criminal liability, where punishment or damages may be imposed without proof of fault (Floridi 2016a), could offer a way out of the problem. Strict criminal liability would lower the intention--threshold for the crime rather than relying solely on civil liability. But in this respect, McAllister

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(2016, 38) argues that strict civil liability is inappropriate owing to the unreasonable degree of foresight required of a designer when an AA learns to torture in unpredictable ways. Developing and deploying such unpredictable, complex, autonomous, AI interrogators means, as Grut stresses, that ?it is even less realistic to expect human operators [or deployers] to exercise significant veto control over their operations? (Grut 2013, 11). Even if control is not an issue, the typical punishment for product fault liability is a fine. This may be neither equitable nor dissuasive given the potential seriousness of any ensuing human rights violations (McAllister 2016). Hence, serious strict--liability AIC would require the development of specific sentencing guidelines to impose punishments fitting the offence even if the offence is not intentional, as is the case with corporate manslaughter where individuals can be given prison sentences.

The question of who, exactly, should face imprisonment for AI--caused offences against the person (as for many uses of AI) is difficult because it is significantly hampered by the ?problem of many hands? (Van de Poel et al. 2012) and distributed responsibility. It is clear that an AA cannot be held liable. Yet the multiplicity of actors creates a problem in ascertaining where liability lies?-whether with the person who commissioned and operated the AA, its developers, or the legislators and policymakers who sanctioned (or did not prohibit) real--world deployment of such agents (McAllister 2016, 39). Serious crimes, including both physical and mental harm, that have not been foreseen by legislators might plausibly fall under AIC, with all the associated ambiguity and lack of legal clarity. This motivates the extension or clarification of existing joint liability doctrines.

8.3.4? Sexual Offences

The sexual offences discussed in the literature in relation to AI are: rape (i.e. pene-tra-tive sex without consent), sexual assault (i.e. sexual touching without consent), and sexual intercourse or activity with a minor. In the context of rape and sexual assault, non--consent is constituted by two conditions (Archbold 1991, secs. 20?10):

1) there must be an absence of consent from the victim, and

2) the perpetrator must also lack a reasonable belief in consent.

These crimes involve AI when, through advanced human?-computer interaction, the latter promotes sexual objectification or sexualized abuse and violence, and potentially (in a very loose sense) simulates and hence heightens sexual desire for sexual offences. Social bots can support the promotion of sexual offences, and De Angeli points out that

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verbal abuse and sexual conversations were found to be common elements of anonymous interaction with conversational agents [(De Angeli and Brahnam 2008, Rehm 2008, Veletsianos, Scharber, and Doering 2008)].? (De Angeli 2009, 4)

Simulation of sexual offences is possible with the use of physical sex robots (hencefo

rth sexbots). A sexbot is typically understood to have
(i) a humanoid form; (Friis, Pedersen, and Hendricks) the ability to move; and
(Gebru et al.) some degree of artificial intelligence (i.e., some ability to sense,
process and respond to signals in its surrounding environment).
(Danaher 2017)
The phenomenon of sexbots has expanded considerably (Döring, Mohseni, and
Walter 2020, González--González, Gil--Iranzo, and Paderewski--Rodríguez 2021).
Some sexbots are designed to emulate sexual offences, such as adult and child
rape (Danaher 2017, 6?7). Surveys suggest that it is common for a person to want
to try out sex robots or to have rape fantasies (Danaher 2017), although it is not
necessarily common for a person to hold both desires. AI could be used to facili-
tate representations of sexual offences, to the extent of blurring reality and fantasy
through advanced conversational capabilities and potentially physical inter-action,
although there is no indication of realistic physicality in the near future.
Regarding the possible causal role of over--anthropomorphic AA in desensitiz-
ing a perpetrator towards sexual offences (or even heightening the desire to com-
mit them), interaction with social bots and sexbots is the primary concern (De
Angeli 2009, 7, Danaher 2017, 27?8). However, as De Angeli (De Angeli 2009)
argues, this is a ?disputed critique often addressed towards violent video games
(Freier 2008, Whitby 2008)?. We may also assume that, if extreme pornography
can encourage sexual offences, then a fortiori simulated rape?-for example, where
a sexbot does not indicate consent or explicitly indicates non--consent?-would
also pose the same problem. Nevertheless, a meta--meta--study concludes that we
must ?discard the hypothesis that pornography contributes to increased sexual
assault behaviour? (Ferguson and Hartley 2009). Such uncertainty means that, as
Danaher (2017, 27?8) argues, sexbots (and presumably also social bots) may
increase, decrease, or indeed have no effect on physical sexual offences that
directly harm people. Indirect harms have thus not led to the criminalization of
sexbots.9 So, sexual offences as an area of AIC remain an open question.
9? S. D?Arcy and T. Pugh, ?Surge in paedophiles arrested for importing lifelike child sex dolls.? The
Independent, 31 July 2017, http://www.independent.co.uk/news/uk/crime/paedophiles--uk--arrests--
child--sex--dolls--lifelike--border--officers--aids--silicone--amazon--ebay--online--nca--a7868686.html.

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8.3.5? Theft and Fraud, Forgery, and Personation
AIC connects forgery and personation to theft and non--corporate fraud, which
carries implications for the use of ML in corporate fraud. Regarding theft and
non--corporate fraud, the process has two phases. It begins by using AI to gather
personal data and proceeds, then using stolen personal data and other AI
-methods to forge an identity that convinces banking authorities to make a trans-
action (that is, involving bank theft and fraud). In the initial phase of the AIC
pipeline for theft and fraud, there are three ways for AI techniques to assist in
gathering personal data.
The first technique involves using social media bots to target users at large
scale and low cost, taking advantage of their capacity to generate posts, mimic
people, and subsequently gain trust through friendship requests or ?follows? on
sites such as T

witter, LinkedIn, and Facebook (Bilge et al. 2009). When a user accepts a friendship request, a potential criminal gains personal information such as the user?s location, telephone number, or relationship history, which are normally only available to that user?s accepted friends (Bilge et al. 2009). Many users add so--called friends whom they do not know, which include bots. Unsurprisingly, such privacy--compromising attacks have a high success rate. Past experiments with a social bot exploited 30?40 per cent of users in general (Bilge et al. 2009) and 60 per cent of users who shared a mutual friend with the bot (Boshmaf et al. 2013). Identity--cloning bots have succeeded, on average, in having 56 per cent of their friendship requests accepted on LinkedIn (Bilge et al. 2009). Such identity cloning may raise suspicion due to a user appearing to have multiple accounts on the same site (one real and one forged by a third party). Hence, cloning an identity from one social network to another circumvents these suspicions. In the face of inadequate monitoring, such cross--site identity cloning is an effective tactic (Bilge et al. 2009), as discussed above.

The second technique to gather personal data, which is compatible with (and may even build on) the trust gained via befriending social media users, makes partial use of conversational social bots for social engineering (Alazab and Broadhurst 2016, 12). This occurs when AI ?attempts to manipulate behaviour by building rapport with a victim, then exploiting that emerging relationship to obtain information from or access to their computer? (Chantler and Broadhurst 2008, 65). Although the literature seems to support the efficacy of such bot--based social--engineering, given the currently limited capabilities of conversational AI, scepticism is justified when it comes to automated manipulation on an individual and long--term basis.

Still, as a short--term solution, a criminal may cast a deceptive social--bot net widely enough to discover susceptible individuals. Initial AI--based manipulation may gather harvested personal data and reuse it to produce ?more intense cases of

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simulated familiarity, empathy, and intimacy, leading to greater data revelations? (Graeff 2013a, 5). After gaining initial trust, familiarity, and personal data from a user, the criminal may move the conversation to another context, such as private messaging, where the user assumes that privacy norms are upheld (Graeff 2013b). Crucially, from here, overcoming the conversational deficiencies of AI to engage with the user is feasible using a cyborg; that is, a bot--assisted human (or vice versa) (Chu et al. 2010). Hence, a criminal may make judicious use of the other-wise limited conversational capabilities of AI as a plausible means to gather personal data.

The third technique for gathering personal data from users is automated phishing. Ordinarily, phishing is unsuccessful if the criminal does not sufficiently personalize the messages towards the targeted user. Target--specific and personalized phishing attacks (known as spear phishing), which have been shown to be four times more successful than a generic approach (Jagatic et al. 2007), are labour intensive. However, cost--effective spear phishing is possible using automation (Bilge et al. 2009). Researchers have demonstrated its feasibility through the use of ML

techniques to craft messages personalized to a specific user (Seymour and Tully 2016).

In the second phase of AI--supported banking fraud, AI may support the forging of an identity, including via recent advances in voice synthesis technologies (Bendel 2019). Using the classification and generation capabilities of ML, Adobe?s software can learn adversarially and reproduce someone?s personal and individual speech pattern from a twenty--minute recording of the replicatee?s voice. Bendel argues that AI--supported voice synthesis raises a unique threat in theft and fraud, which

could use VoCo and Co [Adobe?s voice editing and generation software] for biometric security processes and unlock doors, safes, vehicles, and so on, and enter or use them. With the voice of the customer, they [criminals] could talk to the customer?s bank or other institutions to gather sensitive data or to make critical or damaging transactions. All kinds of speech--based security systems could be hacked.? (Bendel 2019, 3)

For years now, credit card fraud has been predominantly an online offence10 that occurs when ?the credit card is used remotely; only the credit card details are needed? (Delamaire, Abdou, and Pointon 2009, 65). Because credit card fraud typically requires neither physical interaction nor embodiment, AI may drive fraud by providing voice synthesis or helping to gather sufficient personal details.

10? Office for National Statistics, ?Crime in England and Wales, Year Ending June 2016. Appendix Tables no. June 2017: 1?60?. https://www.ons.gov.uk/peoplepopulationandcommunity/crimeandjustice/adhocs/006256estimatesofexperiencesandperceptionsofantisocialbehaviourinenglandcrimesurveyenglandandwalescsewyearendingjune2016andselectedyearsfromtheyearendingdecember1996

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In the case of corporate fraud, AI used for detection may also make fraud easier to commit. Specifically,

when the executives who are involved in financial fraud are well aware of the fraud detection techniques and software, which are usually public information and are easy to obtain, they are likely to adapt the methods in which they commit fraud and make it difficult to detect the same, especially by existing techniques.? (Zhou and Kapoor 2011, 571)

More than identifying a specific case of AIC, this use of AI highlights the risks of over--reliance on AI for detecting fraud, which may actually aid fraudsters. These thefts and frauds concern real--world money. A virtual world threat is whether social bots may commit crimes in massively multiplayer online game (MMOG) contexts. The online games often have complex economies where the supply of in--game items is artificially restricted and where intangible in--game goods can have real--world value if players are willing to pay for them. In some cases, items cost in excess of US$1000 (Chen et al. 2004, 1). So it is not surprising that, from a random sample of 613 criminal prosecutions of online game crimes in Taiwan in 2002, virtual property thieves exploited users? compromised credentials 147 times and stole identities 52 times (Chen et al. 2005). Such crimes are analogous to the use of social bots to manage theft and fraud at a large scale on social media sites. The question is whether AI may become implicated in this virtual crime space.

8.4? Possible Solutions

Having outlined and discussed the potential for AIC, the time has come to

analyse
the solutions that are currently available. This is the task of the present section.

8.4.1? Tackling Emergence

There are several legal and technological solutions that can be considered to address the issue of emergent behaviour. Legal solutions may involve limiting agents? autonomy or their deployment. For example, Germany has created de-regu-lated contexts where the testing of self--driving cars is permitted, if the vehicles remain below an unacceptable level of autonomy, ?to collect empirical data and sufficient knowledge to make rational decisions for a number of critical issues? (Pagallo 2017, 7). The solution is that, if legislation does not prohibit higher levels of autonomy for a given AA, the law obliges coupling this liberty with technological remedies to prevent emergent criminal acts or omissions once deployed in the wild.

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One possibility is requiring developers to deploy AAs only when they have runtime legal compliance layers. These layers take the declarative specifications of legal rules and impose constraints on the runtime behaviour of AAs. Whilst still the focus of ongoing research, approaches to runtime legal compliance include architectures for trimming non--compliant AA plans (Meneguzzi and Luck 2009, Vanderelst and Winfield 2018a) and provably correct temporal logic--based formal frameworks that select, trim, or generate AA plans for norm compliance (Van Riemsdijk et al. 2013, Van Riemsdijk et al. 2015, Dennis et al. 2016). In a MAS setting, AIC can emerge from collective behaviour. So MAS--level compliance layers may modify an individual AA?s plans to prevent wrongful collective actions (Uszok et al. 2003, Bradshaw et al. 1997, Tonti et al. 2003). Essentially, such technical solutions propose regimenting compliance?-making non--compliance impossible, at least to the extent that any formal proof is applicable to real--world settings?-with predefined legal rules within a single AA or a MAS (Andrighetto et al. 2013, 4:105).

But the shift in these approaches from mere regulation (which leaves deviation from the norm physically possible) to regimentation may not be desirable when considering the impact on democracy and the legal system. These approaches implement Lessig?s code--as--law concept (Lessig 1999), which considers ?software code as a regulator in and of itself by saying that the architecture it produces can serve as an instrument of social control on those that use it? (Graeff 2013a, 4). However, Hildebrandt (2008, 175) rightly objects:

while computer code generates a kind of normativity similar to law, it lacks?-precisely because it is NOT law?.?.?.?the possibility of contesting its application in a court of law. This is a major deficit in the relationship between law, technology and democracy.

If code--as--law entails a democratic and legal contestation deficit, then a fortiori addressing emergent AIC with a legal reasoning layer comprised of normative but incontestable code (as compared to the contestable law from which it derives) bears the same problems.

Social simulation can address an orthogonal problem whereby an AA owner may choose to operate outside of the law and any such legal reasoning layer requirements (Vanderelst and Winfield 2018b, 4). The basic idea is to use simula-tion as a test bed before deploying AAs in the wild. In a market context, for ex-ample, regulators would
act as

?certification authorities?, running new trading algorithms in the system--simulator to assess their likely impact on overall systemic behavior before allowing the owner/developer of the algorithm to run it ?live?. (Cliff and Northrop 2012, 19)

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Private corporations could fund such extensive social simulations as a common good and as a replacement for (or in addition to) proprietary safety measures (Cliff and Northrop 2012, 21). But a social simulation is a model of an inherently chaotic system, which makes it a poor tool for specific predictions (Edmonds and Gershenson 2015, 12). Still, the idea may nonetheless succeed as it focuses on detecting the strictly qualitative possibility of previously unforeseen and emergent events in a MAS (Edmonds and Gershenson 2015, 13).

8.4.2? Addressing Liability

Although liability is an extensive topic, four models are frequently discussed in relation with AIC (Hallevy 2011, 13): direct liability, perpetration--by--another, command responsibility, and natural--probable--consequence.

The direct liability model ascribes the factual and mental elements to an AA, representing a dramatic shift away from the anthropocentric view of AAs as tools to AAs as (potentially equal) decision--makers (van Lier 2016). Some argue for holding an AA directly liable because ?the process of analysis in AI systems parallels that of human understanding? (Hallevy 2011, 15). We understand the author to mean that, as Dennett (1987) argues, for practical purposes we may treat any agent as if it possesses mental states. However, a fundamental limitation of this model is that AAs do not have (separate) legal personality and agency. We thus cannot hold an AA legally liable in its own capacity regardless of whether this is desirable in practice. Similarly, it has been noted that AAs cannot contest a guilty verdict; ?if a subject cannot take the stand in a court of law it cannot contest the incrimination, which would turn the punishment into discipline? (Hildebrandt 2008, 178). Moreover, AAs cannot legally meet the mental element. This means the ?common legal standpoint excludes robots from any kind of criminal responsibility because they lack psychological components such as intentions or consciousness? (Pagallo 2011, 349).

The lack of actual mental states becomes clear when considering how an AA?s understanding of a symbol (that is, a concept) is limited to its grounding on further syntactic symbols (Taddeo and Floridi 2005, 2007). This leaves the mens rea in limbo. Lack of a guilty mind does not prevent the mental state from being imputed to the AA. In the same way, a corporation could have the mental state of its employees imputed to it and hence be found liable as an organization. But the liability of an AA would still require it to have legal personality. A further problem is that holding an AA solely liable may prove unacceptable, since it would lead to a deresponsabilization of the human agents behind an AA (e.g. an en-gin-eer, a user, or a corporation). This would likely weaken the dissuasive power of criminal law. To ensure that criminal law is effective, we may shift the burden of liabilities onto the humans?-and corporate or other legal agents?-who made a (criminally bad)

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difference to the system. These human agents could be various engine

ers, users,
vendors, and so forth. If the design is poor and the outcome faulty, then all of the human agents involved are deemed responsible (Floridi 2016a). The next two models discussed in the literature move in this direction, focusing on the li-abil-ity of human or other legal persons involved in producing and using the AA.
The perpetration--by--another model (Hallevy 2011, 4), which uses intention as the standard for mens rea, frames the AA as an instrument of crime whereby ?the party orchestrating the offence (the perpetrator--by--another) is the real perpetrator?. Perpetration--by--another leaves ?three human candidates for responsibility before a criminal court: programmers, manufacturers, and users of robots [AAs]? (Pagallo 2017, 21). Clarifying intent is crucial to applying perpetration--by--another. Concerning social media, ?developers who knowingly create social bots to engage in unethical actions are clearly culpable? (de Lima Salge and Berente 2017, 30). For further clarity, Arkin (2008) argues that designers and programmers should be required to ensure that AAs refuse a criminal order, a command that only the deployer can explicitly override. This would remove ambiguity from intent and therefore liability (Arkin and Ulam 2012). To be liable, then, the deployer of an AA must intend the harm by overriding the AA?s default position of ?can but will not do harm?. Together with technological controls and viewing an AA as a mere instrument of AIC, perpetration--by--another addresses those cases when a deployer intends to use an AA to commit an AIC.
The command responsibility model, which uses knowledge as the standard of mens rea, ascribes liability to any military officer who knew (or should have known) about crimes committed by their forces but failed to take reasonable steps to prevent them; in the future, this could include AAs (McAllister 2016). Command responsibility is compatible (or may even be seen as an instance of) perpetration--by--another for use in contexts where there is a chain of command, such as within the military and police forces. This model is normally clear on how ?liability should be distributed among the commanders to the officers in charge of interrogation to the designers of the system? (McAllister 2016, 39). However, ?issues on the undulating waves of increasing complexity in programming, robo--human relationships, and integration into hierarchical structures, call into question these theories? sustainability? (McAllister 2016, 39).
The natural--probable--consequence liability model uses negligence or recklessness as the standard for mens rea. This model addresses AIC cases where an AA developer and user neither intend nor have a priori knowledge of an offence (Hallevy 2011). Liability is ascribed to the developer or user if the harm is a natural and probable consequence of their conduct and if they recklessly or negligently exposed others to the risk (Hallevy 2011), such as in cases of AI--caused emergent market manipulation (Wellman and Rajan 2017).
Natural--probable--consequence and command responsibility are not new concepts. They are both analogous to the respondent superior principle entailed by

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rules as old as Roman law, according to which the owner of an enslaved person was responsible for any damage caused by that person (Floridi 2017c). However, it might not always be

obvious
which programmer was responsible for a particular line of code, or indeed the extent to which the resulting programme was the result of the initial code or the subsequent development of that code by the ML system.? (Williams 2017, 41)
Such ambiguity means that, when emergent AIC is a possibility, some suggest AAs should be banned ?to address matters of control, security, and accountability? (Joh 2016, 18). This would at least make liability for violating such a ban clear. Still, others argue that a possible ban in view of the risk of emerging AIC should be balanced carefully against the risk of hindering innovation; it will therefore be crucial to provide a suitable definition of the standard of negligence (Gless, Silverman, and Weigend 2016). Such a definition would ensure that an all--out ban is not considered to be the only solution, given that it would end up dissuading the design of AAs that compare favourably to people in terms of safety.

8.4.3? Checking the Monitoring

There are four main mechanisms for addressing AIC monitoring. The first is to devise AIC predictors using domain knowledge. This would overcome the limitation of more generic ML classification methods, that is, where the features used for detection can also be used for evasion. Predictors specific to financial fraud can consider institutional properties (Zhou and Kapoor 2011), such as objectives (e.g. whether the benefits outweigh the costs), structure (e.g. a lack of an auditing committee), and the (lack of) moral values in management. But the authors do not say which, if any, of these values are actually predictive. Predictors for identity theft, such as profile cloning, have involved prompting users to consider whether the location of the ?friend? who is messaging them meets their expectation (Bilge et al. 2009).

The second mechanism uses social simulation to discover crime patterns (Wellman and Rajan 2017). Pattern discovery must contend with the sometimes--limited capacity to bind offline identities to online activities. In markets, for example, it takes significant effort to correlate multiple orders with a single legal entity; consequently, ?manipulative algo[rithm]s may be impossible to detect in practice? (Farmer and Skouras 2013, 17). Furthermore, on social media, ?an adversary controls multiple online identities and joins a targeted system under these identities in order to subvert a particular service? (Boshmaf et al. 2012, 4).

The third mechanism addresses traceability by leaving telltale clues in the components that make up AIC instruments. Examples could include physical traces

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left by manufacturers in AA hardware, such as UUVs used to traffic drugs, or fingerprinting in third--party AI software (Sharkey, Goodman, and Ross 2010). Adobe?s voice replication software takes this approach as it places a watermark in the generated audio (Bendel 2019). However, lack of knowledge and control over who develops AI instrument components (used for AIC) limits traceability via watermarking and similar techniques.

The fourth mechanism aims at cross--system monitoring and utilizes self--organization across systems (van Lier 2016). The idea first originated in Luhmann and Bednarz (1995). It begins with the conceptualization (Floridi 2008a) of one system (e.g. a social media site) taking on the role of a moral agent, and a second system (e.g. a marke

t) taking the role of the moral patient.11 A moral patient is
any receiver of moral actions (Floridi 2012b). The conceptualization chosen by
(van Lier 2016) determines that the following are all systems: at the lowest atomic
level, an artificial or human agent; at a higher level, any MAS such as a social
media platform, markets, and so on; and, generalizing further, any system--of--
systems. Hence, any such human, artificial, or mixed system can qualify as a
moral patient or a moral agent.
Whether an agent is indeed a moral one hinges on whether the agent can
undertake actions that are morally qualifiable, not on whether the moral agent
can or should be held morally responsible for those actions(Floridi 2012b).
Adopting this moral agent and moral patient distinction, van Lier proposes a pro-
cess for monitoring and addressing crimes and effects that traverse systems. The
process involves four steps that we describe generically and exemplify specifically:
information--selection of the moral agent?s internal actions for relevance to the
moral patient (e.g. social media posts by users); utterance of the selected informa-
tion from the moral agent to the moral patient (e.g. notifying a financial market
of social media posts); assessment by the moral patient of the normativity of the
uttered actions (e.g. whether social media posts are part of a pump--and--dump
scheme); and feedback given by the moral patient to the moral agent (e.g. notify-
ing a social media site that a user is conducting a pump--and--dump scheme, after
which the social media site should act).
This final step completes a ?feedback loop [that] can create a cycle of machine
learning in which moral elements are simultaneously included? (van Lier 2016,
11). For example, this could involve a social media site learning and adjusting
to the normativity of its behaviour from the perspective of a market. A similar
self--organization process could be used to address other AIC areas. Creating a
11? The adjective ?moral? is taken from the cited work, which considers unethical behaviour as con-
stituting the act of crossing system boundaries. Here, the focus is on criminal acts or omissions that
may have a negative, neutral, or positive ethical evaluation. The use of ?moral? is to avoid misrepre-
senting the cited work, not to imply that the criminal law coincides with ethics.

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profile on Twitter (the moral agent) could have relevance to Facebook (the
moral patient) concerning identity theft (information selection). After notify-
ing Facebook of newly created Twitter profile details (utterance), Facebook
could determine whether the new profile constitutes identity theft by asking
the relevant user (understanding) and notifying Twitter to take appropriate
action (feedback).
8.4.4? Dealing with the Psychology
There are two main concerns about the psychological element of AIC: manipula-
tion of users and (in the case of anthropomorphic AI) the creation of a desire to
commit a crime in a user. Currently, researchers have elaborated on solutions
only for the latter but not for the former.
If anthropomorphic AAs are a problem, there may be two approaches to solv-
ing it. One approach is to ban or restrict anthropomorphic AAs that make it pos-
sible to simulate crime. This position leads to a call for restricting anthropomorphic
AAs in general, because they ?are precisely the sort of robo

ts [AAs] that are most
likely to be abused? (Whitby 2008, 6). Cases whereby social bots are designed, intentionally or not, with a gender in mind?.?.?.?[the] attractiveness and realism of female agents [raise the question: do] ECA?s [that is, social bots] encourage gender stereotypes [and how] will this impact on real women on--line? (De Angeli 2009)

The suggestion is to make it unacceptable for social bots to emulate anthropomorphic properties, such as having a perceived gender or ethnicity. Concerning sexbots that emulate sexual offences, a further suggestion is to enact a ban as a ?package of laws that help to improve social sexual morality? and make norms of intolerance clear (Danaher 2017, 29?30).

A second suggestion, one that is incompatible with the first, is to use anthropomorphic AAs to push back against simulated sexual offences. Concerning the abuse of artificial pedagogical agents, for example, ?we recommend that agent responses should be programmed to prevent or curtail further student abuse? (Veletsianos, Scharber, and Doering 2008, 8). Darling (2015, 14) argues that this would not only ?combat desensitisation and negative externalities from people?s behavior, it would preserve the therapeutic and educational advantages of using certain robots more like companions than tools?.

Implementing these suggestions requires choosing whether to criminalize the demand or supply side of the transaction, or both. Users may be within the scope of application for punishments. At the same time, one could argue that

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as with other crimes involving personal ?vice?, suppliers and distributors could also be targeted on the grounds that they facilitate and encourage the wrongful acts. Indeed, we might exclusively or preferentially target them, as is now done for illicit drugs in many countries.? (Danaher 2017, 33)

8.5? Future Developments

There is still much uncertainty over what we already know about AIC in terms of area--specific threats, general threats, and solutions. AIC research is still in its infancy but, given the preceding analysis, it is possible to outline five dimensions of future AIC research.

8.5.1? Areas of AIC

Understanding areas of AIC better requires extending current knowledge, particularly knowledge about the use of AI in interrogation, theft, and fraud in virtual spaces (e.g. online games with intangible assets that hold real--world value); and AAs committing emergent market manipulation, which research seems to have only studied in experimental simulations. Social engineering attacks are a plausible concern but, for the time being, there is still insufficient evidence of real--world cases. Homicide and terrorism appear to be notably absent from the AIC literature, though they demand attention in view of AI--fuelled technologies such as pattern recognition (e.g. for identifying and manipulating potential -perpetrators, or when members of vulnerable groups are unfairly targeted as suspects), weaponized drones, and self--driving vehicles?-all of which may have both lawful and criminal uses.

8.5.2? Dual--Use

The digital nature of AI facilitates its dual--use (Moor 1985, Floridi 2010a). This means it is feasible that applications designed for legitimate uses could be implemented to commit criminal offences (I shall return to this point in Chapter 12, when discussing deepfakes and fake news). This is the case for UUVs,

for ex-ample.
The further that AI is developed and the more that its implementations become pervasive, the higher the risk of malicious or criminal uses. Left unaddressed, these risks may lead to societal rejection and excessively strict regulation of AI--based technologies. Technological benefits to individuals and societies may be eroded in turn, as the use and development of AI is increasingly constrained. Such limits have already been placed on ML research in visual discriminators of homosexual and heterosexual men (Wang and Kosinski 2018), which was

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considered too dangerous to release in full (i.e. with the source code and learned data structures) to the wider research community?-a limitation that came at the expense of scientific reproducibility. Even when such costly limitations on AI releases are unnecessary, as Adobe demonstrated by embedding watermarks into voice--reproducing technology, external and malevolent developers may nevertheless reproduce the technology in the future. More research is needed to anticipate the dual--use of AI beyond general techniques and the efficacy of policies for restricting the release of AI technologies. This is especially true with the implementation of AI for cybersecurity.

8.5.3? Security

The literature on AIC shows that, within the cybersecurity sphere, AI is taking on a malevolent and offensive role. This role is unfolding in tandem with the development and deployment of defensive AI systems to enhance their resilience in enduring attacks and robustness in averting attacks, along with countering threats as they emerge (Yang et al. 2018). The 2016 DARPA Cyber Grand Challenge was a tipping point for demonstrating the effectiveness of a combined offensive?-defensive AI approach. There, seven AI systems were shown to be capable of identifying and patching their own vulnerabilities while also probing and exploiting those of competing systems. In 2018, IBM launched the Cognitive Security Operations Center. This was the application of an ML algorithm that uses an organization?s structured and unstructured security data, ?including imprecise human language contained in blogs, articles, reports?, to elaborate information about security topics and threats with the goal of improving threat identification, mitigation, and responses.

While policies will obviously play a key role in mitigating and remedying the risks of dual--use after deployment (by defining oversight mechanisms, for instance), these risks are properly addressed at the design stage. Recent reports on malicious AI, such as that by Brundage et al. (2018), suggest how ?one of our best hopes to defend against automated hacking is also via AI?. On the contrary, the analysis developed in this chapter indicates that over--reliance on AI can be counter--productive, too. All of this emphasizes the need for further research not only on AI in cybersecurity, but also on alternatives to AI such as individual and social factors.

8.5.4? Persons

The current debate raises the possibility of psychological factors, like trust, in the criminal role of AI. But research is still lacking on the personal factors that may create perpetrators, such as programmers and users of AI for AIC, in the future.

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Now is the time to invest in longitudinal studies and multivariate anal

yses
spanning the educational, geographical, and cultural backgrounds of both victims and perpetrators or even benevolent AI developers. This will help predict how individuals come together to commit AIC. It can also help understand the -efficacy of ethics courses in computer science programmes, and the capacity to educate users to be less trustful of potentially AI--driven agents in cyberspace.

8.5.5? Organizations

Already in 2017, Europol?s four--year report on the serious and organized crime threat highlighted how the type of technological crime tends to correlate with particular criminal organization topologies. The literature on AIC indicates that AI may play a role in criminal organizations such as drug cartels, which are well resourced and highly organized. Conversely, ad hoc criminal organization on the dark web already takes place under what Europol refers to as crime--as--a--service. Such criminal services are sold directly between buyer and seller, potentially as a smaller element in an overall crime, which AI may fuel (e.g. by enabling profile hacking) in the future.12 On a spectrum ranging from tightly knit to fluid AIC organizations, there exist many possibilities for criminal interaction. Identifying the organizations that are essential or that seem to correlate with different types of AIC will further our understanding of how AIC is structured and operates in practice. Developing our understanding of these four dimensions is essential if we are to successfully track and disrupt the inevitable future growth of AIC. Hopefully, King et al. (2019) and this chapter will spark further research into very serious, growing, yet still relatively unexplored concerns over AIC. The sooner we understand this new crime phenomenon, the earlier we shall be able to put into place preventive, mitigating, disincentivizing, and redressing policies.

8.6? Conclusion: From Evil Uses of AI to Socially Good AI

This chapter provided a systematic analysis of AIC to understand the fundamentally unique and feasible threats that it poses. It addressed this topic based on the classic counterfactual definition of AI discussed in Chapter 1 and in view of the

12? To this end, a cursory search for ?Artificial Intelligence? on prominent dark web markets returned a negative result. Specifically, while developing the analysis presented in King et al. (2019), we checked: ?Dream Market?, ?Silk Road 3.1?, and ?Wallstreet Market?. The negative result is not indicative of AIC--as--a--service?s absence on the dark web, which may exist under a different guise or on more specialized markets. For example, some services offer to extract personal information from a user?s computer. Even if such services are genuine, the underlying technology (e.g. AI--fuelled pattern recognition) remains unknown.

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hypothesis that AI introduces a new form of agency (not intelligence). In the chapter, I have focused on AI as a reservoir of autonomous smart agency, as presented in the first part of the book, and on the ultimate human responsibility for leveraging such artificial agency unethically and illegally. I described the threats area by area in terms of specifically defined crimes, as well as more generally in terms of AI qualities and the issues of emergence, liability, monitoring, and psychology. I then analysed which solutions are available or devisable to deal wi

th
AIC. I approached this topic by focusing on both general and cross--cutting themes, providing an up--to--date picture of the available societal, technological, and legal solutions along with their limitations. This concludes the analysis of criminal, or ?evil?, uses of AI. We can now turn to something much more positive and constructive: the socially good uses of AI, which is the topic of the next chapter.

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9
Good Practices
The Proper Use of AI for Social Good

9.0? Summary

Previously, in Chapter 8, I reviewed the main issues concerning the illegal use of AI or what may be called AI for social evil. This chapter focuses on AI for social good (AI4SG). AI4SG is gaining traction within information societies in general and the AI community in particular. It has the potential to tackle social problems through the development of AI--based solutions. Yet to date, there is only limited understanding of what makes AI socially good in theory, what counts as AI4SG in practice, and how to reproduce its initial successes in terms of policies. This chapter addresses the gap in understanding by first offering a definition of AI4SG, then identifying seven ethical factors that are essential for future AI4SG initiatives. The analysis is supported by some case examples of AI4SG projects, a topic to which I shall return in Chapter 12, when discussing the use of AI to support the United Nation SDGs. Some of the factors discussed in this chapter are almost entirely novel to AI, while the significance of other factors is heightened by the use of AI. Finally, the chapter formulates corresponding best practices from each of these factors. Subject to context and balance, these practices may serve as preliminary guidelines to ensure that well--designed AI is more likely to serve the social good. The chapter does not offer specific recommendations, which are left to Chapter 10.

9.1? Introduction: The Idea of AI for Social Good

The idea of AI4SG is becoming popular in many information societies and gaining traction within the AI community (Hager et al. 2019). Projects seeking to use AI4SG vary significantly. They range from models to predict septic shock (Henry et al. 2015) to game--theoretic models to prevent poaching (Fang et al. 2016); from online reinforcement learning to target HIV education at homeless youths (Yadav, Chan, Xin Jiang, et al. 2016) to probabilistic models to prevent harmful policing (Carton et al. 2016) and support student retention (Lakkaraju et al. 2015). Indeed, new applications of AI4SG appear almost daily, making possible and facilitating

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the attainment of socially good outcomes that were previously unfeasible, un-afford-able, or simply less achievable in terms of efficiency and effectiveness. AI4SG also offers unprecedented opportunities across many domains. These could be of great significance at a time when problems are increasingly global, complex, and interconnected. For example, AI can provide some much--needed support to improve health outcomes and mitigate environmental risks (Wang et al. 2016, Davenport and Kalakota 2019, Puaschunder 2020, Rolnick et al. 2019, Luccioni et al. 2021, Zhou et al. 2020)

. This is also a matter of synergy: AI4SG
builds on and augments other recent examples of digital technologies adopted to advance socially beneficial objectives, such as ?big data for development? (Hilbert 2016, Taylor and Schroeder 2015). As a result, AI4SG is gaining much traction within both the AI community and policymaking circles.

Perhaps because of its novelty and fast growth, AI4SG is still poorly understood as a global phenomenon and lacks a cogent framework for assessing the value and the success of relevant projects. Existing metrics, like profitability or commercial productivity, are clearly indicative of real--world demand, but they remain inadequate. AI4SG needs to be assessed against socially valuable outcomes, much like it happens for ?B Corporation? certification in the for--profit context, or for social enterprises operating in the non--profit sector. AI4SG should be assessed by adopting human and environmental welfare metrics as opposed to financial ones.

Frameworks for the design, development, and deployment of ?ethical AI? in general have recently emerged (see Chapter 4) to offer some guidance in this respect. However, the ethical and social guardrails around AI applications that are explicitly geared towards socially good outcomes are only partially defined. This is because, to date, there is limited understanding of what constitutes AI4SG (Taddeo and Floridi 2018a, Vinuesa et al. 2020, Chui et al. 2018) and what would be a reliable benchmark to assess its success. The best efforts, especially the annual International Telecommunication Union (ITU) Summit on AI for Good and its associated project database (?AI for Good Global Summit? 2019;1 ?AI Repository? 20182), focus on collecting information about, and describing occurrences of, AI4SG. Still, they disregard normative approaches to this phenomenon and are not meant to offer a systematic analysis.

In this and the following chapter, I intend to fill the gap by formalizing a def-in-ition of AI4SG initiatives and the factors that characterize them. In Chapter 12, I shall return to this point to argue that the United Nations? seventeen SDGs -provide a valid framework to benchmark socially good uses of AI technologies.

1? AI for Good Global Summit, 28?31 May 2019, Geneva, Switzerland. ?AI for Good Global Summit?, https://aiforgood.itu.int/.

2? AI Repository, https://www.itu.int/en/ITU--T/AI/Pages/ai--repository.aspx.

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That chapter supports the analysis provided here, introducing a database of AI4SG projects gathered using this benchmark, and discussing several key insights (including the extent to which different SDGs are being addressed). I shall postpone these tasks to the end of the book because the use of AI in support of the UN SDGs provides a valuable conclusion and way forward regarding the future, ethical development of AI.

The ad hoc approach to AI4SG, which has involved analysing specific areas of application (e.g. famine relief or disaster management) at annual summits for AI industry and government,3 indicates the presence of a phenomenon. But it neither explains it nor suggests how other AI4SG solutions could and should be designed to harness the full potential of AI. Furthermore, many projects that generate socially good outcomes using AI are not (self-)described as such (Moore 2019). These shortcomings raise at least two main risks: unanticip

ated failures and
missed opportunities.
First, consider unanticipated failures. Like any other technology, AI solutions are shaped by human values. If not carefully selected and fostered, such values may lead to ?good AI gone awry? scenarios. AI may ?do more harm than good? by amplifying rather than mitigating societal ills. For example, it could widen rather than narrow existing inequities, or exacerbate environmental problems. AI may also simply fail to serve the social good, such as with the failure of IBM?s oncology--support software that attempted to use ML to identify cancerous tumours. The system was trained using synthetic data and US medical protocols, which are not applicable worldwide. As a result, it struggled to interpret ambiguous, nuanced, or otherwise ?messy? patient health records (Strickland 2019), providing misdiagnoses and suggesting erroneous treatment. This led medical practitioners and hospitals to reject the system (Ross and Swetlitz 2017).
Next, consider missed opportunities. AI outcomes that are genuinely socially good may arise from mere accident, such as through the fortuitous application of an AI solution in a different context. This was the case with the use of a different version of IBM?s cognitive system just discussed. In this case, the Watson system was originally designed to identify biological mechanisms. But when used in a classroom setting, it inspired engineering students to solve design problems (Goel et al. 2015). In this positive instance, AI provided a unique mode of education. But the absence of a clear understanding of AI4SG meant that this success was ?accidental?, which means it may not be possible to repeat it systematically or at scale. For each ?accidental success?, there may be countless examples of missed opportunities to exploit the benefits of AI for advancing socially good outcomes
3? ITU (2017). AI for Good Global Summit 2017. https://www.itu.int/en/ITU--T/AI/Pages/201706--default.aspx; ITU (2018). AI for Good Global Summit 2018. https://www.itu.int/en/ITU--T/AI/2018/Pages/default.aspx; AI for Good Global Summit (2019), 28?31 May 2019, Geneva, Switzerland. AI for Good Global Summit. Accessed 12 April 2019. https://aiforgood.itu.int/.

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in different settings. This is especially true when AI--based interventions are developed separately from those who will be most directly subject to their effects, whether defined in terms of area (e.g. residents of a particular region) or domain (e.g. teachers or medical practitioners).
To avoid unnecessary failures and missed opportunities, AI4SG would benefit from an analysis of the essential factors supporting and underwriting the design and deployment of successful AI4SG. This chapter analyses these factors. The goal is not to document every single ethical consideration for an AI4SG project. For example, it is essential, and hopefully self--evident, that an AI4SG project ought not to advance the proliferation of weapons of mass destruction.4 Likewise, it is important to acknowledge at the outset that there are many circumstances in which AI will not be the most effective way to address a particular social problem (Abebe et al. 2020) and would be an unwarranted intervention. This could be due to either the existence of alternative approaches that are less expensive or more efficacious (i.e. ?Not AI for Social Good?) or the una

ce
Haque et al. (2017)
Health
5
The initiatives are especially representative in terms of scope, variety, impact, and for their potentiality to evince the factors that should characterize the design of AI4SG projects.

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throughout the debate, the factors evinced below are more closely associated with design considerations that are specific to AI4SG. They may be operationalized in the form of the corresponding best practices provided for each. In this way, eth-ic-al analysis informing the design and the deployment of AI4SG initiatives has a central role in mitigating the foreseeable risks of unintended consequences and possible misuses of the technology.

Before discussing the factors, it is important to clarify three general features of the whole set: dependency, order, and coherence. The seven factors are often inter-twined and co--dependent but, for the sake of simplicity, I shall discuss them sep-ar-ate-ly. Nothing should be inferred from this choice. In the same way, the factors are all essential; none is ?more important? than any other. So, I shall introduce them not in terms of priority, but somewhat historically. I start with factors that predate AI yet take on greater importance when AI technologies are used, owing to the particular capabilities and risks of AI (Yang et al. 2018).7 These include fal-sifiability and incremental deployment and safeguards against the manipulation of data. There are also factors that relate more intrinsically to the sociotechnical characteristics of AI as it exists today, like situational fairness and human--friendly semanticization.

The factors are ethically robust and pragmatically applicable in the sense that they give rise to design considerations in the form of best practices that should be ethically endorsed. It is crucial to stress here that the seven factors are not by themselves sufficient for socially good AI. Rather, careful consideration of each is necessary. The set of factors identified in this chapter should not be taken as a sort of checklist that, if merely complied with, guarantees socially good outcomes from the use of AI in a particular domain. In the same vein, there is a need to strike a balance between the different factors (and indeed between tensions that may arise even within a single factor). It follows that seeking to frame a project in a binary way as either ?for social good? or ?not for social good? seems needlessly reductive, not to mention subjective. The goal of this chapter is not to identify, or offer the means to identify, AI4SG projects. It is to show the ethically important characteristics of projects that could feasibly be described as AI4SG.

9.3? Seven Essential Factors for Successful AI4SG

As I anticipated, the factors are (1) falsifiability and incremental deployment; (2) safeguards against the manipulation of predictors; (3) receiver--contextualized intervention; (4) receiver--contextualized explanation and transparent purposes;

7? As noted in the introduction, it is not possible to document every single ethical consideration for a social good project. So even the least novel factors here are those that take on new relevance in the context of AI.

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(5) privacy protection and data subject consent; (6) situational fairness; and (7) human--friendly semanticization. This section el

ucidates each factor separately,
with one or more examples, and offers a corresponding best practice.

9.3.1? Falsifiability and Incremental Deployment

Trustworthiness is essential for the adoption of technology in general (Taddeo 2009, Taddeo 2010, Taddeo and Floridi 2011, Taddeo 2017c, Taddeo, McCutcheon, and Floridi 2019) and for AI4SG applications in particular. It is also required for technology to have a meaningful positive impact on human life and environmental well--being. The trustworthiness of an AI application entails a high probability that the application will respect the principle of beneficence (or at the very least, the principle of nonmaleficence). While there is no universal rule or guideline that can ensure or guarantee trustworthiness, falsifiability is an essential factor to improve the trustworthiness of technological applications in general, and AI4SG applications in particular.

Falsifiability entails the specification of one or more critical requirements along with the possibility of empirical testing. A critical requirement is an essential condition, resource, or means necessary for a capability to become fully operational, such that something could or should not work without it. Safety is an obvious critical requirement. For an AI4SG system to be trustworthy, its safety should be falsifiable.8 If falsifiability is not possible, then the critical requirements cannot be checked. In the end, the system should not be deemed trustworthy. This is why falsifiability is an essential factor for all conceivable AI4SG projects.

Unfortunately, we cannot know for sure that a given AI4SG application is safe unless we can test the application in all possible contexts. In this case, the map of testing would simply equate to the territory of deployment. As this reductio ad absurdum makes clear, complete certainty is out of reach. What is within reach, in an uncertain and fuzzy world with many unforeseen situations, is the possibility of knowing when a given critical requirement has not been implemented or may be failing to work properly. We still would not know whether the AI4SG application is trustworthy. But if the critical requirements are falsifiable, we could at least know when it is not trustworthy.

Critical requirements should be tested with an incremental deployment cycle. Unintended hazardous effects may only reveal themselves after testing. At the same time, software should only be tested in the real world if it is safe to do so.

8? It is of course likely that, in practice, an assessment of the safety of an AI system must also take into account wider societal values and cultural beliefs. These (among other considerations) may necessitate different trade--offs between the demands of critical requirements such as safety, and potentially competing norms or expectations.

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This requires adoption of a deployment cycle whereby developers: (a) ensure that the application?s most critical requirements or assumptions are falsifiable, and (b) undertake hypothesis testing of those most critical requirements and assumptions in safe, protected contexts. If these hypotheses are not disproven over a small set of suitable contexts, then (c) they should undertake conduct testing across increasingly wide contexts and/or test a larger set of less critical requirements. All this should be done while (d) being ready to halt or modify

the deploy-
ment as soon as hazardous or other unwanted effects may appear.
AI4SG applications may use formal approaches to try to test critical require-
ments. For example, they may include the use of formal verification to ensure that
autonomous vehicles, and AI systems in other safety--critical contexts, would
make the ethically preferable choice (Dennis et al. 2016). Such methods offer
safety checks that, in terms of falsifiability, can be proved correct. Simulations
may offer roughly similar guarantees. A simulation enables one to test whether
critical requirements (again, consider safety) are met under a set of formal
assumptions. Unlike a formal proof, a simulation cannot always indicate that the
required properties are necessarily always satisfied. But a simulation often enables
one to test a much wider set of cases that cannot be dealt with formally, for
ex-ample due to the complexity of the proof.
It would be misguided to rely purely on formal properties or simulations to
falsify an AI4SG application. The assumptions of these models cage the real--
world applicability of any conclusions that one might make. And assumptions
may be incorrect in reality. What one may prove to be correct via a formal proof,
or likely correct via testing in simulation, may be disproved later with the real--
world deployment of the system. For example, developers of a game--theoretic
model for wildlife security assumed a relatively flat topography without serious
obstructions. Hence, the software that they developed originally had an incorrect
definition of an optimal patrol route. Incremental testing of the application
en-abled the refinement of the optimal patrol route by proving wrong the assump-
tion of a flat topography (Fang et al. 2016).
If novel dilemmas in real--world contexts require the alteration of prior assump-
tions made in the lab, one solution is to rectify a priori assumptions after deploy-
ment. Alternatively, one may adopt an ?on--the--fly? or runtime system for a
constant update of a program?s processing (?understanding?) of its inputs. Yet
problems also abound with this approach. For example, Microsoft?s infamous
Twitter bot Tay (discussed already in Chapter 8) acquired meanings in a very
loose sense at runtime, as it learned from Twitter users how it should respond to
tweets. After deployment in the real?-and frequently vicious?-world of social
media, the bot?s ability to constantly adapt its ?conceptual understanding? became
an unfortunate bug as Tay ?learned? and regurgitated offensive language and
unethical associations between concepts from other users (Neff and Nagy 2016b).

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The use of a retrodictive approach, or an attempt to understand some aspect of
reality through a priori information, to deal with the falsifiability of requirements,
presents similar problems. This is noteworthy since retrodiction is the primary
method of supervised ML approaches that learn from data (e.g. the learning of a
continuous transformation function in the case of neural networks).
From the previous analysis, it follows that the essential factor of falsifiability
and incremental development comprises a cycle: engineering requirements
that are falsifiable, so it is at least possible to know whether the requirements
are not met; falsification testing for incrementally improving levels of trust-
worthi-ness; adjustment of a priori a

ssumptions; then (and only then) deploy-
ment in an incrementally wider and critical context. Germany?s strategy for regulating autonomous vehicles, which was already mentioned in Chapter 8, offers a good example of this incremental approach. Deregulated zones allow for experimentation with constrained autonomy and, after increasing the -levels of trustworthiness, manufacturers may test vehicles with higher levels of autonomy (Pagallo 2017). Indeed, the creation of such deregulated zones (test-strecken) is one recommendation to support more ethical AI policy at the European level, as we shall see in Chapter 15. The identification of this essential factor yields the following best practice:

1)? AI4SG designers should identify falsifiable requirements and test them in incremental steps from the lab to the ?outside world?.

9.3.2? Safeguards against the Manipulation of Predictors

The use of AI to predict future trends or patterns is very popular in AI4SG contexts ranging from applying automated prediction to redress academic failure (Lakkaraju et al. 2015) and preventing illegal policing (Carton et al. 2016) to detecting corporate fraud (Zhou and Kapoor 2011). The predictive power of AI4SG faces two risks: the manipulation of input data and the excessive reliance on non--causal indicators.

The manipulation of data is neither a new problem nor one that is limited to AI systems alone. Well--established findings, such as Goodhart?s Law (Goodhart 1984) (often summarized as ?when a measure becomes a target, it ceases to be a good measure?, as noted in Strathern (1997), 308), long predate the widespread adoption of AI systems. But in the case of AI, the problem of data manipulation may be exacerbated (Manheim and Garrabrant 2018) and lead to unfair outcomes that breach the principle of justice. As such, it is a noteworthy risk for any AI4SG initiative as it can impair the predictive power of AI and lead to the avoidance of socially good interventions at the individual level.

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Consider the concern raised by Ghani (2016) over teachers who face being evaluated with respect to:

the percentage of students in their class who are above a certain risk threshold. If the model was transparent?-for example, heavily reliant on math GPA?-the teacher could inflate math grades and reduce the intermediate risk scores of their students.? (Ghani 2016)

As Ghani goes on to argue, the same concern applies to predictors of adverse police officer interactions:

these systems [are] very easy to understand and interpret, but that also makes them easy to game. An officer who has had two uses of force in the past 80 days may choose to be a bit more careful over the next 10 days, until the count rolls over to zero again.

The two hypothetical examples make clear that, when the model used is an easy one to understand ?on the ground?, it is already open to abuse or ?gaming? regardless of whether AI is used. The introduction of AI complicates matters due to the scale at which AI is typically applied.9 As we have seen, if the information used to predict a given outcome is known, an agent with such information (one that is predicted to take a particular action) can change the value of each predictive variable to avoid an intervention. In this way, the predictive power of the overall model is reduced. Such loss of predictive power has already been shown by empirical resear

ch into the domain of corporate fraud (Zhou and Kapoor 2011).
The phenomenon could carry over from fraud detection to the domains that AI4SG initiatives seek to address, as we saw in Chapter 5.
At the same time, excessive reliance on non--causal indicators (data that cor-relate with but do not cause a phenomenon) could distract attention from the context in which the AI4SG designer is seeking to intervene. To be effective, any such intervention should alter the underlying causes of a given problem, such as a student?s domestic problems or inadequate corporate governance, rather than non--causal predictors. To do otherwise is to risk addressing only a symptom rather than the root cause of a problem.
These risks suggest the need to consider the use of safeguards as a design factor for AI4SG projects. Such safeguards may constrain the selection of indicators for
9? For the sake of simplicity, the focus is on minimizing the spread of information used to predict an outcome. But this is not meant to foreclose the suggestion, offered in Prasad (2018), that in some cases a fairer approach may be to maximize the available information and hence ?democratize? the ability to manipulate predictors.

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use in the design of AI4SG projects, the extent to which these indicators should shape interventions, and/or the level of transparency that should apply to how indicators affect decision. This yields the following best practice:
2)? AI4SG designers should adopt safeguards that (i) ensure non--causal indicators do not inappropriately skew interventions, and when appropriate, limit know-ledge of how inputs affect outputs from AI4SG systems to prevent manipulation.
9.3.3? Receiver--Contextualized Intervention
It is essential that software intervenes in the lives of users only in ways that respect their autonomy. Again, this is not a problem that arises only with AI--driven interventions. But the use of AI introduces new considerations. A core challenge for AI4SG projects is to devise interventions that balance current and future benefits. The balancing problem, which is familiar to preference--elicitation research (Boutilier 2002, Faltings et al. 2004, Chajewska, Koller, and Parr 2000), boils down to a temporal choice interdependency. An intervention in the present can elicit user preferences that then enable the software to contextualize future interventions to the given user. Consequently, an intervention strategy that has no impact on user autonomy (e.g. one that lacks any interventions) may be in-ef-fective in extracting the necessary information for correctly contextualized future interventions. Conversely, an intervention that overly infringes upon user autonomy may cause the user to reject the technology, making future interventions impossible.
This consideration of balance is a common one for AI4SG initiatives. Take, for example, interactive activity recognition software for people with cognitive disabil-ities (Chu et al. 2012). The software is designed to prompt patients to maintain a daily schedule of activities (e.g. taking medication) whilst minimizing interruptions to their wider goals. Each intervention is contextualized in such a way that the software learns the timing of future interventions from responses to past interventions. While only important interventions are made, all interventions are partially optional

rations and out-
comes of these systems explainable but also to make their purposes transparent. These two requirements are intrinsically linked, of course. That is because the operations and outcomes of AI systems reflect the wider purposes of human designers, which this section addresses in turn.

As we saw in Chapter 4, making AI systems explainable is an important ethical principle. It has been a focus of research since at least 1975 (Shortliffe and Buchanan 1975). And given the increasingly pervasive distribution of AI systems, it has gained more attention recently (Mittelstadt et al. 2016, Wachter, Mittelstadt, and Floridi 2017a, b, Thelisson, Padh, and Celis 2017, Watson and Floridi 2020, Watson et al. 2021). As we saw above, AI4SG projects should offer interventions that are contextualized to the receiver. In addition, the explanation for an intervention should also be contextualized for the sake of adequacy and protecting the autonomy of the receiver.

Designers of AI4SG projects have tried to increase the explainability of decision--making systems in various ways. For example, researchers have used ML to predict academic adversity (Lakkaraju et al. 2015). These predictors used concepts that the school officials interpreting the system found familiar and salient, such as GPA scores and socioeconomic categorizations. Researchers have also used reinforcement learning to help officials at homeless shelters educate homeless youths about HIV (Yadav, Chan, Jiang, et al. 2016, Yadav, Chan, Xin Jiang, et al. 2016). The system learns how to maximize the influence of HIV education. It does so by choosing which homeless youths to educate based on the likelihood that a particular homeless youth would pass on their knowledge. One version of the system explained which youth was chosen by revealing their social network graph. However, officials at the homeless shelter found these explanations were counterintuitive. The finding had the potential to affect their understanding of how the system worked and, hence, user trust in the system. These two cases exemplify the importance of the right conceptualization when explaining an AI--based decision.

The right conceptualization is likely to vary across AI4SG projects as they differ greatly in their objectives, subject matter, context, and stakeholders. The conceptual framework (also called the level of abstraction or LoA) depends on what is being explained, to whom, and for what purpose (Floridi 2008a, 2017b). An LoA is a key component of a theory, and hence of any explanation. A theory is comprised of five components:

1. a system, which is the referent or object analysed by a theory;

2. a purpose, which is the ?what for? that motivates the analysis of a system (note that this answers the question ?what is the analysis for?? It should not

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be confused with the purpose of a system, which answers the question ?what is the system for?? Below, I use the term ?goal? for the purpose of a system whenever there may be a risk of confusion);

3. an LoA, which provides a lens through which a system is analysed, to generate

4. a model, which is some relevant and reliable information about the analysed system that identifies

5. a structure of the system, which is comprised of the features that belong to the system under analysis.

There is an interdependency between the choice of the specific

purpose, the
relevant LoA that can fulfil the purpose, the system under analysis, and the model obtained by analysing the system at a specified LoA for a particular purpose. The LoA provides the conceptualization of the system (e.g. GPA scores and socioeconomic backgrounds). At the same time, the purpose constrains the construction of LoAs. For example, we might choose to explain the decision--making system itself (e.g. the use of particular ML techniques). In that case, the LoA can only conceptualize those AI techniques. The LoA then generates the model, which explains the system in turn. The model identifies system structures, such as the GPA score, poor attendance rate, and socioeconomic background of a particular student, as predictors of academic failure. Consequently, designers must carefully choose the purpose and the corresponding LoA. This is so the explanation model can provide the right explanation of the system in question for a given receiver.
A LoA is chosen for a specific purpose. For example, a LoA may be chosen to explain a decision taken based on outcomes obtained through an algorithmic procedure. The LoA varies depending on whether the explanation is meant for the receiver of that decision, or for an engineer responsible for the design of the algorithmic procedure. This is because not every LoA is appropriate for a given receiver. Appropriateness depends on LoA purpose and granularity, for example a customer--friendly vs. engineer--friendly explanation. Sometimes, a receiver?s conceptual view of the world may differ from the one on which the explanation is based. In other cases, a receiver and an explanation may be conceptually aligned. However, the receiver may not agree on the amount of granularity (Sloan and Warner) in the information (what we called more precisely the model) provided. Conceptual misalignment means that the receiver may find the explanation ir-rele-vant, unintelligible, or, as we shall see below, questionable. In terms of (Cowls, Png, and Au) intelligibility, a LoA may use unknown labels (so--called observables) or labels that have different meanings for different users.
Empirical studies suggest that the suitability of an explanation differs among receivers according to their expertise (Gregor and Benbasat 1999). Receivers may require explanations about how the AI software came to a decision, especially

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when they must act based on that decision (Gregor and Benbasat 1999, Watson et al. 2019). How the AI system came to a conclusion can be just as important as the justification for that conclusion. Consequently, designers must also con-textual-ize the method of explanation to the receiver. The case of the software that used influence--maximization algorithms to target homeless youths for HIV education provides a good example of the relevance of the receiver--contextualization of concepts (Yadav, Chan, Jiang, et al. 2016, Yadav, Chan, Xin Jiang, et al. 2016). The researchers involved in that project considered three possible LoAs when designing the explanation model: the first included utility calculations, the second focused on social graph connectivity, and a third focused on pedagogic purpose.
The first LoA highlighted the utility of targeting one homeless youth over another. According to the researchers, this approach could have led the homeless shelter workers (the rec

eivers) to misunderstand the utility calculations or find
them irrelevant. Utility calculations offer little explanatory power beyond the
decision itself as they often simply show that the ?best? choice was made and how
good it was. Explanations based on the second LoA faced a different problem: the
receivers assumed that the most central nodes in the network were the best for
maximizing the influence of education, while the optimal choice is often a set of
less well--connected nodes. This disjuncture may have arisen from the nature of
the connectivity between members of the network of homeless youths, which
reflects real--life uncertainty about friendships. Since who counts as a ?friend? is
often vague and changes over time, the researchers classified edges in the network
as either ?certain? or ?uncertain? based on domain knowledge. For ?uncertain?
-relationships, the probability of a friendship existing between two youths was
determined by domain experts.11
After subsequent user testing of different explanation frameworks, the third
LoA was eventually chosen. Considering their stated goal to justify decisions in a
way that would be intuitive to homeless shelter officials, the researchers con-
sidered omitting references to the Maximum Expected Utility (MEU) calculations?-
even though this is what underlies the decisions made by the system. Instead, the
researchers considered justifying decisions using concepts that were likely more
familiar and therefore comfortable to officials, such as the centrality of the nodes
(i.e. the youths) that the system recommends officials pri-ori-tize for intervention.
In this way, the researchers sought to provide the most relevant information
contextualized to the receiver.
11? Note that the significance of involving domain experts in the process was not merely to improve
their experience as decision recipients. It was also from their unparalleled knowledge of the domain
that the researchers drew upon in the system design, which helped provide the researchers with what
Pagallo (2015) calls a ?preventive understanding? of the field.

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This example shows how given a particular system, the purpose one chooses to
pursue when seeking an explanation for it, the LoA, and the issuing model that is
obtained are crucial variables impacting the effectiveness of an explanation.
Explainability both breeds trust in and fosters the adoption of AI4SG solutions
(Herlocker, Konstan, and Riedl 2000, Swearingen and Sinha 2002, Bilgic and
Mooney 2005). Therefore it is essential that software use persuasive argumentation
for the target audience. Argumentation will likely include information about both
the general functionality and logic employed by a system along with the reasons
for a specific decision.
Transparency in the goal of the system (the system?s purpose) is also crucial as
it follows directly from the principle of autonomy. Consider, for example, the
development of AI solutions to prompt people with cognitive disabilities to take
their medication (Chu et al. 2012). On its face, this application may seem invasive.
After all, it involves vulnerable users and limits the effectiveness of receiver--
conceptualized explanation. But the system is designed neither to coerce the
patients into a given behaviour nor to resemble a human being. Patients have
the autonomy not to interact with the AI system in question. This c

ase highlights
the importance of transparency in goals, particularly in contexts where explainable operations and outcomes are unworkable or undesirable. Transparency in goals thus undergirds other safeguards around the protection of target populations; it may also help ensure compliance with relevant legislation and precedent (Reed 2018). Conversely, opaque goals may prompt misunderstanding and the potential for harm. For instance, users of an AI system may be unclear about what type of agent they are dealing with (human, artificial, or a hybrid combination of both). They may then wrongly assume that the tacit norms of human--to--human social interaction are upheld, such as not recording every detail of a conversation (Kerr 2003). As ever, the social context in which an AI4SG application takes place impacts the extent to which AI systems should be transparent in their operations. Because transparency is the default but not absolute position, there may be valid reasons for designers to obviate informing users of the software?s goals. For ex-ample, the scientific value of a project or the health and safety conditions of a public space may justify temporarily opaque goals. Consider a study that deceived students into believing they were interacting with a human course assistant who, it was realized over time, was actually a bot (Eicher, Polepeddi, and Goel 2017). The authors argued that the bot?s deception lay in playing the ?imitation game? without causing the students to choose simpler, less human--like, natural--language queries based on preconceptions about AI capabilities. In such cases, the choice between opacity and transparency may be informed by pre--existing notions of informed consent for human--subject experiments that are embedded in the Nuremberg Code, the Declaration of Helsinki, and the Belmont Report (Nijhawan et al. 2013).

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More broadly, the ability to avoid the use of an AI system becomes more likely when AI software reveals its endogenous goals, such as classifying data about a person. For example, AI software could inform staff in a hospital ward that it has the goal of classifying their hygiene levels (Haque et al. 2017). In this case, the staff may decide to avoid such classifications if there are reasonable alternative actions they can take. In other cases, revealing a goal makes it less likely that the goal will be fulfilled.
Making transparent the goals and motivations of AI4SG developers themselves is an essential factor to the success of any project. At the same time, transparency may contrast with the very purpose of the system. Therefore it is crucial to assess what level of transparency (i.e. how much, what kind, for whom, and about what?) the project will embrace, given its overall goal and the context for imple-mentation, during the design stage. Taken together with the need for receiver--conceptualized explanation, this consideration yields the following set of best practices:
4)? AI4SG designers should first choose an LoA for AI explanation that fulfils the desired explanatory purpose and is appropriate to both the system and receivers. Then, designers should deploy arguments that are rationally and suitably persuasive for the receivers to deliver the explanation. Finally, designers should ensure that the goal (the system?s purpose) for which an AI4SG system is developed and deploye

d is knowable to the receivers of its outputs by default.

9.3.5? Privacy Protection and Data Subject Consent

Of the seven factors, privacy has the most voluminous literature. This should not be a surprise as privacy is considered an essential condition for safety and social cohesion, among other things (Solove 2008). Also, earlier waves of digital technology have already had a major impact on privacy (Nissenbaum 2009). People?s safety may be compromised when a malicious actor or state gains control over individuals via privacy infringements (Taddeo 2014, Lynskey 2015). Respect for privacy is a necessary condition for human dignity because we can view personal information as constituting an individual. So, deprivatizing records without -consent will likely constitute a violation of human dignity (Floridi 2016c). The conception of individual privacy as a fundamental right underlies recent legislative action in, for example, Europe (through the GDPR) and Japan (through its Act on Protection of Personal Information), as well as judicial decisions in jurisdictions such as India (Mohanty and Bhatia 2017). Privacy supports people when deviating from social norms without causing offence, and communities in maintaining their social structures. So, privacy also undergirds social cohesion.

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In the case of AI4SG, it is particularly important to emphasize the relevance of user consent to the use of personal data. Tensions may arise between different thresholds of consent (Price and Cohen 2019). The tension is often at its most fraught in life--or--death situations, such as national emergencies and pandemics. Consider the 2014 Ebola outbreak in West Africa, which posed a complex ethical dilemma (The Economist 27 October 2014). In that case, the rapid release and analysis of call--data records from cell phone users in the region may have allowed epidemiologists to track the spread of the deadly disease. However, the release of the data was held up over valid concerns around users? privacy and the value of the data to industrial competitors. Related considerations have emerged during the COVID--19 pandemic (Morley, Cowls, et al. 2020).

In circumstances where haste is not so crucial, it is possible to obtain a subject?s consent for, and before, the data being used. The level or type of consent sought can vary with the context. In healthcare, one may adopt an assumed consent threshold whereby the act of reporting a medical issue to a doctor constitutes an assumption of patient consent. In other circumstances, an informed consent threshold will be more appropriate. But since informed consent requires re-searchers to obtain a patient?s specific consent before using their data for a non--consented purpose, practitioners may choose an explicit consent threshold for general data processing (i.e. for any medical usage). This threshold does not require informing the patient about all of the possible ways that researchers may use their data (Etzioni 1999). Another alternative is the evolving notion of ?dynamic consent? whereby individuals can monitor and adjust their privacy preferences on a granular level (Kaye et al. 2015).

In other cases, informed consent may be waived altogether. This was the case with the recent creation of ML software to predict the prognosis of ovarian cancer sufferers by drawing upon a retrospective analysis of anonymized images (

Lu et al. 2019). The use of patient health data in the development of AI solutions without patient consent has attracted the attention of data protection regulators. In 2017, the UK Information Commissioner ruled that the Royal Free National Health Service (NHS) Foundation Trust violated the Data Protection Act when it provided patient details to Google DeepMind for the purposes of training an AI system to diagnose acute kidney injury (Burgess 2017). The commissioner noted a ?shortcoming? in how ?patients were not adequately informed that their data would be used as part of the test? (Information Commissioner?s Office 2017). However, striking a balance between respecting patient privacy and creating effective AI4SG is still possible. This was the challenge faced by the researchers in Haque et al. (2017), who wanted to create a system for tracking compliance with rules around hand hygiene in hospitals to prevent the spread of infections. Despite the clear technical advantages of taking a computer vision--based approach to the problem, the use of video recording runs up against privacy regulations constraining it. Even in cases where video recording is allowed, access to

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the recordings (to train an algorithm) is often strict. Instead, the researchers resorted to ?depth images? that de--identify subjects, preserving their privacy. While this design choice meant ?losing important visual appearance cues in the process?, it satisfied privacy rules and the researchers? non--intrusive system still managed to outperform existing solutions.

Finally, consent in the online space is also problematic. Users often lack the choice to consent and are presented with a ?take it or leave it? option when accessing online services (Nissenbaum 2011, Taddeo and Floridi 2015). A relative lack of protection or consent for the second--hand use of personal data that are publicly shared online enables the development of ethically problematic AI software. For instance, a recent paper used publicly available images of faces uploaded to a dating website as a way to train AI software to detect someone?s sexuality based on a small number of photos (Wang and Kosinski 2018). While the study received ethics committee approval, it raises further questions around consent as it is implausible that the users of the dating website could or necessarily would have consented to the use of their data for this particular purpose.

Privacy is not a novel problem. However, the centrality of personal data to many AI (and AI4SG) applications heightens its ethical significance and creates issues around consent (Taddeo and Floridi 2018a). From this we can derive the following best practice:

5)? AI4SG designers should respect the threshold of consent established for processing datasets of personal data.

9.3.6? Situational Fairness

AI developers typically rely on data. Data, in turn, could be biased in ways that are socially significant. This bias can carry over to algorithmic decision making, which underpins many AI systems, in ways that are unfair to the subjects of the decision--making process (Caliskan, Bryson, and Narayanan 2017). In doing so, it may breach the principle of justice. Decisions may be based on factors of ethical importance (e.g. ethnic, gender, or religious grounds) and irrelevant to the decision making at hand, or they may be relevant but legally

protected as a non--
discriminatory characteristic (Friedman and Nissenbaum 1996). Moreover, AI--driven decisions can be amalgamated from factors that are not of obvious ethical importance, yet collectively constitute unfairly biased decision making (Pedreshi, Ruggieri, and Turini 2008, Floridi 2012b).
AI4SG initiatives relying on biased data may propagate this bias through a vicious cycle (Yang et al. 2018). Such a cycle would begin with a biased dataset informing a first phase of AI decision--making, which would result in dis-crim-in-atory actions. This would then lead to the collection and use of biased data in

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turn. Consider the use of AI to predict preterm birth in the United States, where the health outcomes of pregnant women have long been affected by their eth-ni-city. Due to harmful historical stereotypes, longstanding bias against African--American women seeking medical treatment contributes to a maternal morbidity rate that is over three times higher than that of white women (CDC 2019). Here, AI may offer great potential to reduce this stark racial divide?-but only if the same historical discrimination is not replicated in AI systems (Banjo 2018). Or consider the use of predictive policing software: developers may train predictive policing software on policing data that contains deeply ingrained prejudices. When discrimination affects arrest rates, it becomes embedded in prosecution data (Lum and Isaac 2016). Such biases may cause discriminatory decisions, such as warnings or arrests, that feed back into the increasingly biased datasets (Crawford 2016) and thereby complete a vicious cycle.
The previous examples involve the use of AI to improve outcomes in domains where data were already collected. Yet in many other contexts, AI4SG projects (or indeed similar initiatives) are effectively making citizens ?visible? in ways that they previously were not, including in Global South contexts (Taylor and Broeders 2015). This increased visibility stresses the importance of protecting against the potential amplification of harmful bias by AI technologies.
Clearly, designers must sanitize the datasets used to train AI. But there is equally a risk of applying too strong a disinfectant, so to speak, by removing important contextual nuances that could improve ethical decision making. So, designers must also ensure that AI decision making maintains sensitivity to fac-tors that are important for inclusiveness. For instance, one should ensure that a word processor interacts identically with a human user regardless of that user?s gender and ethnicity. But one should also expect that it may operate in an unequal--yet--equitable way by aiding people with visual impairments.
Such expectations are not always met in the context of AI--driven reasoning. Compared to the word processor, AI makes possible a far wider range of decision--making and interaction modalities. Many of these are driven by potentially biased data. Training datasets may contain natural language that carries unfair as-so-ci-ations between genders and words that carry normative power in turn (Caliskan, Bryson, and Narayanan 2017). In other contexts and use cases, an equitable approach may require differences in communication based on factors such as gender. Consider the case of the virtual teaching assistant that failed to dis-crim-in-ate sufficiently well between men and w

omen in its responses to being told that a
user was expecting a baby, congratulating the men and ignoring the women
(Eicher, Polepeddi, and Goel 2017). A BBC News investigation highlighted an
even more egregious example: a mental health chatbot deemed suitable for use by
children was unable to understand a child explicitly reporting underage sexual
abuse (White 2018). As these cases make clear, the use of AI in human?-computer
interactions (such as chatbots) requires the correct understanding of both the

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salient groups to which a user belongs and the characteristics the user embodies
when they interact with the software.
Respecting situational fairness is essential for the successful implementation of
AI4SG. To achieve it, AI4SG projects need to remove the factors and their proxies
that are of ethical importance but irrelevant to an outcome. But the projects also
need to include the same factors when these are required, whether for the sake of
inclusiveness, safety, or other ethical considerations. The problem of historical
biases affecting future decision making is an old one. What is new is the potential
that these biases will be embedded in, strengthened, and perpetuated anew by
erroneous reinforcement learning mechanisms. This risk is especially pronounced
when considered alongside the risk of opacity in AI decision--making systems and
their outcomes. We will return to this topic in the next section.
The identification of situational fairness as an essential factor yields now the
following best practice:
6)? AI4SG designers should remove from relevant datasets variables and proxies
that are irrelevant to an outcome, except when their inclusion supports inclusivity,
safety, or other ethical imperatives.
9.3.7? Human--Friendly Semanticization
AI4SG must allow humans to curate and foster their ?semantic capital?, which,
as I wrote in Floridi et al. (2018), is ?any content that can enhance someone?s power
to give meaning to and make sense of (semanticise) something?. This is crucial to
maintain and foster human autonomy. With AI, we may often have the technical
capacity to automate meaning- and sense--creation (semanticization), but mis-
trust or unfairness may also arise if we do so carelessly. From this, two prob-
lems emerge.
The first problem is that AI software may define semanticization in a way that
diverges from our own choices. This is the case when a procedure arbitrarily
defines meanings, for example based on a coin toss. The same problem may arise
if AI software supports some kind of semanticization based on pre--existing uses.
For example, researchers have developed an application that predicts the legal
meaning of ?violation? based on past cases (Al--Abdulkarim, Atkinson, and Bench--
Capon 2015). If one used the software to define the meaning of ?violation?,12 then
one would end up limiting the role of judges and justices. They would no longer
be able to semanticize (refine and redefine the meaning and the possibility of
making sense of) ?violation? when they interpret the law. This is a problem because
12? There is no suggestion that this is the intended use.

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past usage does not always predict how we would semanticize the same concepts
or phenomena in the future.
The second problem is that, in a social setting, it would be impractical for

estions to arise from the factors described in this chapter
are likely to concern this challenge of balancing the competing needs and claims
that the factors and corresponding best practices introduce. This concerns what it
is that legitimates decision making with and about AI. I shall return to this topic
in Chapter 10 as a matter of ethical discussion (for a discussion of its political
implications in terms of digital sovereignty, see Floridi (2020b). Here I offer some
remarks by way of conclusion.
Questions about trade--offs, balances, and their legitimacy are inevitably inter-
twined with wider ethical and political challenges regarding who has the power
or ?standing? to participate in this process of evaluation, as well as how multiple
preferences are measured and aggregated, which Baum outlines in a trichotomic
framework (Baum 2020). If we assume that the challenge of balancing factors
ought to be at least somewhat participatory in nature, the overview of relevant
social choice theorems in Prasad (2018) identifies several background conditions
to support effective group decision making. As these analyses suggest, the in-corp-
or-ation of multiple perspectives into the design of AI decision--making systems is
likely to be an ethically important step both for AI in general and for AI4SG in
particular. So will be efforts to implement forms of co--design.

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There is clearly much work still to be done to ensure that AI4SG projects are
designed in ways that do not merely advance beneficial goals and address societal
challenges, but do so in socially preferable (equitable) and sustainable ways.
In this chapter, we looked at the foundation for good practices and policies. We
also looked at the foundation for further research into the ethical considerations
that should undergird AI4SG projects, and hence the ?AI4SG project? at large. The
time has come to discuss how ethical, legal, and policymaking considerations
interact in such a development, in terms of specific recommendations. This is the
topic of the next chapter.

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10
How to Deliver a Good AI Society
Some Recommendations
10.0? Summary
Previously, in Chapters 4?6, I analysed the foundational concepts that can ground
a future ?Good AI Society?. Chapters 7?9 discussed the challenges, the bad, and
the good practices characterizing the use of AI systems. This chapter, along with
the next, turns to some constructive and concrete recommendations about how
to assess, develop, incentivize, and support good AI. In some cases, these
-recommendations may be undertaken directly by national or supranational
pol-icy-makers. In others, changes may be led by other stakeholders ranging
from civil society to private actors and sectoral organizations. The hope is that,
if adopted, these recommendations may support a firm foundation for the
establishment of a Good AI Society.
10.1? Introduction: Four Ways of Delivering a Good AI Society
AI is not another utility that needs to be regulated once it is mature. We made the
mistake of thinking so when the Web began to develop in the 1990s and today the
mess is obvious. The Web was always going to be a new environment?-part of our
infosp

vy--handed or misconceived regulation,
under--investment, or a public backlash akin to that faced by genetically modified crops and nuclear power. As a result, the benefits offered by AI technologies may not be fully realized by society. These dangers arise largely from unintended consequences and relate typically to good intentions gone awry. And of course, one must also consider the risks associated with inadvertent overuse or wilful misuse of AI technologies, grounded, for example, in misaligned incentives, greed, adversarial geopolitics, or malicious intent. As we saw in Chapters 7 and 8, everything from email scams to full--scale cyber warfare may be accelerated or intensified by the malicious use of AI technologies (Taddeo 2017b). And new evils may be made possible. The possibility of social progress represented by the aforementioned opportunities must be weighed against the risk that malicious manipulation will be enabled or enhanced by AI. Yet a broad risk is that AI may be underused out of fear of overuse, or misused for lack of a clear ethical and legal framework?-as we shall see in the rest of this chapter.

10.2? Who We Can Become: Enabling Human Self--Realization without Devaluing Human Abilities

AI may enable self--realization, that is, the ability for people to flourish in terms of their own characteristics, interests, potential abilities or skills, aspirations, and life projects. Much like inventions such as the washing machine, which liberated -people (particularly women) from the drudgery of domestic work, the ?smart? automation of other mundane aspects of life may free up yet more time for cultural, intellectual, and social pursuits, and more interesting and rewarding work. More AI could easily mean more human life spent more intelligently. The risk in this case is not the obsolescence of some old skills and the emergence of new ones per se, but the pace at which this is happening and the unequal distributions of the costs and benefits that result.

A very fast devaluation of old skills, and hence a quick disruption of the job market and the nature of employment, can be seen at the level of both the individual and society.

At the level of the individual, jobs are often intimately linked to personal identity, self--esteem, and social role or standing. These are all factors that may be adversely affected by redundancy, even putting to one side the potential for severe economic harm. At the level of society, deskilling in sensitive, skill--intensive domains (such as health care diagnosis or aviation) may create dangerous vulnerabilities in the event of AI malfunction or an adversarial attack. Fostering the

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development of AI in support of new abilities and skills while also anticipating and mitigating its impact on old ones will require both close study and potentially radical ideas, such as the proposal for some form of ?universal basic income? (which is already growing in popularity and experimental use). In the end, we need some intergenerational solidarity between those disadvantaged today and those advantaged tomorrow to ensure that the disruptive transition between the present and the future will be as fair as possible for everyone. To adopt a ter-min-ology that we have sadly come to appreciate during the COVID--19 -pandemic, we need to flatten the curve of the social impact of digital tech

nologies?-
AI included.

10.3? What We Can Do: Enhancing Human Agency without Removing Human Responsibility

AI is providing a growing reservoir of ?smart agency?. Put at the service of human intelligence, such a resource can hugely enhance human agency. We can do more, better, and faster, thanks to the support provided by AI. In this sense of ?[human] augmented intelligence?, AI could be compared to the impact that engines have had on our lives. The larger the number of people who will enjoy the op-por-tun-ities and benefits of such a reservoir of smart agency ?on tap?, the better our so-ci-eties will be. Responsibility is therefore essential, in view of what sort of AI we develop, how we use it, and whether we share with everyone its advantages and benefits.

Obviously, the corresponding risk is the absence of such responsibility. This may happen not just because we have the wrong socio--political framework, but also due to a ?black box? mentality according to which AI systems for decision making are seen as being beyond human understanding?-and hence, beyond human control. These concerns apply not just to high--profile cases, such as deaths caused by autonomous vehicles. They also apply to more commonplace but still significant uses, such as in automated decisions about parole or creditworthiness. Yet the relationship between the degree and quality of agency that people enjoy and how much agency we delegate to autonomous systems is not zero sum, either pragmatically or ethically. In fact, if developed thoughtfully, AI offers the oppor-tunity to improve and multiply the possibilities for human agency. Consider examples of ?distributed morality? in human--to--human systems such as peer--to--peer lending (Floridi 2013). Human agency may be ultimately supported, refined, and expanded by the embedding of ?facilitating frameworks? (designed to improve the likelihood of morally good outcomes) in the set of functions that we delegate to AI systems. If designed effectively, AI systems could amplify and strengthen shared moral systems.

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10.4? What We Can Achieve: Increasing Societal Capabilities without Reducing Human Control

AI offers many opportunities for improving and augmenting the capabilities of individuals and society at large. Whether by preventing and curing diseases or optimizing transportation and logistics, the use of AI technologies presents countless possibilities for reinventing society by radically enhancing what humans are collectively capable of. More AI may support better collaboration, and hence more ambitious goals. Human intelligence augmented by AI could find new solutions to old and new problems ranging from a fairer or more efficient distribution of resources to a more sustainable approach to consumption. Precisely because such technologies have the potential to be so powerful and dis-ruptive, they also introduce proportionate risks. Increasingly, there will be pro-cesses in which we may not need to be either ?in or on the loop? (that is, as part of the process or at least in control of it) if we can delegate our tasks to AI. But if we rely on the use of AI technologies to augment our own abilities in the wrong way, we may delegate important tasks. Above all, we may delegate crucial decisions to autonomous systems that should remain at least partly subject to human supervision, choice, and rectification. This in

turn may reduce our ability to
monitor the performance of these systems (by no longer being ?on the loop? either) or prevent or redress the errors and harms that arise (?post loop?). It is also possible that these potential harms may accumulate and become entrenched as more and more functions are delegated to artificial systems. It is therefore imperative to strike a balance between pursuing the ambitious opportunities offered by AI to improve human life and what we can achieve on the one hand, and ensuring that we remain in control of these major developments and their effects on the other.

10.5? How We Can Interact: Cultivating Societal Cohesion without Eroding Human Self--Determination

From climate change and antimicrobial resistance to nuclear proliferation, wars, and fundamentalism, global problems increasingly involve high degrees of co-ord-in-ation complexity. This means they can be tackled successfully only if all stake-holders co--design and co--own the solutions and cooperate to bring them about. AI can hugely help to deal with such coordination complexity with its data--intensive, algorithmic--driven solutions, supporting more societal cohesion and collaboration. For example, we shall see in Chapter 12 efforts to tackle climate change have exposed the challenge of creating a cohesive response, both within societies and between them.

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In fact, the scale of the challenge is such that we may soon need to decide how to strike a balance between engineering the climate directly and designing societal frameworks to encourage a drastic cut in harmful emissions. This latter option could be undergirded by an algorithmic system to cultivate societal cohe-sion. Such a system should not be imposed from the outside; it should be the result of a self--imposed choice, not unlike our choice to not buy chocolate if we had earlier chosen to go on a diet, or to set an alarm to wake up early. ?Self--nudging? to behave in socially preferable ways is the best form of nudging, and the only one that preserves autonomy (Floridi 2015c, 2016f). It is the outcome of human decisions and choices, yet it can rely on AI solutions to be implemented and facilitated. But the risk is that AI systems may erode human self--determination as they may lead to unplanned and unwelcome changes in human behaviours to accommodate the routines that make automation work and people?s lives easier. AI?s predictive power and relentless nudging, even if unintentional, should be at the service of human self--determination. It should foster societal cohesion, not undermine human dignity or human flourishing.

10.6? Twenty Recommendations for a Good AI Society

Taken together along with their corresponding challenges, the four opportunities outlined above paint a mixed picture about the impact of AI on society and the people in it, and the overall environments they share. Accepting the presence of trade--offs and seizing the opportunities while working to anticipate, avoid, or minimize the risks head--on will improve the prospect for AI technologies to promote human dignity and flourishing. Ensuring that the outcomes of AI are socially preferable (equitable) depends on resolving the tension between in-corp-or-at-ing the benefits and mitigating the potential harms of AI?-in short, sim-ul-tan-eous-ly avoiding the misuse and underuse of these technologies.

In thi

s context, the value of an ethical approach to AI technologies comes into starker relief. In Chapter 6, I argued that compliance with the law is merely necessary (the least that is required), but significantly insufficient (not the most that can be done). It is the difference between playing according to the rules and playing well to win the game, so to speak. We need an ethical strategy, a soft ethics for the digital society we wish to build. In line with this distinction, what follows are twenty recommendations for a Good AI Society. There are four general areas for action to assess, to develop, to incentivize, and to support. Some recommendations may be undertaken directly, by national or European policymakers, for example, working in collaboration with stakeholders where appropriate. For other recommendations, policymakers may play an enabling role to support efforts undertaken or led by third parties.

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The assumption is that, to create a Good AI Society, the ethical principles identified in Chapter 4 should be embedded in the default practices of AI. In particular, we saw in Chapter 9 that AI can and should be designed and developed in ways that decrease inequality and further social empowerment, with respect for human dignity and autonomy, to increase the benefits that are equitably shared by all. It is especially important that AI be explicable as explicability is a critical tool for building public trust in, and understanding of, the technology. Creating a Good AI Society requires a multi--stakeholder approach. This is the most effective way to ensure that AI will serve the needs of society by enabling developers, users, and rule makers to all be on board, collaborating from the outset. Inevitably, different cultural frameworks inform attitudes to new technology.

A final comment before presenting the recommendations: the following European approach is meant to complement other approaches. It should not be misunderstood for some sort of endorsement of a Eurocentric perspective. In the same way that I needed to ground ethical considerations in Chapter 8 within a specific legal framework (English criminal law, in that case) without loss of generality, the choice to contextualize the following recommendations within an EU setting is likewise only meant to provide them with a concrete and actionable value. There is nothing in the recommendations that make them particularly EU--only or EU--centred, and they remain universalizable. This is because no matter where we live in the world, we should all be committed to the development of AI technologies in a way that secures people?s trust, serves the public interest, strengthens shared social responsibility, and supports the environment.

1) Assess the capacity of existing institutions, such as national civil courts, to redress the mistakes made or harms inflicted by AI systems. This assessment should evaluate the presence of sustainable, majority--agreed foundations for liability from the design stage onwards to reduce negligence and conflicts (see also Recommendation 5).3

2) Assess which tasks and decision--making functionalities should not be -delegated to AI systems using participatory mechanisms to ensure alignment with societal values and understanding of public opinion. This assessment should consider existing legislation and be supported by ongoing dialogue bet

ween all stakeholders (including government, industry, and civil society), to debate how AI will impact society (in concert with Recommendation 17).

3) Assess whether current regulations are sufficiently grounded in ethics to provide a legislative framework that can keep pace with technological

3? The task of determining accountability and responsibility may be borrowed from lawyers in ancient Rome who would go by the formula ?cuius commoda eius et incommoda? (?the person who derives an advantage from a situation must also bear the inconvenience?). A good 2,200--year--old principle supported by a well--established tradition and elaboration could properly set the starting level of abstraction in this field.

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developments. This may include a framework of key principles that would be applicable to urgent and/or unanticipated problems.

4) Develop a framework to enhance the explicability of AI systems that make socially significant decisions. Central to this framework is the ability for individuals to obtain a factual, direct, and clear explanation of the decision--making process, especially in the event of unwanted consequences. This is likely to require the development of frameworks specific to different industries; professional associations should be involved in this process alongside experts in science, business, law, and ethics.

5) Develop appropriate legal procedures and improve the digital infrastructure of the justice system to permit the scrutiny of algorithmic decisions in court. This is likely to include the creation of a framework for AI explainability (as indicated in Recommendation 4) specific to the legal system. Examples of appropriate procedures could include the applicable disclosure of sensitive commercial information in intellectual property (IP) litigation. Where disclosure poses unacceptable risks (such as to national security), procedures should include the configuration of AI systems to adopt technical solutions by default (such as zero--knowledge proofs) to evaluate their trustworthiness.

6) Develop auditing mechanisms for AI systems to identify unwanted -consequences, such as unfair bias. Auditing should also (perhaps in co-oper-ation with the insurance sector) include a solidarity mechanism to deal with severe risks in AI--intensive sectors. Those risks could be mitigated by multi--stakeholder mechanisms upstream. Pre--digital experience indicates that, in some cases, it may take a couple of decades before society catches up with technology by way of rebalancing rights and protection ad-equate-ly to restore trust. The earlier that users and governments become involved (as made possible by digital technologies), the shorter this lag will be.

7) Develop a redress process or mechanism to remedy or compensate for a wrong or grievance caused by AI. To foster public trust in AI, society needs a widely accessible and reliable mechanism of redress for harms inflicted, costs incurred, or other grievances caused by the technology. Such a mechanism will necessarily involve a clear and comprehensive allocation of accountability to humans and/or organizations. Lessons could be learnt from the aerospace industry, for example, which has a proven system for handling unwanted consequences thoroughly and seriously. The development of this process must follow from the assessment of existin

I for society analysed in Chapters 6?9. They are formulated in the spirit of collaboration and in the interest of creating concrete and constructive responses to the most pressing social challenges posed by AI. With the rapid pace of technological change, it is tempting to view the political process of contemporary liberal democracies as old--fashioned, out of step, and no longer up to the task of preserving the values and promoting the interests of society and everyone in it. I disagree. The recommendations offered here, which include the creation of centres, agencies, curricula, and other infrastructure, support the case for an ambitious, inclusive, equitable, and sustainable programme of policymaking and technological innovation. This will contribute to securing the benefits and mitigating the risks of AI for all people, as well as for the world that we share. This is also the goal pursued by the next two chapters on the use of AI in support of the UN SDGs and the impact of AI on climate change.

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11
The Gambit
AI Impact on Climate Change

11.0? Summary

Previously, in Chapter 4, I presented the ethical principles that provide a framework for AI. One of these principles, beneficence, includes sustaining the planet. This and the following chapter further explore that requirement by analysing the positive and negative environmental impacts of AI. The goal is to provide policy recommendations for a path towards a greener and more climate--friendly development of AI?-development especially in line with EU values and legislation. The timing is critical because AI is already used today to model events related to climate change and contribute to efforts in combating global warming. It can thus be a great positive force for a fair and sustainable society.

11.1? Introduction: AI?s Double--Edged Power

We saw in the previous chapters how AI is a new kind of agency that can be -harnessed to solve problems and perform tasks with unmatched success. But it is equally clear that such a new and powerful agency needs to be developed and governed ethically to avoid, minimize, and rectify any negative impact it may have, and to ensure that it benefits humanity and the planet. This holds true also, or perhaps particularly, when it comes to climate change, which is the biggest threat facing humanity in our century (Cowls et al. 2021a). On the one hand, AI can be an extremely powerful tool in the fight against climate change and we need all the help we can get (Ramchurn et al. 2012, Rolnick et al. 2019). On the other hand, there are significant pitfalls (both ethical and environmental) that must be avoided (Mendelsohn, Dinar, and Williams 2006, Anthony, Kanding, and Selvan 2020, Cowls et al. 2021a, Malmodin and Lundén 2018). Understanding these pitfalls is crucial to this end and concrete policy steps can help, as we shall see in this chapter. More specifically, AI presents two crucial opportunities.

The first opportunity is epistemological. AI can help improve and expand our current understanding of climate change by making it possible to process immense volumes of data. This will permit study of existing climate trends, the forecasting of future developments, and predictions about the impact and success (or la

ck thereof)

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of policies. AI techniques have been used to forecast global mean temperature changes (Ise and Oba 2019, Cifuentes et al. 2020); predict climactic and oceanic phenomena such as El Niño (Ham, Kim, and Luo 2019), cloud systems (Rasp, Pritchard, and Gentine 2018), and tropical instability waves (Zheng et al. 2020); better understand aspects of the weather system like rainfall, both generally (Sønderby et al. 2020, Larraondo et al. 2020) and in specific locales such as Malaysia (Ridwan et al. 2020); and further study their knock--on consequences, such as water demand (Shrestha, Manandhar, and Shrestha 2020, Xenochristou et al. 2020). AI tools can also help anticipate the extreme weather events that are more common due to global climate change. For example, AI tools have been used to anticipate heavy rain damage (Choi et al. 2018), wildfires (Jaafari et al. 2019), and downstream consequences such as patterns of human migration (Robinson and Dilkina 2018). In many cases, AI techniques can help improve or expedite existing forecasting and prediction systems. They can automatically label climate modelling data (Chattopadhyay, Hassanzadeh, and Pasha 2020), improve ap-proxi-ma-tions for simulating the atmosphere (Gagne et al. 2020), and separate signals from noise in climate observations (Barnes et al. 2019). All of this is part of an even more general trend: there is virtually no scientific analysis and evidence--based policy that is not powered by advanced digital technologies. AI is becoming part of the necessary tools to advance our scientific understanding in many fields.

The second opportunity is ethical. AI can help deliver greener and more -ef-fect-ive solutions, such as improving and optimizing energy generation and use. In this sense, it can proactively contribute to the fight against global warming. Combating climate change effectively requires a vast array of responses related to mitigating the existing effects of climate change and reducing emissions through decarbonization to prevent further warming. For example, a 2018 Microsoft/PwC report estimated that using AI for environmental applications could boost global GDP by between 3.1 and 4.4 per cent; at the same time, the use of AI in this con-text would also reduce greenhouse gas (GHG) emissions anywhere from 1.5 to 4 per cent by 2030, compared to a ?business as usual? scenario (Microsoft 2018, 8). In fact, an array of AI--based techniques already plays a key role in many of these responses (Inderwildi et al. 2020, Sayed--Mouchaweh 2020). This includes energy efficiency in industry, for example, especially in the petrochemical sector (Narciso and Martins 2020). Studies have also used AI to understand industrial pollution in China (Zhou et al. 2016), the carbon footprint of concrete used in construction (Thilakarathna et al. 2020), and even energy efficiency in shipping (Perera, Mo, and Soares 2016). Other work has explored the use of AI in electrical grid management (Di Piazza et al. 2020), to forecast building energy usage (Fathi et al. 2020), and to assess the sustainability of food consumption (Abdella et al. 2020). AI can help to predict carbon emissions based on present trends (Mardani et al. 2020, Wei, Yuwei, and Chongchong 2018) along with the impact of interven-tionist policies like a carbon tax (Abrell, Kosch, and Rausch 2019) or carbon

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trading systems (Lu et al. 2020). AI could further be used to help monitor the active removal of carbon from the atmosphere through sequestration (Menad et al. 2019). Beyond this indicative evidence, the growing use of AI to fight climate change can also be seen from the higher vantage point of major institutions and large--scale initiatives. The European Lab for Learning & Intelligent Systems (ELLIS) has a Machine Learning for Earth and Climate Sciences programme that aims to ?model and understand the Earth system with Machine Learning and Process Understanding?.1 The European Space Agency has also established the Digital Twin Earth Challenge to provide ?forecasting on the impact of climate change and responding to societal challenges?.2 Several European universities have initiatives and training programmes dedicated to unlocking the power of AI for climate.3,?4 In 2020, a search of Cordis (the European database for funded research) for current projects addressing climate change and AI returned a total of 122 results (Cowls et al. 2021a).5 Analysis of these 122 projects suggested they represent both geographic and disciplinary breadth. The projects were well spread across the continent, albeit with a clear skew towards western Europe in terms of where they were coordinated. A large majority of projects were related to the natural sciences and/or engineering and technology, but a considerable number were also anchored in social sciences. The breadth of subjects that these projects touched on was vast, spanning domains as diverse as viticulture, mycology, and galactic astronomy.

There is also considerable evidence of private and non--profit initiatives using AI to combat climate change around the world. Microsoft?s AI for Earth is a five--year, $50 million initiative established in 2017 to support organizations and researchers using AI and other computational techniques to tackle various aspects of the climate crisis. It currently has sixteen partner organizations.6 The initiative has released relevant open--source tools7 and provided grants in the form of cloud computing credits to projects using AI for a variety of purposes ranging from monitoring climate change in the Antarctic to protecting bird populations after hurricanes. Google?s AI4SG programme supports twenty organizations using AI to pursue various socially beneficial goals with funding and cloud computing credits. This includes projects seeking to minimize crop damage in India, better manage waste in Indonesia, protect rainforests in the US, and improve air quality

1? https://ellis.eu/programs/machine--learning--for--earth--and--climate--sciences.
2? https://copernicus--masters.com/prize/esa--challenge/.
3? https://www.exeter.ac.uk/research/environmental--intelligence/.
4? https://ai4er--cdt.esc.cam.ac.uk.
5? Conducted on 30 November 2020, the search of the Cordis research project database used the search string [(?climate change? OR ?global warming?) AND (?artificial intelligence? OR ?machine learning?)], (n?=?122).
6? https://www.microsoft.com/en--us/ai/ai--for--earth--partners.
7? https://microsoft.github.io/AIforEarth--Grantees/.

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in Uganda.8 Meanwhile, the development company ElementAI?s AI for Climate program provided (the company was acquired by ServiceNow in 2020) expertise and partnership opportunities to impr

ove the energy efficiency of manufacturing
and business operations. Finally, we shall see in Chapter 12 how there are many AI projects that support the UN SDGs.
However, there are also two main challenges associated with developing AI to combat climate change. One pertains to AI in general: it is the possible exacerbation of some social and ethical problems already associated with AI, such as unfair bias, discrimination, or opacity in decision making. I discussed these problems in Chapters 7 and 8. The other challenge is specific to climate change. It is less understood, and thus one of the main topics of this chapter. It concerns the contribution to global climate change from GHGs emitted by training data and computation--intensive AI systems: the current lack of information about the source and amount of energy used in researching, developing, and deploying AI models makes it difficult to define exactly the carbon footprint of AI.
To address this latter challenge, at the Digital Ethics Lab we specifically studied the carbon footprint of AI research and the factors that influence the GHG emissions of AI at the research and development stage (Cowls et al. 2021a). We also looked at the lack of scientific evidence concerning the trade--off between the emissions required to research, develop, and deploy AI, and the energy and resource efficiency gains that AI can offer. The conclusion was that leveraging the opportunities offered by AI for global climate change is both feasible and de-sirable. But it will involve a sacrifice (in terms of ethical risks and a potential increase in carbon footprint) in view of a very significant gain (a more effective response to climate change). It is, in other words, a green gambit. As I highlighted in Floridi (2014a), this gambit requires responsive and effective governance to become a winning strategy. For this reason, we offered some recommendations for policy-makers and AI researchers and developers in the aforementioned work. These recommendations were designed to identify and harness the opportunities of AI for combating climate change while reducing the impact of its development on the environment. This chapter is based on and updates that work.
11.2? AI and the EU?s ?Twin Transitions?
In 2020?1, European Commission President Ursula von der Leyen embraced the digital and ecological ?twin transitions? that will shape our future. High--level policy documents highlighted how EU digital initiatives (e.g. ?A Europe Fit for the Digital Age?), including AI, and ecological goals such as the ?European Green
8? https://impactchallenge.withgoogle.com/ai2018.

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Deal? can converge. This suggests a helpful starting point for policymaking. The need for financial stimulus resulting from the coronavirus pandemic also opened avenues for tailoring public assistance to promote these ?twin transitions?. The commission saw the two transitions as being deeply interrelated, to the point that the ?European Green Deal? proposal was full of references to the role of digital tools. A roadmap for the Green Deal released in December 2019 noted that:
digital technologies are a critical enabler for attaining the sustainability goals of the Green Deal in many sectors?.?.?.?[and that technologies] such as artificial intelligence?.?.?.?can accelerate and maximise the impact of policies to deal with climate change and protect th

e environment specifically in the areas of energy
grids, consumer products, pollution monitoring, mobility, and food and agriculture. The approach of the commission resonates with the idea, which I shall articulate in Chapter 13, of a new marriage between the Green of all our habitats and the Blue of all our digital technologies (Floridi and Nobre 2020). The use of AI to fight climate change is a leading example of such a Green & Blue marriage (Floridi 2019c). Finally, many of the documents produced by the commission to flesh out its digital and environmental vision refer to AI as a key tool, particularly in reference to ?Destination Earth? (an ambitious plan to create a digital model of the Earth and simulate human activity to test the potential effectiveness of Europe?s environmental policies). AI is explicitly mentioned as a key component of the Destination Earth initiative. The draft of the proposal for EU legislation on AI also identified AI as a potentially great tool to support sustainable policies (European Commission 2021). In short, the European focus on ?good AI? is in line both with EU values and funding and with scientific efforts around the world. But for all the documents that have been produced by the European Commission in which AI has been identified as a key tool, and for all the opportunities they highlight, they tend to overlook the challenges that must be addressed to ensure the successful and sustainable adoption of AI tools.

11.3? AI and Climate Change: Ethical Challenges

From ethical and privacy concerns to the significant amount of energy that goes into training and deploying AI tools, understanding and taking steps to address those challenges is crucial to ensure the sustainability of AI deployment in the battle against global warming.

A difficulty facing any effort to explain the role that AI can play in combating climate change is determining where the technology is already being used equit-ably and sustainably. There have been several efforts to create an accurate overview of how AI is being harnessed around the world in climate projects.

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However, the rapid pace of technological development has inevitably limited the accuracy of any single survey. Some approaches have focused on the United Nations SDGs (particularly those dealing most specifically with climate--related issues) as a starting point to identify AI projects. I shall analyse this approach in Chapter 12. Here, it is sufficient to anticipate that the University of Oxford?s Research Initiative on AIxSDGs9 includes tens of projects related to Goal 13, which most specifically focuses on climate change. But other databases using the SDGs as a starting point contain far fewer projects. This is a testament to the work that still needs to be done to improve the understanding of the many climate--related areas where AI has been deployed.

Against this background, it is important to recall that a key challenge involves the ethical risks associated with AI more broadly. We have seen in Chapters 7 and 8 that these risks are far more significant in areas such as healthcare and criminal justice, where data privacy is crucial, and decisions have a far greater impact on individuals. Even so, it is nevertheless vital to minimize the ethical risks that may arise when applying AI solutions to climate change. Because AI models are ?trained? using existing d

atasets, there is a potential for introducing
bias into such models due to the datasets chosen for training. Imagine, for
ex-ample, deploying AI to determine where to set up charging stations for electric
vehicles. Using the existing driving patterns of electric cars could skew the data
towards a wealthier demographic due to the relatively higher prevalence of elec-
tric vehicle use among higher income brackets. That, in turn, would create an
additional hurdle to increasing electric car use in less wealthy areas.
Another potential ethical pitfall concerns the erosion of human autonomy.
Tackling climate change requires large--scale coordinated action, including systemic
changes to individual behaviour. A careful balance must be struck here between
protecting individual autonomy and implementing large--scale, climate--friendly
policies and practices.
Finally, there is the problem of protecting individual and group privacy (Floridi
2014c). In control systems designed to decrease carbon footprints in a range of
contexts, such as energy storage (Dobbe et al. 2019), industrial heating and cool-
ing (Aftab et al. 2017), and precision agriculture (Liakos et al. 2018), the effective-
ness of AI systems depends on granular data about energy demands often
available in real time. While many AI solutions deployed in the battle against cli-
mate change rely on non--personal data, such as meteorological and geographical
data, some strategies for limiting emissions may require data pertaining to human
behaviour. This tension was highlighted in a survey10 of thirteen EU countries
produced by the Vodafone Institute for Society and Communications. The survey
9? This is a project I directed in collaboration with the Saïd Business School to provide a database of
AI projects that support the SDGs; see https://www.aiforsdgs.org.
10? https://www.vodafone--institut.de/wp--content/uploads/2020/10/VFI--DE--Pulse_Climate.pdf.

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showed that while Europeans are broadly willing to share their data to help
-protect the environment, a clear majority (53 per cent) would do so only under
strict conditions of data protection.
11.4? AI and Climate Change: Digital Carbon Footprint
Perhaps the greatest challenge associated with the use of AI in addressing climate
change is assessing and accounting for the carbon footprint of the technology
itself. After all, it is of little help if AI solutions contribute to alleviating one
aspect of climate change while exacerbating another. In the sense of both training
-models and uses, AI can consume vast amounts of energy and generate GHG
emissions (García--Martín et al. 2019, Cai et al. 2020). A significant share of the
carbon footprint generated by AI is associated with the computing power neces-
sary to train ML systems.
Such systems are trained by being fed vast quantities of data, which requires
correspondingly powerful data centres. Those data centres need energy to oper-
ate, in turn. I recall reminding people in the past that the heat of a laptop on your
legs was a clear sign of environmental impact. Today, after the advent of deep
learning, a type of ML that involves algorithms learning from huge amounts of
data, the computing power required for model training has doubled every 3.4
months. This has resulted in growing energy demand. The increase in energy
consumption associated with training larger models and with the widespread
a

doption of AI has been in part mitigated by hardware efficiency improvements. But depending on where and how energy is sourced, stored, and delivered, the rise of compute--intensive AI research can have significant negative environmental effects.

Many factors contribute to the carbon footprint of AI. One of these is the computing power necessary to train ML models. With the emergence of deep learning as a central technique for AI systems since 2012, the computing power required for model training has seen an exponential rise (Amodei and Hernandez 2018). The result is growing energy demand and hence carbon emissions. Its carbon footprint remains relative to where and how electricity is sourced, stored, and delivered. As an example, let us consider GPT--3, the third--generation, autoregressive language model that uses deep learning to produce human--like text. Its development is problematic from an environmental perspective. According to documentation published in May 2020, GPT--3 required an amount of computational power (compute) several orders of magnitude higher than its predecessor GPT--2, which was published only one year prior. Due to lack of information pertaining to the training conditions and development process of GPT--3, a number of researchers using different methodologies have attempted to estimate the cost of a single training run (Anthony, Kanding, and Selvan 2020). Several factors

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need to be tracked to determine the carbon footprint of an AI system, including the type of hardware used, the duration of a training run, the number of neural networks trained, the time of day, memory usage, and energy resources used by the energy grid supplying the electricity (Henderson et al. 2020). Missing some of these data can skew carbon assessments. Thus, one can estimate the cost of a single training run for GPT--3, but the publication does not reveal how many models were trained to achieve publishable results. Consider, too, that it is common for AI researchers to train thousands of models first (Schwartz et al. 2020).

Despite all these constraints, using information pertaining to the amount of computing power and type of hardware that was used by researchers at OpenAI to train GPT--3 (Brown et al. 2020), making assumptions as to the rest of the model training conditions (for more information, see Cowls et al. (2021a)), and using Lacoste et al.?s (2019) carbon impact calculator, we estimated that a single training run of GPT--3 would have produced 223,920 kg of CO2 (or equivalent, CO2eq). Had the cloud provider been Amazon Web Services, the training run would have produced 279,900 kg CO2eq. That is without considering the providers? accounting and offsetting techniques to achieve ?net zero? emissions. By comparison, a typical passenger car in the United States emits about 4,600 kg of CO2eq per year, meaning that a single training run using Microsoft Azure would emit as much as forty--nine cars do in a year (EPA 2016). Furthermore, geography is important. It is ten times more costly in terms of CO2eq to train a model using energy grids in South Africa than in France, for example.

The increasing availability of massive quantities of data has been a major factor fuelling the rise of AI. So, too, have new methods designed to leverage Moore?s law, according to which microchip performance doubles every two years. The intr

oduction of chips with multiple processor cores massively accelerated that development (Thompson et al. 2020). This innovation has enabled the development of increasingly complex AI systems, but it has also significantly increased the amount of energy needed to research, train, and operate them. The trend is well known and major data centre operators, such as cloud providers like Microsoft Azure and Google Cloud, have taken significant steps to reduce their carbon footprint by investing in energy--efficient infrastructure, switching to renewable energy, recycling waste heat, and other similar measures. Both pro-viders have leveraged AI to reduce the energy consumption of their data centres?-in some cases by up to 40 per cent (Evans and Gao 2016, Microsoft 2018).

The demand for data centres, which are key to the ICT sector and the op-er-ation of AI in research and production settings, has also grown substantially in recent years. On the one hand, the energy consumption of data centres has remained relatively stable (Avgerinou, Bertoldi, and Castellazzi 2017, Shehabi et al. 2018, Jones 2018, Masanet et al. 2020). The International Energy Agency reports that, if current efficiency trends in hardware and data centre infrastructure can be maintained, global data centre energy demand (currently 1 per cent of global

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electricity demand) can remain nearly flat through 2022 despite a 60 per cent increase in service demand. On the other hand, even in the EU, where energy--efficient cloud computing has become a primary issue on the political agenda, the European Commission estimates a 28 per cent increase in the energy consumption of data centres by 2030 (European Commission 2020). A source of uncertainty is the lack of transparency regarding the data required to calculate GHG emissions of on--premises data centres as well as cloud vendors. Also, calculating the carbon footprint of AI involves more than just data centres. It thus remains unclear whether increased energy efficiency in data centres will offset the rapidly rising demand for computational power. It is further unclear whether such efficiency gains will be realized equally around the world.

For all these reasons, it is crucial to assess the carbon footprint of various AI solutions used in different aspects of understanding climate change or in developing strategies to address specific aspects of it. But this, too, is problematic. Easy--to--use techniques for monitoring and controlling the carbon emissions of AI research and development are only now beginning to appear. Still, some approaches seem promising. The goal is to track several factors during model training phases to help assess and control emissions. The factors include the type of hardware used, the training duration, the energy resources used by the energy grid supplying the electricity, memory usage, and others.

Nevertheless, even the lower hurdles to reducing AI?s carbon footprint are -difficult to clear due to the lack of widespread adoption of such approaches and thus to the lack of information in many AI research publications. That can also lead to unnecessary carbon emissions when other researchers seek to reproduce the results of AI studies. Some papers do not provide their code, while others provide insufficient information about the training conditions of their models. Additionally, the number of experiment

s run by researchers before achieving
publishable results is underreported. Some experiments require the training of thousands of models during research and development phases only to achieve modest performance improvements. Massive amounts of computational power can go into fine--tuning. One example here is a research paper by Google Brain, which described the training of over 12,800 neural networks to achieve a 0.09 per cent improvement in accuracy (Fedus, Zoph, and Shazeer 2021). Modern AI research has tended to focus on producing deeper and more accurate models at the detriment of energy efficiency.

Focusing heavily on accuracy over efficiency improvements tends to create a high barrier to entry, since only research groups that are very well funded can afford the computational power required. Research teams from smaller organizations or those in developing countries are thus sidelined. It also institutionalizes an attitude of ?the bigger, the better? and incentivizes incremental improvements even if they are negligible in terms of practical utility. Some researchers are seeking to lower the computational burden and energy consumption of AI through

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algorithmic improvements and building more efficient models. And it is also vital to consider how even if training AI models is energy intensive, many of these models alleviate (or replace altogether) tasks that would otherwise require more time, space, human effort, and energy.

What the field needs to reconsider is its commitment to computer--intensive research for its own sake. It should move away from performance metrics that focus exclusively on accuracy improvements while ignoring their environmental costs. The EU has a vital role to play here. Given the positive role that AI can have in the battle against climate change and given Europe?s goals for both climate change and digitization, the EU would be a perfect sponsor in addressing the complexities associated with the technology?s own contribution to the problem and in meeting the need for coordinated, multilevel policymaking to ensure the successful adoption of AI solutions. That is why this chapter and the following recommendations are written from an EU perspective. As I already stressed before, it is not because the EU is the only or even the best actor in this area. Rather, it is a sufficiently significant actor that can make a difference and lead by example.

11.5? Thirteen Recommendations in Favour of AI against Climate Change

The following recommendations focus on the two broad objectives suggested above. First, there is the objective of harnessing the opportunities presented by AI to combat climate change in ways that are ethically sound. And second, there is the objective of minimizing the size of AI?s carbon footprint. The recommendations urge all relevant stakeholders to assess existing capacities and potential opportunities, incentivize the creation of new infrastructure, and develop new approaches to allow society to maximize the potential of AI in the context of climate change?-all while minimizing ethical and environmental drawbacks.

11.5.1? Promoting Ethical AI in the Fight against Climate Change

Performing comprehensive surveys and holding global conferences does not seem sufficient to gather, document, and analyse the use of AI to combat climate change. More needs to be done to identify and promote suc

on
systematic and accurate measurements along with continued energy--efficiency gains as global demand rises. In the fight against climate change, the positive impact of AI can be very significant. But it is crucial to identify and mitigate the potential ethical and environmental pitfalls associated with the technology. The EU?s commitment to defending human rights, combating climate change, and promoting the digital transition opens vast opportunities to make sure that AI can realize its potential. The right policies are crucial. If they are properly designed and implemented by policymakers, it will be possible to harness the power of AI while mitigating its negative impact?-paving the way for a fair, sustainable society, and a healthier biosphere. This is a point that holds true not only for climate change but for all the seventeen SDGs set by the UN, as I shall argue in the next chapter.

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12
AI and the UN Sustainable Development Goals

12.0? Summary

Previously, in Chapter 11, I analysed the positive and negative impact of AI on climate change and offered some recommendations to increase the former and decrease the latter. I mentioned that climate change is one of the areas where AI is being used to support the UN SDGs. As we saw in Chapter 9, initiatives relying on AI to deliver socially beneficial outcomes (so--called AI4SG) are on the rise. However, existing attempts to understand and foster AI4SG initiatives have so far been limited by the lack of normative analyses and a shortage of empirical evidence. Following the analyses provided in Chapters 9?11, this chapter addresses these limits by supporting use of the United Nations? SDGs as a benchmark for tracing the scope and spread of AI4SG. The chapter also presents in more detail a database of AI4SG projects (already mentioned in Chapter 11) collected using this benchmark. I discuss several key insights, including the extent to which -different SDGs are being addressed. The goal of the chapter is to facilitate the identification of pressing problems that, if left unaddressed, risk hampering the effectiveness of AI4SG initiatives.

12.1? Introduction: AI4SG and the UN SDGs

The SDGs were set by the United Nations General Assembly in 2015 to integrate the economic, social, and environmental dimensions of sustainable development.1 They are internationally agreed priorities for socially beneficial action, and thus constitute a sufficiently empirical and reasonably uncontroversial benchmark to evaluate the positive social impact of AI4SG globally. Using the SDGs to evaluate AI4SG applications means equating AI4SG with AI that supports the SDGs (AI?×?SDGs). This move, AI4SG = AI×SDGs (Cowls et al. 2021b), may seem restrictive because there is undoubtedly a multitude of examples of socially good

1? United Nations Development Programme 2015, ?Sustainable Development Goals?, https://www.undp.org/content/undp/en/home/sustainable--development--goals.html.

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uses of AI outside the scope of the SDGs. Nonetheless, the approach offers five significant advantages.

First, the SDGs offer clear, well--defined, and shareable boundaries to positively identify what is socially good AI (what

should be done as opposed to what should
be avoided). However, they should not be understood as indicating what it is not
socially good AI.
Second, the SDGs are internationally agreed--upon goals for development.
They have begun informing relevant policies worldwide, so they raise fewer ques-
tions about the relativity and cultural dependency of values. While inevitably
improvable, they are nonetheless the closest thing we have to a humanity--wide
consensus on what ought to be done to promote positive social change, advance
better standards of living, and conserve our natural environment.
Third, the existing body of research on SDGs already includes studies and metrics
on how to measure progress in attaining each of the seventeen goals and the 169
associated targets defined in the 2030 Agenda for Sustainable Development.2 These
metrics can be applied to evaluate the impact of AI?×?SDGs (Vinuesa et al. 2020).
Fourth, focusing on the impact of AI--based projects across different SDGs can
improve existing efforts. It can also lead to new synergies between projects address-
ing different SDGs, further leveraging AI to gain insights from large and diverse data
sets. Ultimately, it can pave the way to more ambitious collaborative projects.
Finally, understanding AI4SG in terms of AI?×?SDGs enables better planning and
resource allocation once it becomes clear which SDGs are under--addressed and why.
12.2? Assessing Evidence of AI?×?SDGs
In view of the advantages of using the UN SDGs as a benchmark for assessing
AI4SG, at the Digital Ethics Lab we conducted an international survey of AI?×?SDG
projects. Running between July 2018 and November 2020, the survey involved
collecting data on AI?×?SDG projects that met the following five criteria:

1) only projects that addressed (even if not explicitly) at least one of the
seven-teen SDGs;

2) only real--life, concrete projects relying on some actual form of AI (symbolic
AI, neural networks, ML, smart robots, natural language processing, etc.)
rather than merely referring to AI, which is a not insignificant problem
among AI start--ups more generally;3
2? United Nations Development Programme 2015, ?Sustainable Development Goals?, https://www.
undp.org/content/undp/en/home/sustainable--development--goals.html.
3? Ram, Allya, 4 March 2019, ?Europe?s AI Start--ups Often Do Not Use AI, Study Finds?, Financial
Times, https://www.ft.com/content/21b19010--3e9f--11e9--b896--fe36ec32aece.

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3) only projects built and used ?in the field? for at least six months, as opposed
to theoretical projects or research projects yet to be developed (e.g. patents
or grant programmes);

4) only projects with demonstrably positive impact, for example documented
through a web site, a newspaper article, a scientific article, an NGO report,
etc.; and

5) only projects with no or minimal evidence of counter--indications, negative
side--effects, or significant counter--effects.
Requirements (c) and (d) were crucial to unearth concrete examples of
AI?×?SDG, that is, projects with a proven record of robust, positive impact (as
opposed to identifying research projects or tools developed in laboratories and
trained on data that may prove to be inadequate or unfeasible when the technol-
ogy is deployed outside controlled environments). No constraints were assumed
about who developed the project, where or by whom it was used,

who supported
it financially, or whether the project was open source, except for projects conducted solely by commercial entities with entirely proprietary systems. These latter projects were excluded.
The projects were discovered via a combination of resources, including academic databases (ArXiv and Scopus), government press releases, patent filings, report tracking, public commitments to the UN SDGs made by organizations, and existing databases (including those at ITU and the Inter--American Development Bank?s fAIr LAC partnership). This approach built on existing work by Vinuesa et al. (2020), who used an expert elicitation process to ascertain which of the SDGs could potentially be affected by AI, by offering empirical evidence of actual benefits already being felt with respect to various SDGs.
From a larger pool, the survey identified 108 projects matching these criteria.
The data about the AI?×?SDG projects collected in this study are publicly available in the database (https://www.aiforsdgs.org/all--projects). This is part of the Oxford Research Initiative on Sustainable Development Goals and Artificial Intelligence (https://www.aiforsdgs.org/) that I directed in collaboration with Oxford?s Saïd Business School. We presented the first results from our research in September 2019 at a side event during the annual UN General Assembly, then again in 2020. The database we created was merged in 2023 with the database of AI for Sustainable Development Goals (AI4SDGs) Think Tank (available here: https://ai--for--sdgs.academy/)> The outcome of that research was published in Mazzi and Floridi (2023).
The analysis (see Figure 12.1) shows that every SDG is already being addressed by at least one AI--based project. It indicates that the use of AI?×?SDGs is an increasingly global phenomenon with projects operating from five continents.
But also, the phenomenon may not be equally distributed across the SDGs. Goal 3 (?Good Health and Well--Being?) leads the way, while Goals 5 (?Gender Equality?),

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16 (?Peace, Justice, and Strong Institutions?), and 17 (?Partnerships for the Goals?) appear to be addressed by fewer than five projects.
It is important to note that use of AI to tackle at least one of the SDGs does not necessarily correspond to success. Note also that a project could address multiple SDGs simultaneously or on different timescales and in different ways. Moreover, even a complete success for a given project would be exceedingly unlikely to result in the eradication of all the challenges associated with an SDG. This is chiefly because each SDG concerns entrenched challenges that are widespread and structural in nature. This is well reflected by the way SDGs are organized: all seventeen goals have several targets, and some targets in turn have more than one metric of success. The survey shows which SDGs are being addressed by AI at a high level. However, a more finely grained analysis is required to assess the extent of the positive impact of AI--based interventions with respect to specific SDG indicators, as well as possible cascade effects and unintended consequences. As I shall indicate in the conclusion, much research still needs to be done.
The unequal allocation of efforts detected by the survey may be due to the constraints of our survey criteria. But given the degree of systematic analysis and search conducted, it mor

theless, this finding highlights the risk that projects based (and funded) in one part of the world may not necessarily be responsive to actual needs elsewhere. Overall, this case study provides promising preliminary evidence that AI is in fact being used to tackle climate change and associated problems. As the cross--referencing effort above shows, this dovetails with wider research suggesting that AI could and should be developed and used for this purpose. As Rolnick et al. (2019) show, AI could support efforts to mitigate climate change in thirteen existing and potential domains ranging from CO2 removal, transport optimization, to forestry protection. The potential for AI to support climate action has also been recognized by a consortium of academics, NGOs, and energy companies who wrote to the UK government in 2019 to call for the establishment of an international centre for ?AI, Energy and Climate?.4

The previous overview indicates that there are already ?boots on the ground? using AI to tackle the climate crisis, even if such efforts are only at an early stage. Given the strict criteria applied in our sampling process (e.g. that projects must have had evidence of positive impact), our evidence shows positive direction. At the same time, it highlights that there is much more to be done with several existing gaps towards which prospective initiatives could orient their efforts.

12.4? Conclusion: A Research Agenda for AI?×?SDGs

A growing number of projects are using AI4SG by addressing the UN SDGs. AI technologies are not a panacea. But they can be an important part of the solution and help address major challenges, both social and environmental, facing humanity today. If designed well, AI technologies can foster the delivery of socially good outcomes at an unprecedented scale and efficiency. Therefore, it is crucial to provide a coherent structure within which new and existing AI?×?SDG projects can thrive.

4? Priyanka Shrestha, 2019, ?Leading Energy and Tech Groups Call for International Centre for AI, Energy and Climate?, Energy Live News, 20 August 2019, https://www.businessgreen.com/news/3080630/essential--ai--top--business--groups--call--for--creation--of--international--centre--for--ai--energy--and--climate

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The next steps in understanding AI4SG in terms of AI?×?SDGs are to analyse what factors determine the success or failure of AI?×?SDG projects, particularly with respect to their specific impacts on the ground; to explain gaps and discrepancies between the use of AI to tackle different SDGs and indicators, as well as mismatches between where projects are based and where SDG--related need is greatest; and to clarify what roles key stakeholders could play to advance the success of AI?×?SDG projects and address important gaps or discrepancies. Inevitably, all of this will need to be based on a multidisciplinary approach. It will have to involve deeper investigation of the AI?×?SDGs projects in the locations and communities where they are both developed and deployed. And it will require a broader perspective, at least in terms of what sort of human project we want to design and pursue in the twenty--first century. This is the topic of the next and final chapter.

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13
Conclusion
The Green and the Blue
13.0? Summary
Previously, in Chapters 1?12, I analysed the nature of AI, its positive and negative impact, the ethical issues arising from it, and what measures may be taken to ensure that a good AI society is more likely to develop. This final chapter draws some general conclusions and offers a view of what may be coming: a move from the ethics of artificial agency to the politics of social actions, to pursue a human project based on the marriage between the green of all our environments and the blue of all our digital technologies.
13.1? Introduction: From the Divorce of Agency and Intelligence to the Marriage of Green and Blue
Sometimes we forget that life without good politics, reliable science, and robust technology soon becomes ?solitary, poor, nasty, brutish, and short?, to borrow the phrase from Thomas Hobbes?s Leviathan. The COVID--19 crisis has tragically reminded us that nature can be merciless. Only human ingenuity and goodwill can improve and safeguard the living standards of billions of people and their environments. Today, much of this ingenuity is busy delivering an epochal revo-lution: the transformation of an exclusively analogue world into one that is also increasingly digital. The effects are already widespread. This is the first pandemic when a new habitat, the infosphere, has helped overcome the dangers of the biosphere. We have been living onlife (both online and offline) for some time, but the pandemic has made the onlife experience a reality of no return for the whole planet.
In the previous chapters, I have argued that the development of AI is an im-port-ant factor in this epochal revolution. AI should be interpreted as the engineering of artefacts to do things that would require intelligence if we were able to do them at all. This is illustrated by a classic example that I have used more than once: while only as intelligent as a toaster, a smartphone can beat almost anyone at chess. In other words, AI is an unprecedented divorce between the ability to complete tasks or solve problems successfully in view of a goal and any need to

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be intelligent in doing so. This successful divorce has become possible only in recent years, thanks to skyrocketing quantities of data, gigantic computational power at ever lower costs, very sophisticated statistical tools, and the transformation of our habitats into increasingly AI--friendly places, through a process I defined as enveloping. The more we live in the infosphere and onlife, the more we share our every-day realities with engineered forms of agency, and the more AI can deal with an increasing number of problems and tasks. There is no limit to AI but human ingenuity.
From this historical and ecological perspective, AI is an amazing technology, a new form of agency that can be a powerful force for good in two main ways. It can help us know, understand, and foresee more and better the many challenges that are becoming so pressing, especially climate change, social injustice, and global poverty. AI?s successful management of data and processes can accelerate the virtuous circle between more information, better science, and improved pol-icies. Yet knowledge is power only when translated into action. AI can be a remarkable force for good here, too, helping us improve the world and

not just
our interpretation of it. The pandemic has reminded us how the problems we face are complex, systemic, and global. We cannot solve them individually. We need to coordinate so we do not get in each other?s way, collaborate so we each do our part, and cooperate so we work together more, better, and internationally. AI can enable us to deliver these 3Cs more efficiently (through more results with fewer resources), efficaciously (through better results), and innovatively (through new results).
There is a ?but? here. We have seen that human ingenuity without goodwill can be dangerous. If AI is not controlled and steered equitably and sustainably, it can exacerbate social problems due to bias or discrimination; erode human autonomy and responsibility; and magnify past problems ranging from the digital divide and the unfair allocation of wealth to the development of a culture of mere distraction?-that of panem et digital circenses. AI risks going from part of the solution to part of the problem. Therefore ethical initiatives such as those described in Chapter 4 and, ultimately, good international regulations are essential to ensure that AI remains a powerful force for good.
AI4SG is part of a new marriage between the Green of all our habitats (whether natural, synthetic, and artificial, from the biosphere to the infosphere, from urban environments to economic, social, and political circumstances) and the Blue of our digital technologies (from mobile phones to social platforms, from the Internet of Things to Big Data, from AI to future quantum computing). With all its advantages, the marriage between the Green and the Blue counterbalances the divorce between Agency and Intelligence with all its risks. It is our responsibility to design and manage both successfully. The pandemic has made clear that what is at stake is not so much digital innovation as it is good governance of the digital. Technologies multiply and improve daily. To save our planet and ourselves

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(from ourselves), we can and must use them much better. Just think of the spread of COVID--19--related disinformation on social media, or the ineffectiveness of the so--called coronavirus apps. The pandemic could also be interpreted been the dress rehearsal of what should be the human project for the twenty--first century: a strong and fruitful marriage between the Green and the Blue. We can make it a success together by relying on more and better philosophy, not less.
13.2? The Role of Philosophy as Conceptual Design
For some time, the frontier of cyberspace has been the human/machine divide. Today, we have moved inside the infosphere. Its all--pervading nature also depends on the extent to which we accept its digital nature as integral to our reality and transparent to us (in the sense that it is no longer perceived as present). What matters is not moving bits instead of atoms. This is an outdated, communication--based interpretation of the information society that owes too much to mass--media sociology. Instead, what matters is the far more radical fact that our understanding and conceptualization of the essence and fabric of reality is changing. Indeed, we have begun to accept the virtual as partly real and the real as partly virtual. Thus, the information society is better seen as a neo--manufacturing society in which raw materials and energy have been superseded by dat

a and
information, the new digital gold, and the real source of added value. Aside from communication and transactions, the creation, design, and management of information are all key to a proper understanding of our predicament and the development of a sustainable infosphere. Such understanding requires a new narrative. That is, it requires a new sort of realistic and reliable story that we tell ourselves to make sense of our predicament and design the human project we wish to pursue. This may seem like an anachronistic step in the wrong direction. Until recently, there was much criticism of ?big narratives? ranging from Marxism and Neo--liberalism to the so--called ?end of history?. But the truth is that such a criticism, too, was just another narrative, and it did not work. A systematic critique of grand narratives is inevitably part of the problem that it tries to solve. Understanding why there are narratives, what justifies them, and what better narratives may replace them is a less juvenile and more fruitful way ahead.
Information and communication technologies (ICTs) are creating the new informational environment in which future generations will spend most of their time. Previous revolutions in the creation of wealth (especially the agricultural and the industrial ones) led to macroscopic transformations in our social and political structures and architectural environments. These occurred often without much foresight, and usually with deep conceptual and ethical implications. Whether understood in terms of wealth creation or in terms of a reconceptualization of our own selves, the information revolution is no less dramatic. We shall be in serious

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trouble if we do not take seriously the fact that we are constructing new -environments that will be inhabited by future generations.
In view of this important change in the sort of ICT--mediated interactions that we will increasingly enjoy with other agents (whether biological or artificial) and in our self--understanding, an ethical approach is vital to tackle the new challenges posed by ICTs. This must be an approach that does not privilege the natural or untouched, but treats as authentic and genuine all forms of existence and behaviour, even those based on artificial, synthetic, hybrid, and engineered artefacts. The task is to formulate an ethical framework that can treat the infosphere as a new environment worthy of the moral attention and care of the human agents inhabiting it.
Such an ethical framework must address and solve the unprecedented challenges arising in the new environment. It must be an e--nvironmental ethics for the whole infosphere, as I have argued for some time with little success (Floridi 2013). This sort of synthetic (both in the sense of holistic or inclusive, and in the sense of artificial) e--nvironmentalism will require a change in how we perceive ourselves and our roles with respect to reality, what we consider worthy of our respect and care, and how we might negotiate a new alliance between the natural and the artificial. It will require serious reflection on the human project and a critical review of our current narratives at individual, social, and political levels. These are all pressing issues that deserve our full and undivided attention.
Unfortunately, I suspect it will take some time and a whole new kind of education and sensitivity befo

re we realize that the infosphere is a common space?-one
that needs to be preserved to the advantage of all. Philosophy as conceptual
design (Floridi 2019d) should contribute not only with a change in perspective
but also with constructive efforts that may be translated into policies. It should
help us improve our self--understanding and develop a society that can take care
of both humanity and nature.

13.3? Back to ?the Seeds of Time?

In Chapter 2, we saw how Galileo thought nature was like a book written in mathematical symbols that could be read by science. This might have been a metaphorical stretch at the time, but today the world we live in is certainly becoming more and more a book written in digits, one that can be read and expanded by computer and data science. Within this book, digital technologies are increasingly successful because they are the true natives of the infosphere, like fish in the sea. This also explains why AI applications are better than us at an increasing number of tasks: we are mere analogue organisms trying to adapt to such a new habitat by living onlife. The epochal change of our environment into a mixed infosphere that is both analogue and digital, along with the fact that we are

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sharing the infosphere with increasingly smart, autonomous, and social AAs,
has profound consequences. Some of these ?seeds of time? (to rely again on
Shakespeare?s metaphor, introduced in Chapter 3) are still undetectable. We shall
discover them only in time. Others are barely discernible. And still others are in
front of our eyes. Let me start with these.

AI agents can be soft (apps, webots, algorithms, software of all kinds) or hard
(robots, driverless cars, smart watches, and gadgets of all kinds). These agents are
replacing human ones in areas that were thought off limits for any technology just
a few years ago. Artificial agents are cataloguing images, translating documents,
interpreting radiographies, flying drones, extracting new information from huge
masses of data, and doing many other things that only white--collar workers were
supposed to be able to do. Brown--collar workers in agriculture and blue--collar
workers in industry have been feeling the digital pressure for decades. Services
are just the new target. So, many white collar jobs will also disappear or be completely transformed. How many and how quickly, we can only reasonably guess?-
but the disruption is likely to be profound. Wherever humans work like interfaces
today, that job is at risk. For example, humans currently work as interfaces
between a GPS and a car, between two documents in different languages, between
some ingredients and a dish, between symptoms and the corresponding disease.
At the same time, new jobs will appear. New interfaces will be needed between
services provided by computers, between web sites, between AI applications,
between the outcomes of AI, and so forth. Someone will have to check that an
approximately good translation is a sufficiently reliable translation, and take
care of it. Many tasks will also remain too expensive for AI applications, even
assuming they will be doable by AI. Consider Amazon, for instance. We saw in
Chapter 2 that the company provides ?access to more than 500,000 workers from
190 countries?,1 so--called Turks, which are also defined by Amazon as ?artificial
artificial [sic] intelligence?. The repetit

ion here is indicative: these are mindless
jobs paid pennies. This is not the sort of job you may wish for your children, yet it is still a job of some sort that many people need and cannot turn down. Clearly, AI risks polarizing our society even further, unless we provide better legal and ethical frameworks. Polarization will occur especially between the few people above the machines (the new capitalists) and the many below the machines (the new proletariat). Taxes will also go with jobs, although this is a bit further in the future. No workers means no taxpayers?-this is obvious, and the companies that take advantage of the AI--fication of tasks will not be as generous as with their former employees when it comes to supporting social welfare. Something will have to be done to make companies and well--off people pay more taxes. Legislation will thus play an influential role also in determining which jobs will have to be
1? https://requester.mturk.com/create/projects/new.

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kept ?human?. Driverless trains will remain a rarity also because of le-gis-la-tive reasons,2 even though they are vastly easier to manage than driverless taxis or buses. Inevitably, regulations will contribute significantly to how we design the future of our infosphere and protect both humanity and the planet.
On a brighter note, many tasks that disappear will not make the corresponding jobs vanish. Gardeners helped by one of the many robot lawnmowers already available will have more time to do other things instead of cutting the grass. And many tasks will not actually disappear. Instead, they will simply be relocated onto our shoulders as users. We already push the buttons in the elevator (that job is gone). We are growing increasingly used to scanning goods at the supermarket (cashier jobs are going). And we will certainly do more jobs ourselves in the future. More generally, there will be a growing need for managers, carers, and curators of digital environments and smart agents. These new herds will need human shepherds (Heideggerian pun intended) or what I like to call ?green collars?. The point deserves further analysis.
13.4? Green Collars Needed
On 26 April 2015 the House of Commons? Science and Technology Committee of the British Parliament published online the responses to its inquiry on ?algorithms in decision--making?.3 The responses varied in length, detail, and approach, but appeared to share at least one feature: acknowledgement that human intervention may be unavoidable, even welcome, when it comes to trusting algorithmic decisions. This was sensible and it remains good news.
Automation has transformed agriculture and industry. Today, brown and blue collars are a minority. About 90 per cent of American jobs are in services and government. Most of us deal with data and software, not with bioware or hardware. The trouble is that computers eat data and software for breakfast. So, the digital revolution is challenging white--collar jobs everywhere. This is not because the digital makes technology intelligent?-if that were so, I would still have a chance playing chess against my idiotic smart phone. It is because the digital makes tasks stupid by decoupling them from any intelligence needed to perform them successfully. And wherever the digital divorces a task from intelligence, an algorithm can step in and replace us. The result is rapid and widespr

ead un-employ-ment
today, but also new jobs tomorrow. The long term is brighter. According to the World Bank, for example, the world will need 80 million healthcare workers by 2030, which is double the number in 2013.
In a society increasingly pervaded by algorithms and automated processes, the question addressed by the committee was how we can trust such brainless
2? https://www.vice.com/en/article/wnj75z/why--dont--we--have--driverless--trains--yet
3? https://publications.parliament.uk/pa/cm201719/cmselect/cmsctech/351/351.pdf

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technologies when they regularly make decisions for us and instead of us. Note that this really is about you and me. Since automation replaced brown collars, potatoes might face the risk of mishandling. Once blue collars go, robots might paint cars the wrong colour. Now that white collars are being replaced, everyone may be at the mercy of an algorithm?s mistake, an unfair identification of responsibility, a biased decision, or any other Kafkian mess concocted by a computer. Examples abound. We saw in the previous chapters that the risk is real and serious. The solution is to put some human intelligence back in place. This could be done in at least four ways: before, in, on, and after the loop.
Trust is based on deliverance, transparency, and accountability. You trust your doctor when she does what she is supposed to do. You are, or at least can be, informed about what she does, and she is responsible if anything goes wrong. The same holds true for algorithms. We can trust them when it is clear what they are supposed to deliver, and it is transparent whether they are delivering it. If there is any problem, we can trust that someone is causally accountable (or at least morally responsible), if not legally liable. Therefore we need humans before, in, on, and after the loop. We need humans before the loop because we want to design the right sort of algorithms and minimize risks. Humans should be in the loop because sometimes even the best algorithm can go wrong, get fed the wrong data, or be misused. We need control so that some consequential decisions are not entirely left to mindless machines. Humans should be on the loop because while crucial decisions may be too complex or time--sensitive for any human to make, it is our intelligent supervision that should check and manage processes that are both complex and fast. Humans should also be after the loop because there are always issues that we would like to redress or rectify, or different choices to make. The fact that a decision was made by an algorithm provides no grounds for giving up our right to appeal to human insight and understanding. So, we need design, control, transparency, and accountability operationalized by humans. And all this with all the help we can get from digital technologies themselves. You see why the responses by the committee were good news. There is plenty of intelligent work to be done in the future, but it will not be white collars who fill the new positions. It will be experts who can take care of the new digital environments and its AAs. Algorithms are the new herd. Our future jobs will be in the shepherding industry. The age of green collars is coming.
13.5? Conclusion: Humanity as a Beautiful Glitch
What about the ?seeds? less obvious, the consequences barely discernible, when AI will no longer be in the hands of researchers,

technicians, and managers
but ?democratized? in the pockets of billions of people? Here I can only be even more abstract and tentative. Along with predictive tools that can anticipate and ma-nipu-late human decisions, AI offers a historic opportunity to rethink human

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exceptionalism as not mistaken, but rather misdirected. Our intelligent behaviour will be challenged by the smart behaviour of AIs, which can adapt more successfully in the infosphere. Our autonomous behaviour will be challenged by the predict-abil-ity and manipulability of our rational choices, and by the development of artificial autonomy. And our sociability will be challenged by its artificial counterpart, as represented by artificial companions, holograms or mere voices, 3D servants or human--like sex robots. These counterparts can be both attractive to humans and sometimes indistinguishable from them. How all this will play out is unclear. But one thing seems predictable: the development of AAs will not lead to any scaremongering realization of some dystopian sci--fi scenario, a possibility that is irresponsibly distracting. Terminator is not coming. In fact, in this book I argued that AI is an oxymoron: smart technologies will be as stupid as our old technologies. But AI will invite us to reflect more seriously and less complacently on who we are, could be, and would like to become, and thus our responsibilities and self--understanding. It will deeply challenge our identity and ?exceptionalism?, in terms of what we mean by considering ourselves somewhat ?special? even after the fourth revolution (Floridi 2014a), according to which we are not at the centre of the universe (Copernicus), of the biosphere (Darwin), of the mental space (Freud), and now of the infosphere (Turing). Here, I am not arguing that our exceptionalism is incorrect. I am suggesting that AI will make us realize that our exceptionalism lies in the unique and perhaps irreproducible way that we are successfully dysfunctional. We are as unique as a hapax legomenon in Galileo?s book of nature, that is, as an expression that occurs only once in a text, like the expression ?gopher wood?, which refers to the primary building material for Noah?s ark and appears only once in the whole Bible. With a more digital and contemporary metaphor, we are a beautiful glitch in the great software of the universe, not the ultimate application. We shall remain a bug, a uniquely successful mistake, while AI will become more and more of a feature in Galileo?s mathematical book of nature. Such a beautiful glitch will be increasingly responsible for nature and history.
In short, Shakespeare was right:
Men at some time are masters of their fates.
The fault, dear Brutus, is not in our stars
But in ourselves, that we are underlings.
Shakespeare, Julius Caesar, 1.2
But we shall not be able to become the masters of our fate without rethinking another form of agency, the political one. Hence The Politics of Information, which is the topic of the next volume (see also Floridi 2023).

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